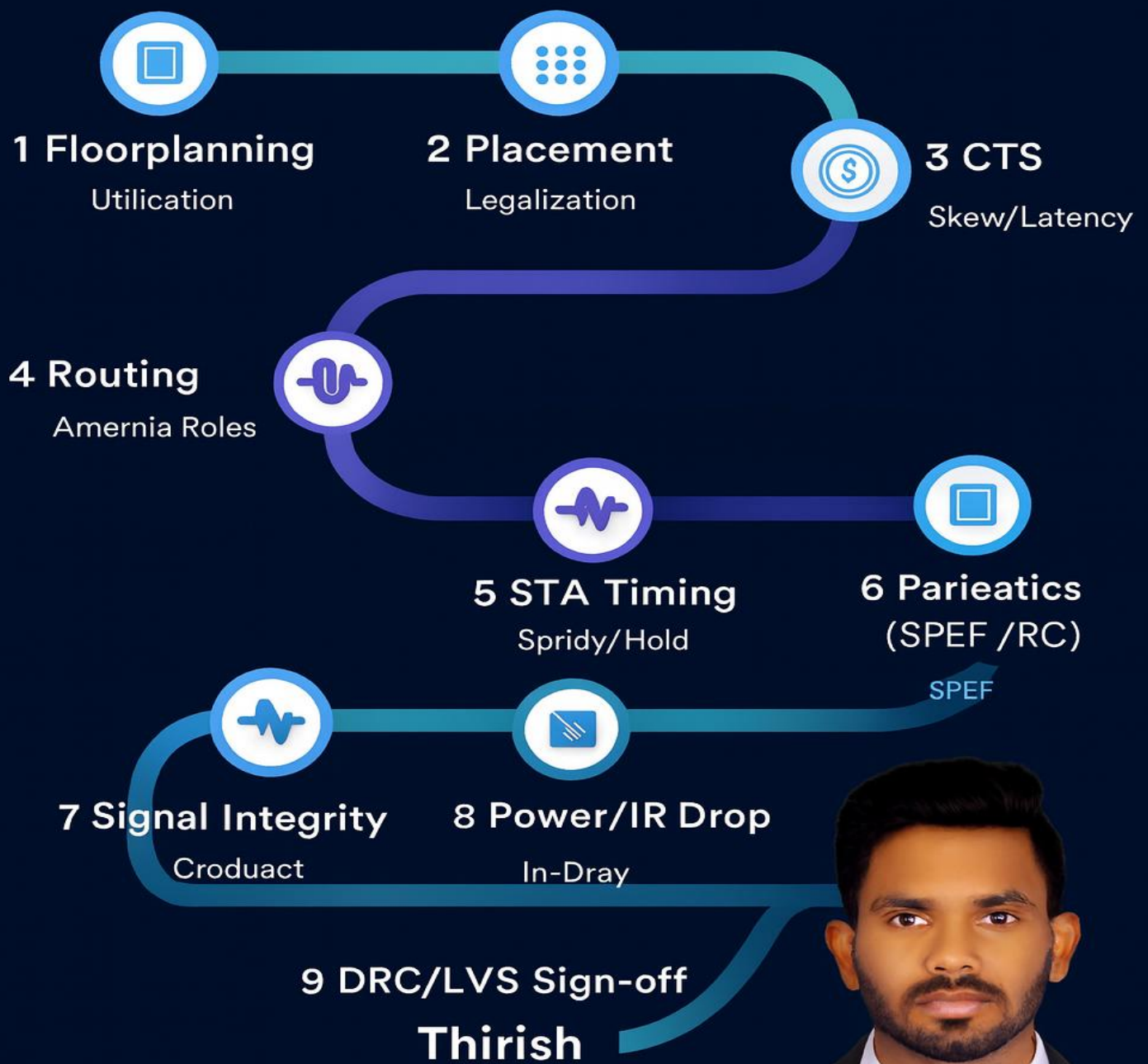


PHYSICAL DESIGN Q&A

METRO – 300 Interview Questions



300+ Must-Know Physical Design Interview Quesbons

Shared by:Thirish

Questions 1-10

1. What are the inputs required by any Physical Design tool and the output generated from it?

Input data required by physical design tools

- Technology file
- Physical Libraries
- Timing, Logical and Power Libraries
- TDF file
- Constraints
- Physical Design Exchange Format -PDEF (optional)
- Design Exchange Format -DEF (optional)

Output data required by physical design tools

- Standard delay format (.sdf)
- Parasitic format (.spef , .dspf)
- Post routed Netlist (.v)
- Physical Layout (.gds)
- Design Exchange format (. def)

Input data required by physical design tools

1. Technology file

(synopsys format: .tf cadence format: .techlef):

Cells, drawing patterns, layer design rules, vias, and parasitic resistances and capacitances during fabrication are described.

2. Physical Libraries

Overall, Lef files of GDS files are used for all design elements like macros, std cells, I/O PAD, etc. and above synopsys format .CEL, .FRAM views:

Contains complete layout information and abstract models for placement and routing, such as pin accessibility, blockages , etc.

3. Timing , Logical and Power Libraries

(.lib or LM view -.db for all design elements):

Contains timing and power information

4. TDF file (.tdf or .io) :

Contains pads or pin arrangements, such as the same order and location. For the full chip, the instantiated VDD and VSS PAD provide power to the Cut diode and so on. (Not in the Verilog netlist)

5. Constraints (.sdc):

Contains all design-related constraints such as area, power, and timing.

6. Physical Design Exchange Format -PDEF (optional):

Contains row (row), unit placement position, etc.

7. Design Exchange Format -DEF (optional):

Contains row (row), unit placement position, etc.

Output data required by physical design tools

1. **Standard delay format (.sdf)** : Timing details (except load information)
 2. **Parasitic format (.spef , .dspf)** : resistance and capacitance information of cells or nets
 3. **Post routed Netlist (.v)** can be flat or hierarchical: it contains connection information for all cells.
 4. **Physical Layout (.gds)** : physical layout information
 5. **Design Exchange format (. def)** : includes row, cell, net placement locations, etc.
-

2. What are sanity checks?

Sanity Checks mainly check the quality of the netlist in terms of timing. It also includes checking issues related to library files, timing constraints, IO and optimization instructions.

Some netlist sanity checks:

Floating Pins

Unconstrained Pins (unconstrained pins)

Un -driven i/ p Ports

Unloaded o/p Ports (unloaded o/p port)

- Pin orientation mismatch
- Multiple drivers, etc.

Other possible questions

- Unconnected/wrongly connected Tie-high/Tie-low pins
- Power pin (because Tie-up or Tie-down connections are always through Tie-Cells)

3. What do I need to do to start a Floor plan?

(1) First do data input: input .v, .lib, .lef, .SDC and other data. **[This is the first important step in completing the floor plan.]**

(2) Define the chip/block size, allocate power routing resources, place hard macros, and reserve space for standard cells. **[Floor plan determines chip quality]**

4. How to implement PD (physical design)?

Flat

1. Small and medium-sized application-specific integrated circuits
2. Better area usage since there is no spare space around each sub-design for power/ground

Hierarchical

1. for very large designs
2. When subsystems are designed individually
3. Only possible if there is a design hierarchy

5. What are the guidelines for placing macros?

Place macros around the chip

If you don't have a valid reason to place macros inside the core area, then place macros around the periphery of the chip.

Since there are many cases of detour routing (many lines around), placing macros inside the core may have serious consequences during the routing process, because macros are a big obstacle to routing.

Another advantage of placing hard macros on the periphery of the core is that it is easier to power them and reduces the IR drop problem of macros that consume a lot of power.

When placing macros, consider connections to fixed cells:

When determining the location of macros, attention must be paid to connections to fixed elements such as I/O and preset macros. Place the macro near its associated fixed element. **[Check connections by displaying flight lines in GUI]**

Directional macros to minimize distance between pins

When determining the direction of your macro, you must also consider pin locations and their connections.

Leave enough space around macros

Leave enough space for wiring

As with regular network cabling and power networks, you must leave adequate cabling space around the macro. In this case, it is important to accurately estimate routing resources. Use the congestion map in the trial Route to identify hot spots of congestion between macros and adjust their locations as needed.

Reduce open fields as much as possible:

In addition to reserved routing resources, remove dead zones to increase the area for random logic. Selecting a different aspect ratio (if that option is available) eliminates open fields.

Reserve space for the power network:

The amount of power wiring required can vary based on one's power consumption. You must estimate power consumption and allow enough space for the grid. If you underestimate the space required for power cabling, you may encounter cabling problems.

6. What happens if pins are assigned to the left and right sides. (if you have IO pins on top and bottom)?

The top chip will actually be divided into several blocks, and the IO pins will be placed according to the communication between the surrounding blocks.

If we assign the pins to the left and right sides instead of the top and bottom, we will face wiring problems at a later stage.

7. Allocate spacing between two macros?

channel spacing= no of pins*pitch/ (total number of metal layers/2)

8. What happens if we align macros?

There are two situations

- . Macros can be aligned if they only communicate with each other.
- . If the macro communicates with other cells (std cells and IO ports), then proper channel spacing must be provided between the macros, otherwise, there will be wiring issues.

9. Can we place macros in 90 and 270 ° directions?

It depends on which technology you are looking into.

Foundries of 45nm and below have orientation requirements. Poly orientation should be the same throughout the chip. Therefore, the poly direction of the macro should match the poly direction of the standard cell.

10. In power planning, which metal layers are used for ring and stripe , and why?

For rings and stripes, we use the top metal layer because the top metal layer has low resistivity.

☆ High-rise buildings are more suitable for global routing. The low-level usage rate is relatively high. If used for power, it will occupy some useful resources. For example, std cell is usually m1 Pin.

☆ EM capabilities are different. Generally, the top layer is 2~3 times that of the lower layer. Better suited for power wiring. The top metal is usually thicker and can pass larger currents

☆ Generally, the layers occupied by IP addresses are close to the lower layers. If the upper layer is not prohibited from routing, the top layer can be traversed, but it is impossible for the lower layers, and the noise impact of the upper layers on the lower layers is much smaller.

Questions 11-20

11. Can we place cells in the space between IO and core boundaries?

No, we cannot place cells between the space of IO and core boundaries, because between IO and core boundaries there will be a power ring placed and there will be cabling issues.

12. What type of congestion have you seen after placement ?

- . Due to insufficient placement blockage, there is congestion near the Macro corner.
- . Placing standard cells in narrow aisles can lead to congestion.
- . Macros of the same module may cause timing violations if they are far apart.
- . The placement of Macros or the channels of Macros are inappropriate.
- . No placement blockage is given.
- . No macro-to-macro channel space is given.
- . High cell density.
- . High local utilization rate.
- . A large number of complex cells (such as AOI/OAI cells with many pins) are placed together.
- . Place standard cells near macros.
- . Logic optimization was not completed correctly.
- . The pin density is more on the edges of the block.
- . Too much Buffer is added during optimization.
- . The IO ports are crisscrossed; it needs to be properly aligned in order.

13. What are physical cells?

End Cap cells:

- . These cells prevent cell damage during the manufacturing process.
- . Used for row connections and specifies the end of row.
- . Avoid drain-source short circuit.
- . These are used to solve the boundary N-Well problem of DRC cleaning.

Well tap cells:

- . These are used to connect VDD and GND to the substrate and N-Well respectively as it creates less drift to prevent latch-up.
- . If the specified distance between well taps is maintained, the N-Well potential will produce normal electrical function.
- . Limit the resistance between power and ground connections to the substrate well.

De-cap Cells:

They are temporary capacitors added between the power and ground rails in the design to cope with functional failures due to dynamic IR drops.

Avoid triggers that are far away from the power supply from entering a metastable state.

Filler Cells:

Fills empty areas and provides connectivity to the N-well and implant layers.

14. What are the related contents of Non Default Rules (NDR)?

Double width and double space. (Double width and double space.)

After the PNR stage, if you will encounter timing / crosstalk / noise violations that are difficult to fix in the ECO stage, we can try this NDR option in the route stage .

NDR usage and examples

When we route special nets like clocks, we want to give them greater width and greater spacing. Replace the default 1 unit spacing and 1 unit width in the technical file;

But NDR has double spacing and double width. When the clock network uses NDR wiring, the signal integrity is better, the crosstalk is smaller, and the noise is smaller, but we cannot increase the spacing and width because it will affect the chip area.

Double spacing : used to avoid crosstalk.

Double width: used to avoid EM.

15. What are setup and hold?

SETUP: The minimum time required for data to stabilize before the clock edge.

HOLD: The minimum time required for data to stabilize after the clock edge.

16. Can setup check be performed during the placement stage?

Yes, we will check the setup in the placement phase, and we won't worry about the hold because the clock is the idea in the placement phase.

17. What are the methods to repair setup and hold violation ?

A. Setup:

- . Upsizing the cells (enlarging the cells)
- . Replace buffer with two inverters (use 2 inverters for 1 buffer)
- . HVT to LVT (high voltage threshold cell is replaced by low voltage threshold cell)
- . If the net delay is more than break the net and insert the buffer
- . Pin swapping
- . Pulling the launch and pushing the capture
- . Cloning

B. Hold:

- . Inserting the buffers
- . Downsizing the cells
- . LVT to HVT (low voltage threshold cell is replaced by high voltage threshold cell)
- . Pushing the launch and pulling the capture

18. How do you know if you have a max cap violation?

```
report_timing - all violators
```

19. How to use High Vt and Low Vt to reduce power consumption (Power Dissipation) in design?

1. Use HVT units for timing paths with +ve slacks.
2. Use LVT units for timing paths with -ve slacks.
3. HVT cells have greater latency but less leakage. In design, +ve slack is useless because only some paths work faster and doesn't help the overall design. If slack is 0, good. In this case, by using the HVT unit to give up slack, gain in power consumption.
4. LVT units are very fast but have high leakage. Limit the use of LVT units to paths where timing closure is difficult.

20. What is electromigration (Electromigration EM) and how to solve it?

Explanation 1:

Electromigration (EM) refers to the phenomenon of movement of metal atoms due to momentum transfer from conducting electrons to metal atoms. Due to EM effects, current conduction in metal paths over a period of time can lead to open or short circuits. Electromagnetic effects are unavoidable.

To minimize the effects of electromigration (EM), we use wider traces so that even with EM effects, the traces remain wide enough to conduct electricity throughout the life of the IC.

Explanation 2:

Due to the high current flow in metal atoms in metals can be displaced from their original positions. When it happens in large quantities, the metal opens up or the metal layer expands. This effect is called electromigration.

Impact: The signal line or power line is short-circuited or open-circuited.

Questions 21-30

21. Why is IR drop analysis important?

The IR drop determines the voltage level on the standard cell pins. The acceptable IR derating will be determined at the beginning of the project and is one of the factors used to determine the derating value. If the value of IR drop is greater than the acceptable value, it requires changing the derating value. Without this change, timing calculations become optimistic. For example, the setup slack calculated by the tool is smaller than the actual value.

22. If you encounter IR drop and congestion problems at the same time, how will you fix them ?

Spread Marcos out

spread out standard units

Increase the width of the strap

Increase the number of straps

Proper use of blockage

23. In the Reg to Reg path, if you have a setup problem, where would you insert the buffer - near the start trigger or the capture trigger? Why?

1. Buffers are inserted to fix fanout violations, so they reduce setup violations; otherwise we would try to fix setup violations by the size of the cell; now assume you have to insert buffers!)
2. Close to the capture path.
3. Because there may be other paths through or originating from a trigger that is closer to the initiating trigger. Therefore inserting the buffer may also affect other paths. It may improve all these paths or downgrade them. If all these paths are violated, then you can insert the buffer closer to the launch trigger if it improves slack.

24. Why are Buffers used in Clock Tree?

In order to balance skew (such as flop to flop delay)

25. What is Cross Talk?

Switching signals in one network can interfere with adjacent networks due to cross coupling capacitance . This effect is called crosstalk. Crosstalk may cause setup or hold violation.

26. How to avoid Cross Talk?

1. Double spacing → Wider spacing → Smaller capacitance → Smaller crosstalk effect
2. Porous (via) → smaller resistance - smaller RC delay
3. Shielding → constant cross-coupling capacitance value → known crosstalk value
4. Insert Buffer → Enhance victim driving force

27. How does Shielding avoid Crosstalk problems? What exactly happens?

1. Because the shielded layers are connected to VDD or VSS. , high-frequency noise (or glitch) couples to VSS (or VDD)
2. The coupling capacitance is constant with VDD or VSS.

28. How does spacing help reduce Crosstalk Noise?

The larger the width → the larger the spacing between the two conductors → the smaller the cross-coupling capacitance → the smaller the cross talk

29. How is Buffer used in Victim to avoid Crosstalk?

Buffer increases the victim signal drive strength;

buffers interrupt net length → victims are more tolerant of coupled signals from aggressors.

30. Why is Setup checked at the max corner and Hold at the min corner?

For setup,

required time should be greater than arrival time. When the arrival time is large, there are setup violations.

Therefore, when the arrival time is large or when the launch clock arrives later than the capture clock, the setup check is more pessimistic.

This means the delay is larger. Therefore, the setup will check at max delays.

For hold,

Arrival time should be greater than required time. When the required time is large, there are hold violations.

Therefore, when the required time is large or when the launch clock arrives earlier than the capture clock, the hold check is more pessimistic.

This means less data arrival time. Therefore, hold will be checked at min delays.

Questions 31-40

31. Why is hold not checked before CTS?

Before CTS, clock was ideal. This means there is no exact skew. All clocks arrive at flops at the same time. Therefore, we do not have the skew and transition values of the clock path, but this information is enough to perform setup analysis because setup violations depend on the data path delay.

Only after CTS, the clock is propagated (the actual clock tree has been built, clock buffers have been added to the clock tree, and there is already a clock tree hierarchy, clock skew, insertion delay).

That's why hold violations are fixed after CTS.

32. Can Setup and Hold violations appear in the same start points and end points at the same time?

Yes, if they have different combinational logic paths.

33. What is the derate value that can be used?

1. For setup check, reduce the data path by 8% to 15%, with no derate in the clock path.
2. For the hold check, the clock path is reduced by 8% to 15%, with no derate in the data path.

34. What are the corners checked for timing sign-off? Is there any change in the derate value of each corner?

- . Corners: Worst , Best, Typical.
- . For the best and worst corners, the derating value is the same and has no change; for the typical corner, the value may become smaller.

35. Where to get WLM? Do you create WLMs? How to specify WLM?

1. Wire Load Models (WLM) are available from library vendors.
2. We don't create WLM.
3. WLM can be specified based on area.

36. Where do you get the derating value? What are the factors that determine the derating factor?

1. Determine derating values based on library vendors' guidelines and recommendations and previous design experience.
2. PVT change is the factor that determines the derating factor.

37. How to repair Setup during placement? How to fix Setup and hold during CTS?

How to fix setup Violation

Placement stage:

- **Timing path groups:**

In the placement stage, we can use the group path option to solve Setup timing.

Group a set of paths or endpoints for latency cost function calculations. The delay cost function is the sum of all groups (weight * violation), where violation is the number of setup violations for all paths in the group. If there is no violation in a group, its delay cost is 0.

group enables you to specify a set of paths to optimize even though there may be larger violations in another group.

When endpoints are specified, all paths leading to those endpoints are grouped.

ICC syntax:

```
group_path [-weight weight_value] [-critical_range range_value] -name group_name  
[-from from_list] [-through through_list] [-to to_list]
```

Example:

```
group_path -name "group1" -weight 2.0 -to {CLK1A CLK1B}
```

- **Create Bounds:**

We can constrain the placement of relative placement cells by defining move bounds using fixed coordinates .

Related placement units support soft bounds and hard bounds, as well as rectangular bounds and rectilinear bounds (rectangular bounds and rectilinear bounds).

To constrain relative placement by using move bounds, use the command

ICC command:

```
create_bounds -coordinate {100 100 200 200} "U1 U2 U3 U4"-name bound1
```

- ♦ **place_opt**

If the design has a timing violation, you can rerun the **place_opt command with the -**

timing and **-effort high** options .

ICC command:

```
place_opt -mtiing-driven -effort high
```

Timing driven placement strives to place cells together along the timing critical path to reduce net RCs and meet setup timing.

- **Change the Floorplan**

In order to better meet the timing, change the Floorplan (macros placement, macro spacing and pin direction)

CTS stage

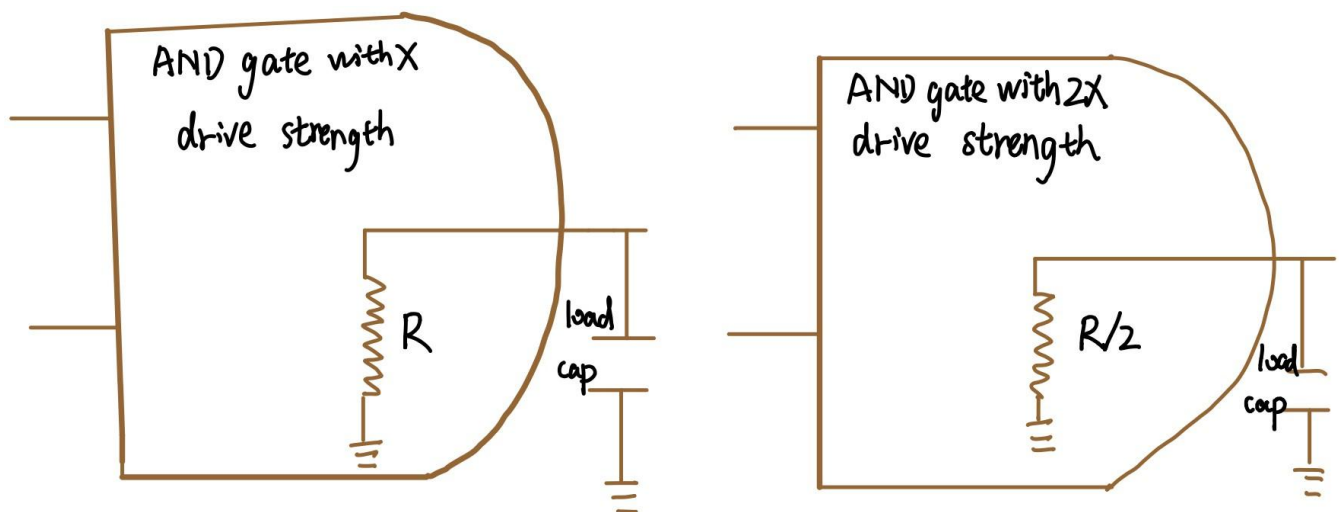
- **Increase the drive strength of data-path logic gates:**

Cells with better drive strength can quickly charge the load capacitor, which means smaller propagation delay.

Moreover, the output transition should be improved and there should be a better delay in the program phase.

A gate with good drive strength will have a smaller resistance, which can effectively reduce the RC time constant; therefore, it can provide less delay. This is illustrated in the figure below.

If an AND gate with 'X' drive strength has a pull down resistance of 'R', the other AND gate has '2X' drive strength with a resistor value of $R/2$. Well, a bigger AND gate with better drive strength has smaller delay.



- **Use data-path cells with lower threshold voltage:**

Replace HVT. Means changing HVT unit to SVT/RVT or LVT. Low V_t reduces conversion time and therefore propagation delay is reduced. Therefore, replacing HVT with RVT or LVT will speed up timing.

- **Buffer insertion**

If the net length is very long, then we insert the Buffer. It reduces transition time and thus wire delay. If the amount of line delay is reduced due to a reduction in conversion time > the buffer's unit delay, then the overall delay is reduced.

- **Reduce the number of buffers in path**

This will reduce cell delay but increase wire delay.

Therefore, if we can reduce the cell delay more than the wire delay, the active stage delay will increase.

1. **Use higher metal layers to route nets**

2. **Replace 1 buffers with 2 inverters:**

Adding an inverter reduces the transition time by 2 times compared to the existing buffer gate . Therefore, the RC delay of the wire is reduced.

Cell delay: 1buffer gate = 2 inverter gate

1. **Use clock skew (useful skew):**

Positive skew helps improve setup slack.

Therefore, in order to fix the setup violation, we can choose to increase the clock latency of the capture flip-flop, or reduce the clock latency of the transmit flip-flop.

However, in doing so , we need to carefully consider the setup and hold slack of other timing paths formed from/to these flip-flops.

This is called Useful Skew. Therefore, generally speaking , Useful skew is to deliberately add delay intentionally in the clock path to meet better timing.

How to fix Hold Violation

Hold violation is the opposite of setup violation.

A Hold violation occurs when the data is too fast compared to the clock speed.

To fix the hold violation, you should increase the delay in the data path.

Increase the drive strength of data path logic gates

Use data-path cells with high threshold voltages

Insert/remove Buffer

Use higher metal layers to route nets

Add the delay from the transmit flip-flop clock port to the data output port (clk ->q)

38. Why not derate the clock path by -10% for worst corner analysis ?

can do. But it may not be accurate because the data path is derated.

39. What is the importance and requirement of MMMC files in VLSI physical design?

- . Multi-Mode Multi corner (MMC) files during physical design analyze the design with different modes & corners.
- . VLSI designs can be modeled in modes such as functional or test mode, each located at different process corners.
- . We need to ensure that the design is stable in all corners, specifically the technical term PVT Corners (Process, Voltage & Temperature process, voltage and temperature).
- . During the physical design process, (prescribed Tool-Cadence, synopsys, etc.) the MMMC file captures all relevant details to obtain the desired design.

40. What is Timing DRV's, explain its causes and how to fix it?

Timing Drvs:

1. Max Tran
2. Max Cap
3. Max Fanout

Causes:

- **HVT cells have slower transitions**

Compared with LVTs and RVTs, HVT cells have larger threshold voltages. Therefore, it takes more time to turn on the cell, which results in a larger transition time.

- **Weak Driver**

The driver cannot drive load, which will cause the driven cell to have poor transitions. Therefore, delay will be increased.

- **Load a lot**

A driving cell cannot drive a load that exceeds its characteristics. This is the maximum cap value set in .lib to use. If the load on a cell is increased beyond its maximum capacitance value, then it will cause poor switching, thereby increasing latency.

- **net length is very long**

The larger the net, the larger the resistance, and the worse the transition. This leads to trans violation. Long net RC values also increase the load on the unit, resulting in max cap violations.

1. **Fanout is too large:**

If the number of fanouts increases beyond the characteristic limits of the drive unit, it will lead to max fanout violations. The increased load leads to max cap violation, which also indirectly leads to max tran violation.

Fixes:

Max Tran:

1. Use LVT cells instead of HVT cells
2. Increase driver size
3. By increasing the buffer, reduce the net length. The larger the net, the greater the resistance. Put a buffer in the middle of a long net and divide the resistor into two halves.
4. Reduce load by reducing the size of fanout and driven cells.

Max Cap:

1. Increase driver size
2. By increasing the buffer, reduce the net length.
3. Reduce the load by reducing fanout (via load splitting) or downsizing the drive unit.

Max Fanout:

1. Reduce fanout by splitting the load through buffer insertion or cloning.

Questions 41-50

41. Why do we emphasize setup violation before CTS and hold violation after CTS?

The setup time of a valid timing path depends on: the maximum data network calculation time and the time when the clock edge reaches the sink.

Before the POST CTS stage, we assume that all clocks are in the ideal network and that it can reach every possible clock sink of the chip in 0 time!

What we need to focus on is implementing the data path in such a way that it should not take at least more than one clock cycle from start to end. (Assuming a full cycle valid timing path).

Of the two components of the setup timing check, one is always a constant (the period of the clock) and the other variable is the data path delay. We have all options to use this variable until the CTS stage is completed.

If we cannot achieve this stretch goal before CTS, it will be difficult to converge timing later.

Therefore, until the CTS stage, we focus on obtaining data path synthesis or data network physical implementation alone.

I hope it's clear why we focus on setup timing before the CTS stage.

Let's look at it from another perspective, why don't we just focus on hold time?

The hold time of the path depends on: the minimum data path delay and the clock edge time.

Since the clock arrives at each receiver of the chip in 0 seconds, and at least, the data path delay will always be larger than the hold request (hold req) of the flip-flop/timing path endpoint.

That's it, unless the clock path network delay changes, there is no need to analyze the hold timing of the valid paths. (But at least, you can look at the total hold timing path to see if it's FP/MCP.)

42. After placement, there is a setup violation. What should we do? Even though we have completed the optimization.

Setup violation after placement is not worth worrying about. Well, unless it comes from improper module placement. Look at the macro layout and module layout to see if there are any problems.

For example, if there is a module for instruction fetch and it is split and placed in two or three different clusters, then we might want to constrain it with module placement guidelines or boundaries.

During the placement phase, let the tool have the correct constraints, mark the timing effort flag as high, and perform another round.

Carry out multiple rounds of CTS and routing phase optimization. Each of these will try to revisit the problem and will make some improvements.

I've seen some bad slacks, like -500 ps and 30000 plus path failures, but these were actively handled by STA's timing team. (Utilize things like upsize, fix max cap, max fanout and max transition, put in lvt units, etc.)

****Additional note:**

The routing engine and timing engine used in the place phase are not signoff quality and are far from what tools like tempus or primetime can evaluate.

43. What does the insertion delay in VLSI physical design mean?

1. The concept of insertion delay comes from clock tree synthesis.
2. When building the clock tree, cts starts building the clock from the clock source to the sink.
3. Once the clock is built, now the clock signal must be transmitted from the source to the sinks. The time it takes for a clock signal to travel from source to receiver is called insertion delay.

example:

The clock source is at point A, so the clock is built from point A and it must reach points B, C, and D receivers (flip-flops).

So from point A to points B, C, and D, the clock signal must propagate. But in between it will set up some logic to balance all three receivers, since the signal has to reach 3 receivers BCD at once, this is called Skew Balance (the main goal of CTS).

The time it takes for the clock signal to travel from point A to BCD is called insertion delay. You can refer to the LATENCY concept for more in-depth information.

44. In VLSI Physical Design, why don't we route before CTS?

1. Once your design is at the stage where all data and clock logic networks have been properly balanced and synthesized, it's time to route them. Laying the actual metal wire requires placing all design objects (units) in legal locations. The post-placement phase is when we reach this point. But this doesn't mean your design is ready for routing, you should consider other high fanout nets and clock net signals after placement. Before this stage, the clock is an ideal network (assuming it can drive any amount of load without any buffering).
2. During logic synthesis, we do not balance HFN and clock networks, so a single clock port may drive thousands of flip-flops (even after placement there is virtual routing). CTS is the stage that synthesizes this loading into a balanced tree to have minimum bias and latency for all receivers (flip-flops).
3. You should not route anything until you have completed the logic synthesis of the clock. After completing the CTS, you can begin routing the design clock first and then the data signals. Let me know if any clarifications are needed.

45. What is path group in VLSI and why should we use it?

As the name suggests, it is a set of paths.

The reason for path grouping is to guide the work of the synthesis engine .

For example let's assume you start with all paths in a single path group.

In this case, the synthesis engine will spend most of its time optimizing the logic of the worst-case violators . Once the time requirement is met, it moves on to the next worst-case violators , and so on.

Now review the initial timing report that you may have identified .

Some paths require architectural changes (e.g. cascaded adders/multipliers will be replaced by pipeline logic), so you don't want the synthesis engine to spend too much time optimizing this logic. Make it a separate path group with lower priority.

Because all the effort is spent on high violation Paths , no low violation Paths are optimized . Make separate path groups for these two groups.

46. What are the benefits of setting up separate path groups for I/O logical paths in VLSI?

1. Path groups form the basis of optimization functionality in tools that perform synthesis and PnR. Now, more realistic path groups make the tool easier to optimize in all aspects.
2. Now most of the time our I/O constraints are budgeted and cannot be actual. Additionally, they may not be clean from a clock domain perspective. Therefore, they may affect qor if they remain in the same group as internal paths. Furthermore, the tool works on the most critical path and tries to optimize below a specific range called the critical range. If the IO path is the most critical path, then the tool may not work on the internal path and is therefore a suboptimal design.

47. When repairing timing, how do I find the false path in VLSI design?

false path is a very commonly used term in STA. It refers to a timing path that never needs to be captured within a limited time when the chip is working normally, so there is no need to optimize the timing. Under normal circumstances, a signal from a flip-flop must be captured by another flip-flop within one clock cycle.

However, in some cases it doesn't matter when the signal from the transmit flip-flop reaches the receive flip-flop. The timing path leading to this condition is marked as a false path, and the optimization tool does not optimize for timing.

48. On the clock gating path, what makes meeting timing very challenging? What makes it more important than the regular setup/hold flop to flop timing path?

1. When building the clock tree, we try to balance all flip-flops. This allows the clock gate (CG) in the early clock tree to drive a bunch of flip-flops through its own delay. This results in the clock gating latch clock period minus the delay resulting in the time available to satisfy the setup, thus making it more tightly satisfied.
2. Now if the clock gate's fanout exceeds its drive capability, a larger treelet (or perhaps 2 parallel buffers) will appear, causing the clock to reach the clock gate earlier, making it more difficult to meet the setup.

49. What is the difference between static IR drop and dynamic IR drop analysis?

Static IR drop is the voltage drop when a constant current flows through a power network with varying resistance. This IR drop occurs when the circuit is in steady state.

Dynamic IR voltage drop is the voltage drop when high current consumption of the power network occurs due to high switching of cell data. Due to reduced static electricity, you should increase the width of the grid or have to design a robust grid to reduce dynamic IR drop, lower the trigger rate or place decapacitance units near high switching units.

50. What is required for static IR drop analysis?

IR drop is the voltage drop in the metal wire from the grid before it reaches the VDD pin of the standard cells. Due to IR drop, timing issues may occur due to changes in VDD value.

Questions 51-60

51. What is GDSII file?

GDS (Graphic Data Stream) is a file developed by calma in 1971 and GDS II in 1978.

It is a binary file format that represents layout data in a hierarchical format.

There are data such as labels, shapes, layer information and other 2D and 3D layout geometry data.

This file is then provided to the fabrication house who uses the file to etch the chip according to the parameters provided in the file.

52. What are SDF files related to VLSI physical design?

SDF stands for Standard delay format. It provides information on timing data that is widely used in back-end VLSI design flows.

SDF provides information on:

- path delay
- Interconnect delays
- Timing constraints
- Technical parameters that affect latency
- Cell delays

SDF files are also used for delayed back annotation in gate-level simulations to simulate accurate Si behavior.

53. In VLSI, what is a DEF file?

The Design Exchange File is an industry standard file used to represent the logic and connectivity of an IC in ASCII format.

It usually defines die size , connectivity , pin placement and power domain information .

54. Explain the types of metal programmable ECO cells?

There are 2 types of programmable ECO cells :

- ECO filler
- Functional ECO cells

ECO Fill Cell: Built on a base layer called Front-end-of-line (FOBL), FEOL is an implant , diffusion, and poly layer . This allows any functional ECO to be performed using back-end-of-line layers

. **Functionally programmable ECO unit:** including various combinations and timing units, achieving various drive strengths by using multiples of the width of filler cells . Their cells have the same FEOL footprint as ECO filled cells . **The only difference:** a functional ECO will use an ECO filler FEOL layout and have contact connections to poly - layers and diffusion layers as well as internally connected metal layers to build the functional gate.

55. What are +ve unateness, -ve unateness & non-unate?

+ve unateness: If the output signal direction is the same as the input signal direction or the output signal does not change, use +ve unate to represent a timing arc [Example - AND, OR]

-ve Unateness: If the output signal direction is the same as the input signal. If the direction is opposite or the output signal remains unchanged, then a timing arc is called -ve unate [Example: NOR, NAND, Inverter]

Non-Unate: In a non-unate timing arc, the output transition cannot be determined only based on the direction of change of the input, but also depends on the state of other inputs. [Example: XOR]

56. Is there any problem that we can get 0 skew?

If skew is 0, then all flops will fire at the same time. So the power consumption will be more.

57. If an inverter is inserted into the capture clock pin, what impact will it have on the timing?

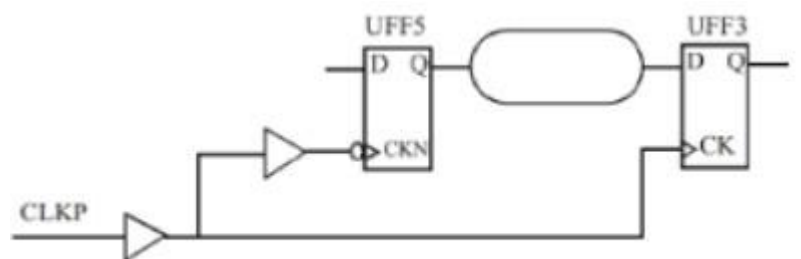
Before inserting the inverter, a full clock cycle is available for Setup.

After inserting the inverter, the timing calculation for the setup becomes a half-cycle path.

And therefore setup timing will be very critical. But we don't see any hold timing issues because the capture clock arrives half a cycle early (for example, on the -ve edge), and the launch clock arrives after the capture clock (for example, on the +ve edge). The Hold path will add an extra half cycle and therefore become less important.

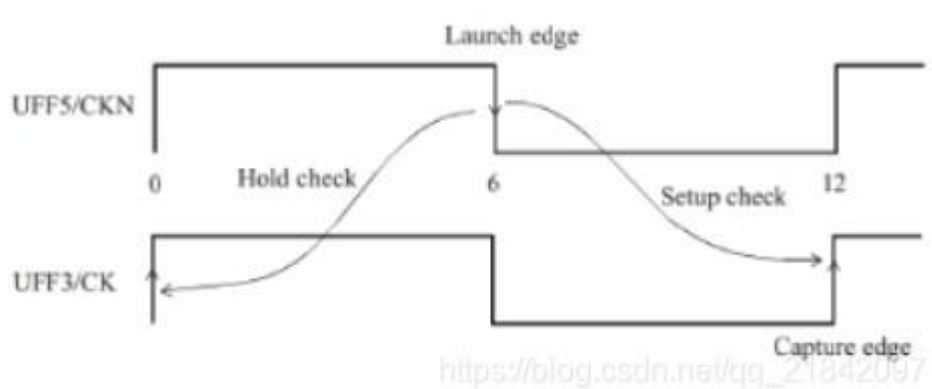
If there are both positive clock edge triggered and negative clock edge triggered flip-flops in a circuit design, a half-cycle check is required in this circuit.

Example: Insert inverter on launch clock pin



https://blog.csdn.net/qq_21842097

If the setup time is checked, the clock edge of the check is as follows:



58. What is the difference between clock skew and clock latency?

Clock skew: is when the clock arrives at clocked elements (such as flip-flops) at different times.
clock latency: is the clock arriving at the clock input pin from where it is generated. The clock is only supplied to the different flip-flops from this pin.

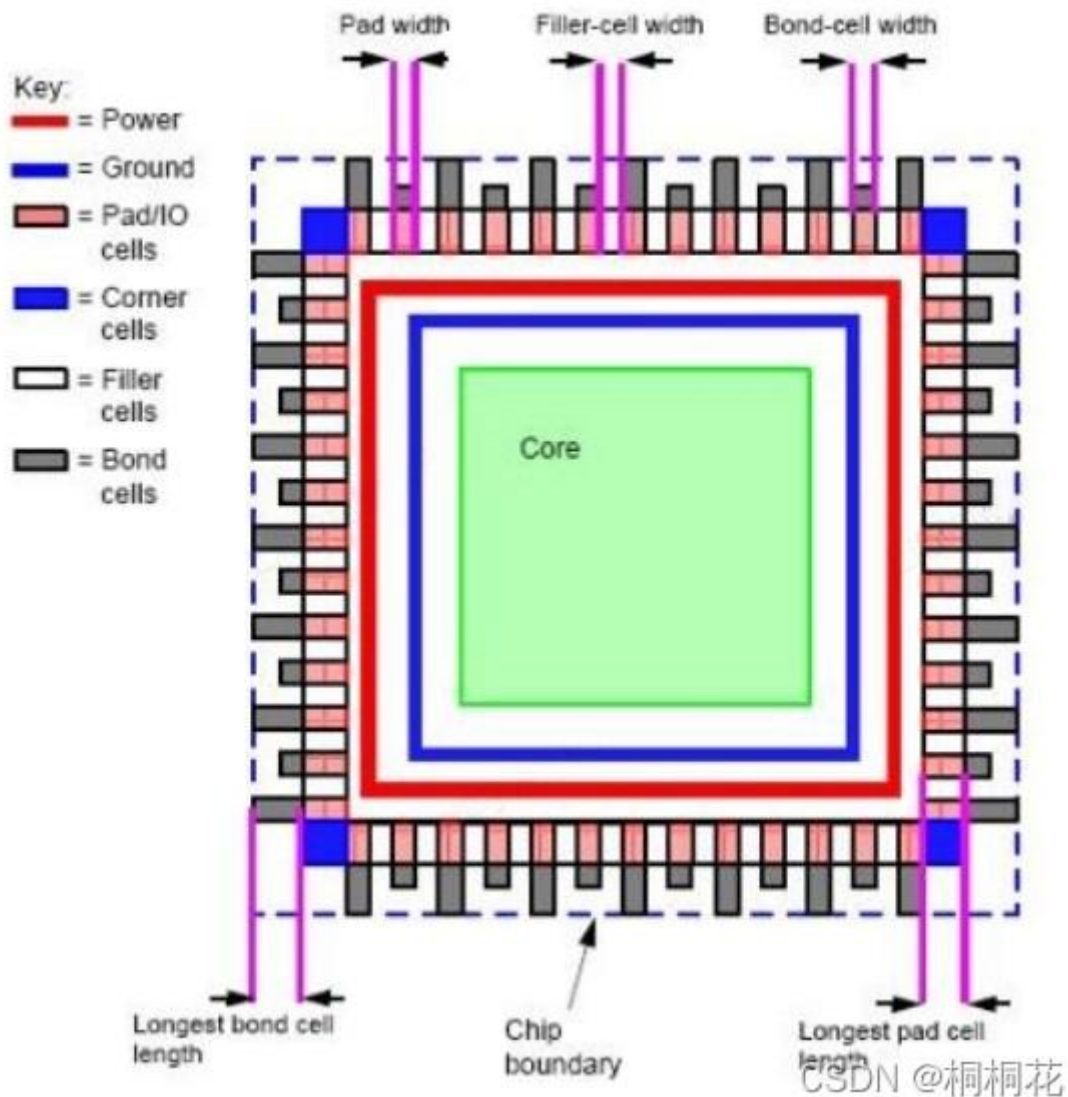
59. What are Pad limited design and core limited design? Is there any difference in the way you approach these two?

- Pad limited design:**

The pad area limits the size of the die. The number of IO pads may be larger. If die area is a constraint, we can choose staggered IO Pads.
- Core limited design:**

The core area limits the size of the die. The number of IO pads may be less. In these designs, in line IOs can be used

60. How do we decide the chip core area?



$$\text{Core Size} = \frac{\text{Standard Cell Area}}{\text{Standard Cell Utilization}} + (\text{Macro Area} + \text{Halo})$$

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Die Size = Core Size + IO to Core Clearance + Area of Pad (Including IO Pitch Area)

芯片尺寸 = 内核尺寸 + IO 到 Core 间隙 + Pad 面积 (包括 IO 间距面积) + Pad 最长焊盘面积

IO 到 Core 间隙 (IO to Core Clearance):

是从 Core 边界到 I/O Pad 内侧的空间 (设计边界)

Questions 61-70

61. How to obtain the utilization factor and aspect ratio values during the initial floorplan?

Utilization Percentages

Assume that standard cells occupy 70% of the Base Layers and the remaining 30% is used for wiring. If the area of the macro is greater than the utilization, the utilization can be increased

accordingly.

$$\text{Chip Utilization} = \frac{\text{Area of [Standard Cell + Macro + (Pad, PadFiller, CornerPad)]}}{\text{Area of Chip}}$$

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Blockages, macros, and pads are combined in the denominator of effective Utilization.

Effective utilization definition :

All standard units are placed outside the blockage area. This includes buffers, which (for utilization calculation purposes) are assumed to be placed outside the non-buffer blockage area.

Best Aspect Ratio :

- Consider a five-story design in which floors 1, 3, and 5 are horizontal and floors 2 and 4 are vertical. Typically, layer 1 is occupied by standard cell geometry and cannot be used for routing. Normally, Metal layer 2 is connected to metal layer 1 pin via vias. These vias tend to block about 20% of the potential vertical routing on metal layer 2. If the routing spacing is the same on all layers, the ratio between horizontal and vertical layers is approximately 2:1.8. This means that there are fewer vertical routing resources available than horizontal routing resources, which determines the chip aspect ratio to be wider than tall.
- Using the ratio of horizontal to vertical wiring sources, the optimal aspect ratio is 1.11; therefore, the chip aspect ratio is rectangular rather than square, and the aspect ratio is tall:

$$\text{AspectRatio} = \frac{W}{H} = \frac{\text{Horizontal Routing Resources}}{\text{Vertical Routing Resources}}$$

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- Next, consider a four-story design. Metal layer 1 is not available for routing, and metal layer 2 is blocked by 20% of the vias connecting layer 1 and layer 2. Layer 3 is horizontal and fully usable, layer 4 is vertical and fully usable. For this case, vertical routing resources are 80% more than horizontal resources. Therefore, the ratio of horizontal to vertical routing resources is 0.56, and the vertical size of the chip is larger than its horizontal size. Aspect Ratio = $W/H = 1/1.8 = 0.56$

62. What is HALO? What is the difference between it and blockage?

Block halos can be specified for hard macros, black boxes, or committed partitions.

When you add a halo to a block, it becomes part of the block's property. If you move the block, the halo will move with it. Blockages can be specified for any part of the design. If we move a block, blockage will not.

63. What is the utilization ratio used in the design?

There are no hard and fast rules, even if you keep the following values, the design can be converged without too much congestion.

Floor Plan - 70%

Placement-75%

CTS - 80%

CTS - 85%

During GDSII generation - 100%

64. What is the difference between standard cells and IO cells? Is there any difference in IR working voltage? If so, why?

1. Standard cells (Std Cells) are logical cells. But IO units interact between the core and the outside world. The IO unit contains some protection circuits such as short circuit, overvoltage.
2. There will be a difference between Core operating voltage and IO operating voltage. This depends on the technology library used. For the 130 nm universal library, the Core voltage is 1.2V and the IO voltage is 2.5/3.3V.

65. What is the importance of simultaneous switching output (SSO) file?

- SSO:

Abbreviation for "Simultaneously Switching Outputs", indicating that a certain number of I/O buffers are switched in the same direction at the same time (H !L, HZ !L or L !H, LZ !H).

This simultaneous switching will generate noise on the power/ground lines due to the large di/dt value and the parasitic inductance of the bonding wires on the power/ground cells.

- SSN:

Noise generated by switching output buffers at the same time. "Simultaneously Switching noise"

It will change the voltage level of the power/ground node, the so-called "Ground Bounce Effect".

Test this effect on the device output by holding one stable output at a low "0" or high "1" while all other outputs of the device switch simultaneously. The noise that occurs at the stable output node is called "Quiet Output Switching (QOS)". If the input low voltage is defined as V_{il} , then the QOS of " V_{il} " is considered to be the maximum noise that the system can withstand.

- DI:

When applying a single ground cell, instantiations (copies) of the specified I/O cell simultaneously switch from high to low without causing the voltage on the static output "0" to be higher than " V_{il} ". We use the QOS of " V_{il} " as the criterion for defining DI because "1" has a greater noise margin than "0".

For example, in the LVTTTL specification, the margin from " V_{ih} " (2.0V) to V_{D33} (3.3V) is 1.3V at the typical corner, which is higher than the margin from " V_{il} " (0.8V) to ground (0V).

DF: " Drive Factor ". It is the contribution of the specified output buffer to SSN

on power/ground rail .

The DF value of the output buffer is proportional to di/dt , which is the derivative of the current on the output buffer .

We can get DF as: $DF = 1 / DI$

66. Is there any checklist received from the front-end, and is this checklist related to the switching activity of any nets that needs to be processed during the floor plan stage ?

Yes. The switching activities of the macro will be provided in the checklist ; it contains the power consumption at different frequencies available for each macro

67. What is power trunk?

The power trunk is the metal piece that connects the IO pad and the Core ring.

68. How to deal with hot spots in the chip?

Increasing the number of power straps or increasing the width of the power strap will help reduce hot spots caused by voltage drops and keep voltage drops below 10%.

69. What is power gating?

Power gating is a power reduction technology. This helps shut down specific areas of the chip to prevent power usage.

70. Is the macro power ring necessary or optional?

For hierarchical designs, macro power loops are necessary. For flat designs, macro power rings are optional.

Questions 71-80

71. If you have IR drop and congestion problems at the same time, how to fix them?

1. spread out macros
2. spread out standard cells
3. Increase strap width
4. Increase the number of straps
5. Proper use of blockage

72. Are increasing the power line width and providing more straps the only way to solve IR drop ?

1. spread out macros
2. spread out standard cells
3. Use appropriate blocking

73. What are tie-high cells and tie-low cells, and where are they used?

tie cell: voltage clamp unit

Tie cell function:

1. Protect cells from ESD. The unit input pins will be connected to TIEH/TIEL, not to PG. If they are connected directly to the PG, the cells will be damaged if there are power fluctuations.

2. Some signal ports have fixed logic levels.

Some signal ports or idle signal ports in digital circuits need to be clamped at fixed logic levels. The voltage clamping unit connects these clamped signals to **VDD** through tie high and VSS through tie low according to the logic function requirements. maintain it at a fixed potential.

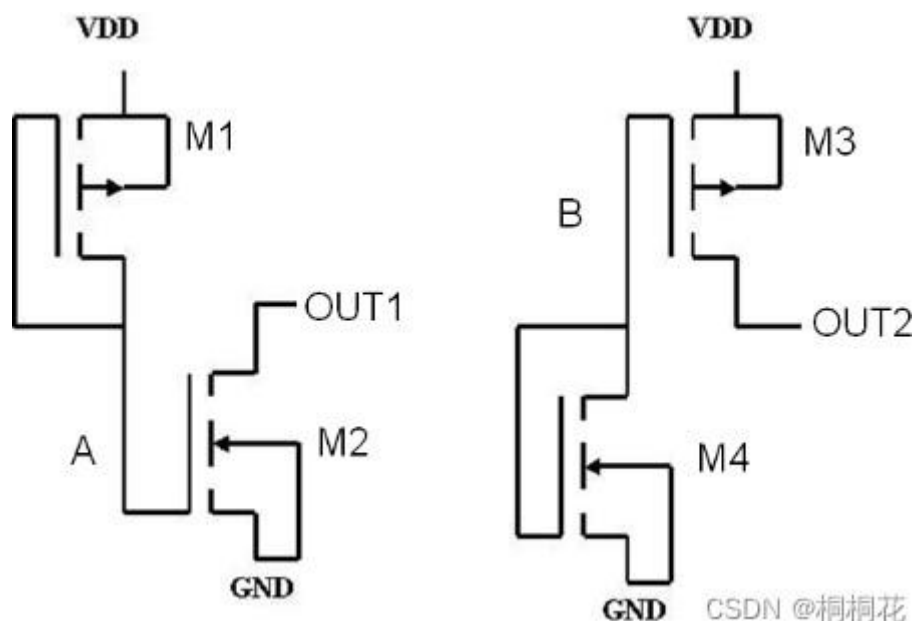
3. Isolate ordinary signals

The tie cell also plays the role of isolating ordinary signals (VDD, VSS), so as not to cause logical confusion when doing LVS analysis or formal verification.

tie cell structure:

M1 is connected to high potential, the gate and source are connected together, and the mos tube works in the saturation region. It acts as an active resistor, making the potential of point A high. M2 works in the linear zone.

M1 and M2 together form Tie-Low . M3 and M4 together form Tie-high .



Tie-high cells, Tie-Low cells Connect the gate of the transistor to power or ground.

If the gate is connected to power/ground, the transistor may turn on/off due to power or ground.

The foundry factory's recommendation is to use tie cells for this purpose. These cells are part of the standard-cell library. A Vdd cell is required to connect to Tie high (so tie high is a power supply cell). And the cell that wants Vss connects itself to Tie-low.

74. What placement optimization methods are used in SOCE and Astro Tool Design?

1. PreplaceOpt
2. Inplace Opt
3. Post Place opt.
4. Incremental Opt
5. Timing Driven
6. Congestion Driven

75. What is Scan chain reordering? How does it affect Physical Design?

Grouping cells belonging to the same area of the chip together to allow scan connections only between cells in the same area is called scan clustering. Clustering also allows the elimination of congestion and timing violations.

Types of scan cell ordering

1. Cluster based scan cell order (cluster-based scan cell order)
2. power - driven scan cell order (power driven scan cell order)
3. Power optimized routing constrained scan cell order (power optimized routing constrained scan cell order)

Power driven scan cell order

1. Determine the linkage of scan units to minimize the toggling rate in the scan chain during shifting operations.
2. Identifies the inputs and outputs of the scan units of the scan chain to limit the propagation of transitions during scan operations.
3. If the scan chain length is reduced, this increases wire ability or reduces chip die area , while also increasing signal speed by reducing the capacitive loading effect of sharing scan chains with register pins.
4. After scan synthesis , connecting all scan units together may cause routing congestion during PAR. This can lead to area overhead and timing closure issues.
5. Scan chain optimization - The task of finding a new order (order) of connecting scan elements so that the line length of the scan chain is minimized.

76. In scan chains, if some flip flops are +ve edge triggered, and the rest of the flip flops are -ve edge triggered, how does it behave?

1. For designs with both positive and negative clock flip-flops , the scan insertion tool will always route the scan chain so that the negative clock flip-flop precedes the positive edge flip-flop in the scan chain. This avoids the need for a lockup latch.
2. For the same clock domain , negedge flop will always capture the data just captured into the posedge flop of the clock posedge.

3. With multiple clock domains, it all depends on how the clock tree is balanced. If the clock domain is fully asynchronous, the ATPG must mask the receive flip-flop.

77. What does scan chain reordering mean?

Answer 1:

Based on timing and congestion, the tool places standard cells in the best way. While doing this, if the scan chain is detached, it can break chain ordering (this is done by scan insertion tools like Synopsys' DFT compiler) and can reorder it to optimize it... it stays in chain The number of triggers.

Answer 2:

During layout, optimizations may make scan chains difficult to route due to congestion. Therefore, the tool will reorder the chain to reduce congestion. This sometimes increases hold time issues in the chain. To overcome these buffers, it may be necessary to insert them into the scan path. It may not maintain scan chain length accurately. It cannot swap units from different clock domains.

78. What is JTAG?

Answer 1:

JTAG is an acronym for "Joint Test Action Group". This is also known as the IEEE 1149.1 standard for Standard Test Access Ports and Boundary Scan Architecture. This is used as one of the DFT techniques.

Answer 2:

JTAG (Joint Test Action Group) boundary scan is a method of testing ICs and their interconnections. This uses a shift register built into the chip so the input can be shifted in and out of the resulting output. JTAG requires four I/O pins called clock, input data, output data, and state machine mode control.

The use of JTAG extends to debugging software for embedded microcontrollers. This eliminates the need for a more costly in-circuit emulator. JTAG is also used to download the configuration bitstream to the FPGA.

The JTAG unit, also known as the boundary scan unit, is a small circuit placed inside the I/O unit. The purpose is to enable data to be passed in/out of I/O via a boundary scan chain. The interfaces of these scan chains are called TAP (Test Access Port). The operations of the scan chain and TAP are controlled by the JTAG controller inside the chip that implements JTAG.

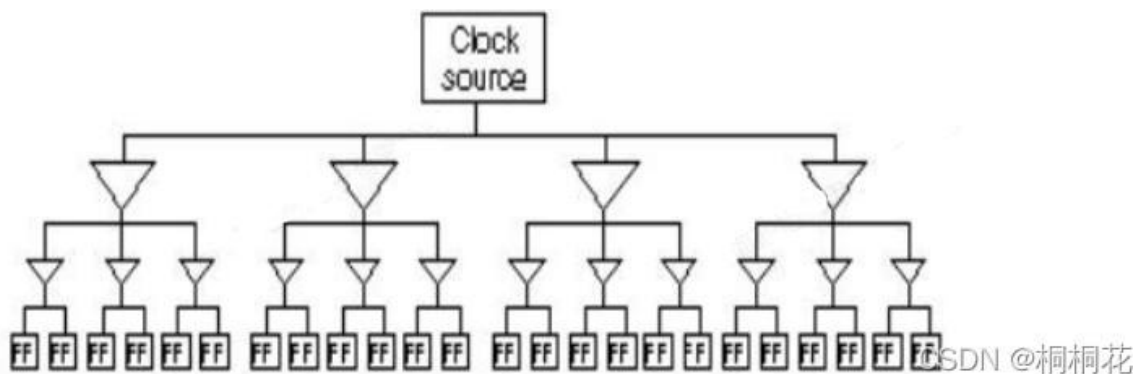
79. What is CTS?

Clock tree synthesis is the process of balancing clock skew and minimizing insertion delays to meet timing, power requirements, and other constraints.

Clock tree synthesis provides the following features to achieve timing closure:

- ♦ Global skew clock tree synthesis (global skew clock tree synthesis)
- ♦ Local skew clock tree synthesis (local skew clock tree synthesis)
- ♦ Real clock useful skew clock tree synthesis (real clock useful skew clock tree synthesis)

- Ideal clock useful skew clock tree synthesis (ideal clock useful skew clock tree synthesis)
- Interlock delay balance
- Splitting a clock net to replicate the clock gating cells
- Clock tree optimization
- High-fanout net synthesis
- Concurrent multiple corners (worst-case and best-case) clock tree synthesis
- Concurrent multiple clocks with domain overlap clock tree synthesis (Concurrent multiple clocks and domain overlap clock tree synthesis)



80. What are the SDC constraints related to the clock tree?

If there is no `create_clock` statement in the loaded SDC file , CTS will not run. Make sure you have at least one `create_clock` in your SDC file.

If you define `create_clock` on a pin that does not physically exist and only exists in the hierarchical netlist , CTS will not run.

It is best to define `set_clock_transition`, `set_clock_latency` and `set_clock_uncertainty` at the same time

Clock tree synthesis has the following clock tree constraints:

Maximum transition delay

Maximum load capacity

Maximum fanout

Maximum buffer level

Questions 81-90

81. During CTS, how is the number of Buffer (logical) layers determined?

1. spread out macros
2. spread out standard cells
3. Increase strap width
4. Increase the number of straps
5. Proper use of blockage

82. Which one is better, buffer or inverter? If so, why?

1. Inverters, due to the shorter conversion time of Inverters. It reduces the current flow between VDD and VSS rail, thereby reducing power consumption. It's best to use both with all drive strengths to get good skew and insertion delay.
2. Another benefit of using inverters in the clock tree is the reduction of duty cycle distortion. A cell library's delay models are typically characterized by three different operating conditions, or corners : worst, typical, and best . However, there are some other effects not modeled in these corners . You may encounter clock jitter introduced by the PLL , differences in PFET or NFET doping, and other known physical effects of the manufacturing process.

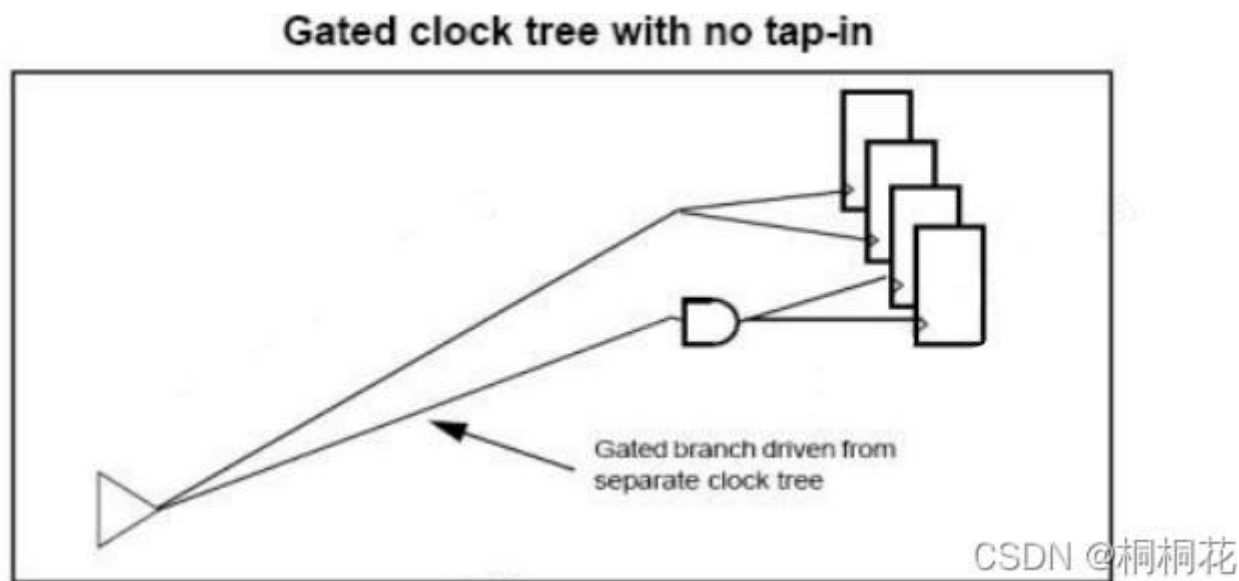
83. When making CTS, what buffers and inverters are used in the design?

Clock tree synthesis uses buffers or inverters in clock tree construction. If boolean functions are defined in the library preparation, the tool recognizes buffers and inverters.

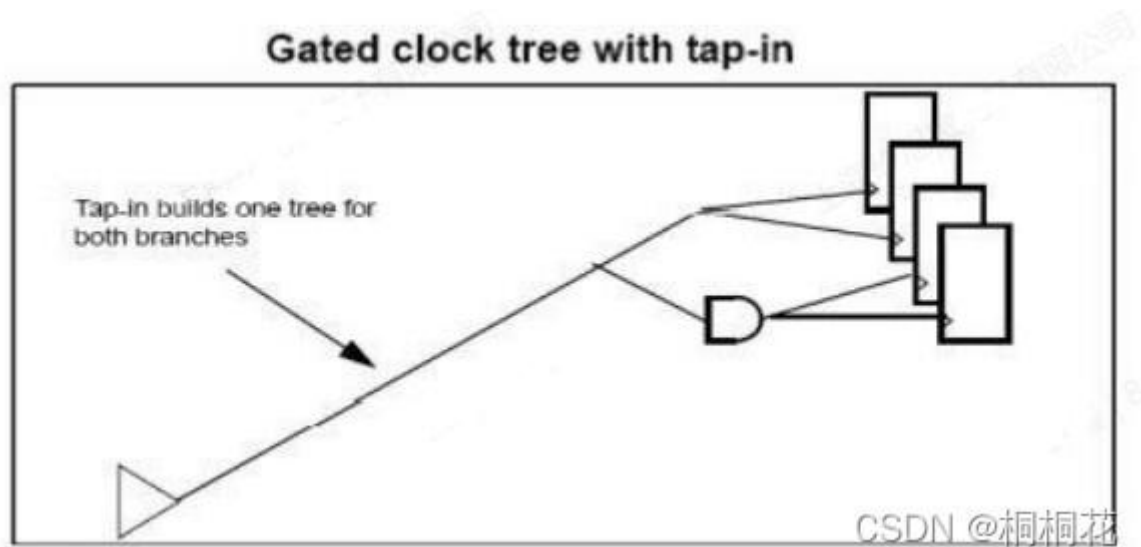
By default, clock tree synthesis synthesizes clock trees with all buffers and inverters available in the library. There is no need to specify all of these explicitly in Buffers/Inverters.

84. How would you build a clock tree for gated clocks ?

Historically, separate trees were built for any net that drove clock gating elements and clock leaves . The two trees bifurcate at the net root. This often results in excessive insertion latency and makes the clock tree more susceptible to failure due to on-chip variation (OCV).



By default, the clock tree synthesizer attempts to connect gated branches to lower points in the clock tree, sharing more of the clock tree topology with non-gated branches. It attempts to insert negative offset branch points earlier in the main tree.



In many cases this results in fewer **buffers** being inserted and lower clock **insertion** delays . Sharing the clock tree topology between gated and ungated branches often also reduces the impact of **local OCV** on timing. If too many **buffers** are inserted or the clock tree insertion delay is too large, the clock tap-in feature should be disabled .

85. Explain Clock Tree Options to build better clock trees?

There are five special clock options available to solve this situation. They greatly expand your ability to control clock construction.

Clock phase:

1. The clock phase is a timer event related to a specific edge of the source clock.
2. Each clock domain is created from two clock phases:
 1. The rising edge
 2. The falling edge
3. The clock phase is named after timing clock, with R or F indicating the rising or falling phase of the clock.
4. These phases are propagated through the circuit to endpoints, so events on the clock pins can be traced back to events driven by the defined clock.
5. Because the tool is able to propagate multiple clocks through the circuit, any clock pin can have two or more clock phases associated with it.
6. For example, if CLKA and CLKB are connected to the i0 and i1 inputs of a 2:1 MUX, then all clock pins in that MUX fanout have four clock phases associated with them - CLKA:R, CLKA:F, CLKB: R and CLKB:F. (This assumes you allow multiple clock phases to be propagated).

skew phase:

1. The skew phase is a collection of clock phases.
2. Each clock phase is placed in a skew phase with the same name .
3. .When you define a clock, **skew phases** are also automatically created . They are created with the same name as the clock phases they were created with .

Skew group

1. Clock tree skew balancing is completed on the basis of each skew group .
 2. The skew group is a subdivision of the clock phase.
 3. Under normal circumstances, all pins of a clock phase are in group 0 and are balanced as a group . .
 4. If you create a group of pins labeled group 1 , for example :
- The skew phase containing these pins will then be split into two skew groups : one containing the user-specified group, and the other containing the "normal " clock pins .
 - This feature is useful if we want to isolate certain groups of clock pins without balancing them with the default group . We can now define multiple sets of pins and balance them independently.

Skew anchor or Sink Point

1. The skew anchor is the clock endpoint that controls the downstream clock tree .
2. For example, a register that is a divide-by-2 clock generator has a clock input pin , which is a skew anchor because the time the clock arrives at that clock pin affects everything in the generated domain starting at the register's Q pin. The arrival time of the clock.

Skew offset

1. Skew offset (offset) is a floating point number used to describe the existence of a certain phase relationship when multiple clocks with different periods or different edges of different phases of the same clock are put into the same skew phase.
2. You can use skew offset to adjust the arrival time of a specific clock phase when you want to compare with another clock phase in the same group .

86. What is the relationship between skew group, clock phase and skew phase?

A skew group is a set of clock pins declared as a group . By default, all clock pins are placed in group 0 . Therefore each skew phase contains a group .

For example, if the user creates a set of pins labeled with the number 1, then the skew phase containing these pins will be divided into two skew groups :

- The “normal” clock pins
- The user-specified group.

This is useful for isolating groups of clock pins that have special cases and that you don't want to balance with the default group .

Skew optimization is performed based on the skew-group that occurs after inserting the basic clock.

87. Why should we reduce Clock Skew?

1. Reducing clock skew is not only a performance issue, but also a manufacturing issue.

2. Scan-based testing, currently the most popular method of structurally testing chip manufacturing defects, requires a minimum **skew** to allow error-free movement of scan vectors to detect stuck and delayed faults in circuits.
3. **Best-case PVT Corner Hold failures** are common in these circuits because there are usually no logic gates between the output of one flip-flop and the scan input of the next flip-flop in the scan chain .
4. Managing and reducing clock skew in this situation can often resolve these hold failures .

88. What tests should be done before CTS?

- Hierarchical pins should not be defined as clock sources .
- The generated clock should have a valid main clock source. The generated clock does not have a valid main clock source in the following situations:

```
create_generated_clock 中指定的主时钟 ( master clock) 不存在。
create_generated_clock 中指定的主时钟不驱动生成时钟的源引脚。
生成时钟的源管脚由多个时钟驱动, 有些主时钟没有用create_generated_clock 指定。
```

- Clocks without sinks (master clock or generated clock)
- Looping clock
- Cascaded clock , which has an unsynthesized clock tree in its fanout
- **Multiple-clocks-per-register propagation** is not enabled but the clock tree exception should not be ignored if the design contains overlapping clocks.
- Stop pin or float pin defined on output pin is a problem.

89. How will you synthesize the clock tree?

1. **Single clock-** normal synthesis and optimizationSingle clock-normal synthesis and optimization
2. **Multiple Clocks** - synthesize each clock separately
3. **Multiple clocks with domain Crossing** - Each clock is independent and has balanced skew.

90. How many clocks are there in this project?

1. Depends on your project
2. More clocks make it more challenging!

Questions 91-100

91. How do you deal with all these clocks?

Multiple clocks (multiple clocks) → synthesize separately → balance skew → optimize the clock tree

Does the clock come from a separate external source or the PLL?

- If it comes from different clock sources (i.e. asynchronously from different pads or pins) then balancing the skew between these clock sources becomes challenging.
- If it comes from PLL (ie Synchronous), then skew balancing is easier.

92. Why use buffer in clock tree?

To balance skew (e.g., flop to flop delay)

93. When you have 48 MHz and 500 MHz clock designs, which one is more complicated?

500 MHz; it is more constrained (i.e. smaller clock period) than the 48 MHz design.

94. What is congestion?

If there are fewer routing tracks available for routing than the number of tracks required, this is called congestion.

95. In a typical timing analysis report, what types of timing violations are there?

1. Setup time violations - Hold time violations
2. Minimum delay - Maximum delay
3. Slack - External delay

96. Can I use STA to analyze latch-based designs (latches)?

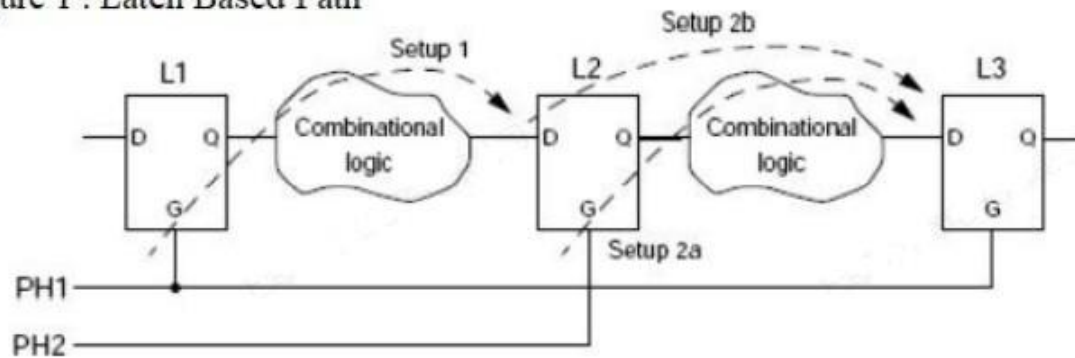
Latch Setup and Hold Checks Latch-based designs typically use two-phase non-overlapping clocks to control consecutive registers in the data path.

In these cases, the Timing Engine can use time borrowing to reduce constraints on successive paths.

For example, consider the two-phase latch-based path shown in Figure 1. All three latches are level-sensitive, and the gate is active when the G input is **high**. L1 and L3 are controlled by PH1, and L2 is controlled by PH2. The rising edge emits data from the latch output and the falling edge captures data at the latch input.

For this example, consider the setup time and delay time to be zero.

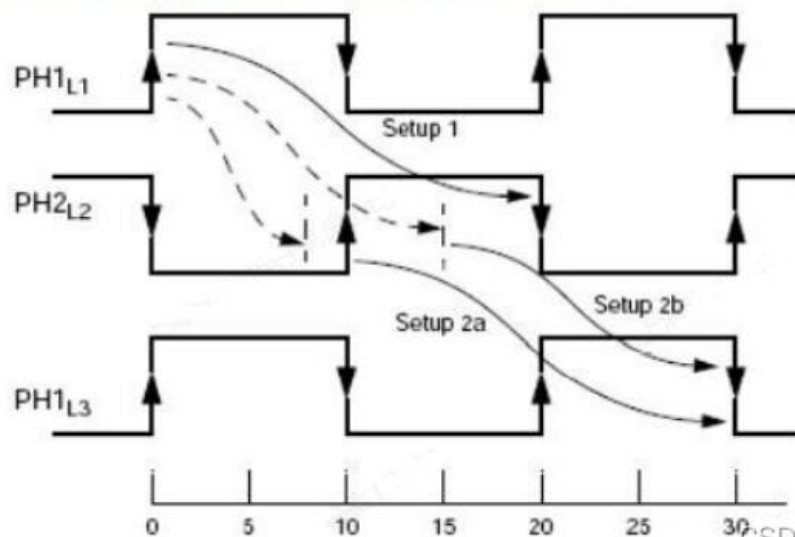
Figure 1 : Latch Based Path



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Figure 2 shows how the Timing Engine performs setup checks between these latches. For the path from L1 to L2, data is transmitted on the rising edge of PH1. The data must arrive at L2 before the PH2 closing edge at time=20. This timing requirement is labeled Setup 1. Depending on the amount of delay between L1 and L2, the data may arrive before or after the opening edge of PH2 (time=10), as indicated by the dotted arrow in the timing diagram. Arrival after time=20 will be a timing violation.

Figure 2: Time Borrowing in Latch Based Path

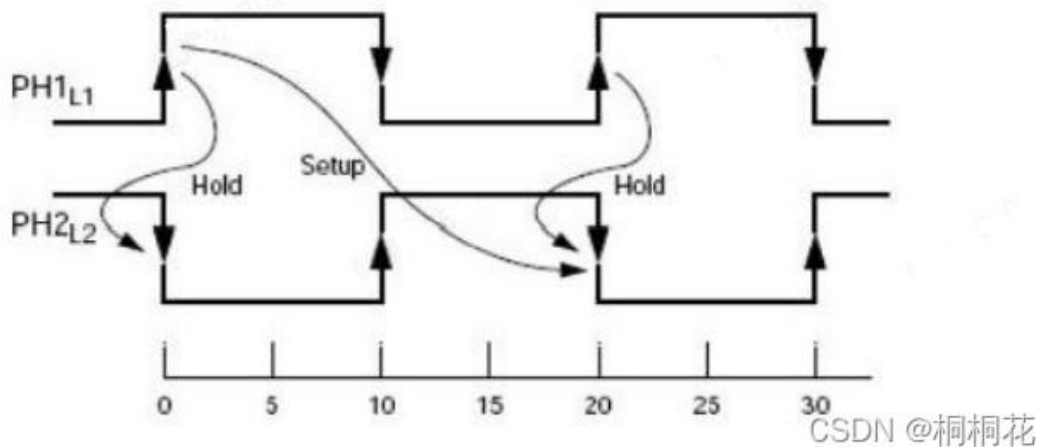


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If the data reaches L2 before the opening edge of PH2 at time=10, the data on the next path from L2 to L3 is started by the opening edge of PH2 at time=10, just like a synchronous trigger. This timing requirement is labeled **Setup 2a**. If data arrives after the open edge of PH2, the first path (from L1 to L2) borrows time from the second path (from L2 to L3). In this case, the data emission for the second path does not occur at the start edge, but at the time when the data arrives at L2, somewhere between the opening and closing edges of PH2. This timing requirement is labeled **Setup 2b**. When a borrow occurs, the path originates from the D pin instead of the G pin of L2. For the first path (from L1 to L2), if a borrow occurs, the Timing Engine reports the setup slack as zero. If the data arrives before the starting edge at time=10, slack is positive, and if the data arrives after the ending edge at time=20, slack is negative (violation). To perform hold checks, the Timing Engine considers startup and capture edges relative to setup checks. It verifies that the data transmitted at the starting point does not reach the end point too quickly, thereby ensuring that the

data transmitted in the previous cycle is latched and not overwritten by new data. This is depicted in Figure 3.

Figure 3: Hold Checks in Latch-Based Paths



97. How does delay change under different PVT conditions?

1. P increases->delay increases
2. P decrease->decrease
3. V increases -> delay decreases
4. V decreases -> delay increases
5. T increases->delay increases
6. T decreases->delay decreases

98. What are cell delay and net delay?

Gate delay

1. Gate delay = (i/p transition time , Cnet+Cpin) function.
2. Cell delay is the same as Gate delay.

cell delay

For any gate, it is measured between 50% of the input transition and 50% of the corresponding output transition.

Intrinsic delay:

Intrinsic delay is the internal delay of gate. The input pin of the unit to the output pin of the unit.

It is defined as the delay between a unit's input and output pair when a near-zero slew (slew) is applied to the input pin and the output does not see any load conditions. It is mainly caused by the internal capacitance associated with its transistor.

This delay is largely related to the size of the transistors that form the gate, as increasing transistor size increases internal capacitance.

Net Delay (or wire delay)

The difference between the time a signal is first applied to a net and the time it reaches other devices connected to that net.

This is due to the finite resistance and capacitance of the net. It is also called wire delay.

Wire delay = $f_n(R_{net}, C_{net} + C_{pin})$

99. What are delay models and what are the differences between them?

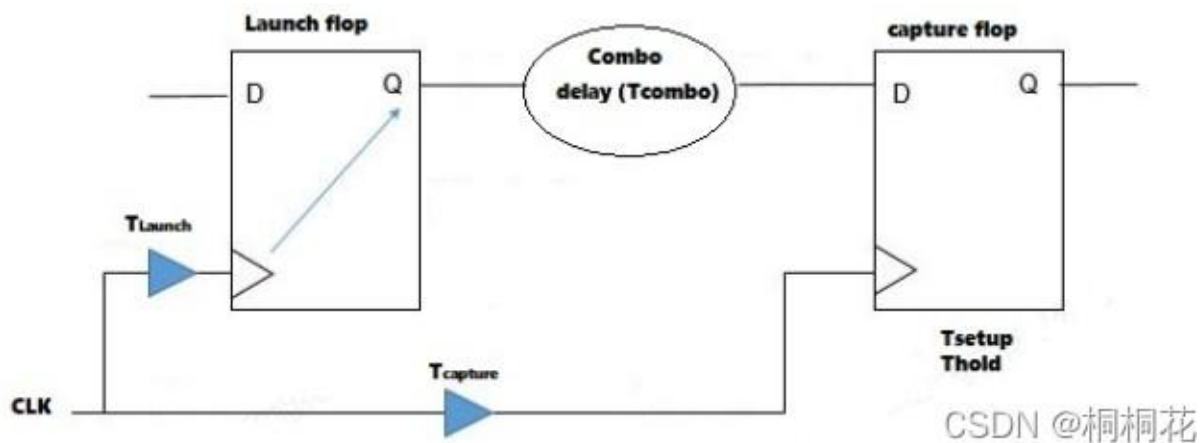
1. Linear Delay Model (LDM)
2. Non Linear Delay Model (NLDM)
3. Composite current source modeling (CCS)

100. What is wire load mode?

Wire load model is NLDM, used to estimate the values of R and C of net.

Questions 101-110

101. Write the Setup and Hold equations?



```
//Setup equation:  
Tlaunch + Tclk-q_max + Tcombo_max <= Tcapture + Tclk - Tsetup  
//Hold equation:  
Tlaunch + Tclk-q_min + Tcombo_min >= Tcapture + Thold
```

102. What are the factors that determine trigger setup time?

D- pin transition (transition time of D port pin)

clock transition (clock transition time)

103. What is latency? What are its types?

Source Latency

1. Source delay is defined in the design as "the delay from clock origin point to clock definition point " .
2. The delay from the clock source to the start of the clock tree (i.e. the clock definition point).
3. The time it takes for a clock signal to propagate from its ideal waveform origin to the clock definition point in the design.

Network latency

1. It is also called insertion delay (Insertion delay) or network latency (Network latency). It is defined as "the delay from the clock definition point to the register clock pin".
2. The clock signal (rising or falling) needs to propagate from the clock definition point to the register clock pin.

104. What violations were resolved in the DRC?

Includes the following 65 and 90nm design rules:

- ♦ Fat metal width spacing rule
- ♦ Fat metal extension spacing rule
- ♦ Maximum number minimum edge rule
- ♦ Metal density rule (requires a Hercules license)
- ♦ via density rule (requires a Hercules license)
- ♦ Fat metal connect rule
- ♦ via corner spacing rule
- ♦ Minimum length rule
- ♦ via farm rule
- ♦ Enclosed via spacing rule
- ♦ Minimum enclosed spacing rule
- ♦ Fat poly contact rule
- ♦ extendMacroP intoBlockage (new parameter)
- ♦ Special end-of-line spacing rule
- ♦ Special notch rule
- ♦ U-shaped metal spacing rule
- ♦ Maximum stack level for via (for array)
- ♦ stud spacing
- ♦ Multiple fat spacing
- ♦ Enclosure

105. What is the difference between Magma and Caliber for solving DRC LVS problems?

Magma is an implementation tool, which only does metal level DRC , but Caliber is a sign off tool, which does DRC at POLY and Diffusion levels .

106. What are the violations solved in LVS?

Open Error

short error

Device Mismatch

Port Mismatch

Instance Mismatch

Net Mismatch

Floating Nets

107. During power analysis, if you face the IR drop problem, what can you do to avoid it?

Increase power metal layer width

- Choose a higher metal layer.
- Spread out macros or standard cells
- Provide more straps.

108. Why use double spacing and multiple vias related to clocks?

Why a clock? -- Because more than any other signal, it is a signal that changes its state regularly.

If any other signal switches quickly then we can use double space as well.

Double spacing -> Wider width -> Smaller capacitance -> Less crosstalk

Multiple vias -> Resistors in parallel -> Smaller resistance -> Smaller RC delay

109. What do the antenna rules related to the ASIC backend mean? How are these violations handled?

Generally speaking, fixing antenna problems is quite expensive. Therefore, routing should be completed with few or 0 DRC violations before repairing antenna violations.

Antenna repair can be done before or after Optimize Routing.

After the antenna is repaired, running "Optimize Routing" will generate a good layout for the first time.

However, in most cases, if both steps are required, running Optimize Routing first can improve overall turnaround time.

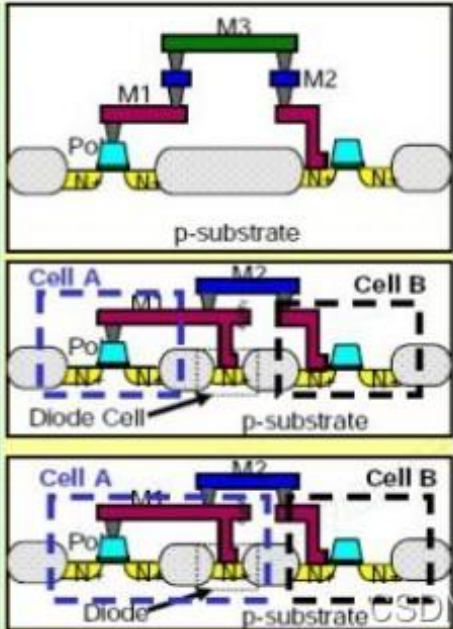
110. What are Antenna effect and antenna ratio? How to eliminate this situation? Why does it only appear in deep sub-micron technology?

Antenna effect : Antenna

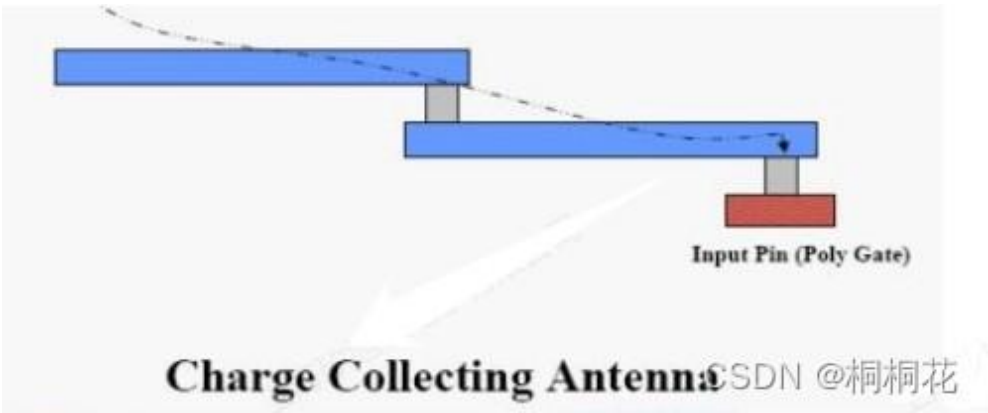
effect occurs during the chip manufacturing process, causing chip failure. During the metallization process (when metal lines are laid across the device), some lines connected to the transistor's polysilicon gate may be left floating (unconnected) until the upper metal layer is deposited. Long flying wire interconnects can act as temporary capacitors, collecting charge during fabrication steps such as plasma etching. If the energy accumulated on the floating node is suddenly released, the logic gate may suffer permanent damage due to breakdown of the transistor gate oxide. This is called the Antenna Effect.

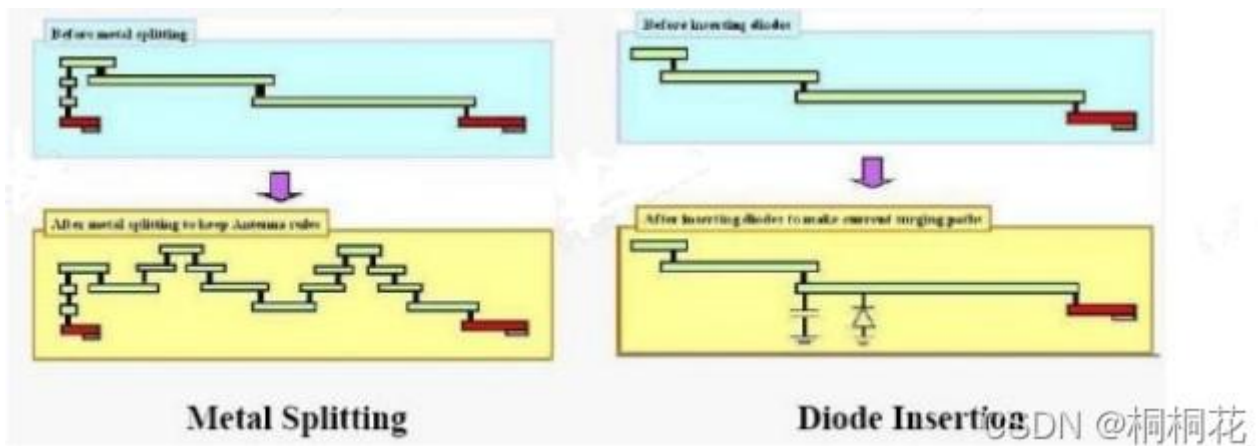
Because in nanotechnology, the oxide thickness underneath the transistor gate is very thin. This problem does not exist in 0.35u technology. Even if this charge is not released to the body. It remains in the oxide as a heat carrier. There by changing the threshold, there is also a big issue with antenna elimination and its effect:

Approach	Rationale	Impact
Layer Hopping	Top metal completes connection	Longer routes More RCs More congestion
Net diode insertion	Discharge path through diode	Added capacitance Area penalty LVS (new cell) longer routes(Place and route to diode)
Diode in cell	Cell protected by construction	Added capacitance, Area penalty



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Antenna Ratio :

Antenna Ratio is defined as **the ratio between the physical area of the conductor and the total gate oxide area to which the antenna is electrically connected.**

A higher ratio means a greater tendency to fail due to antenna effects.

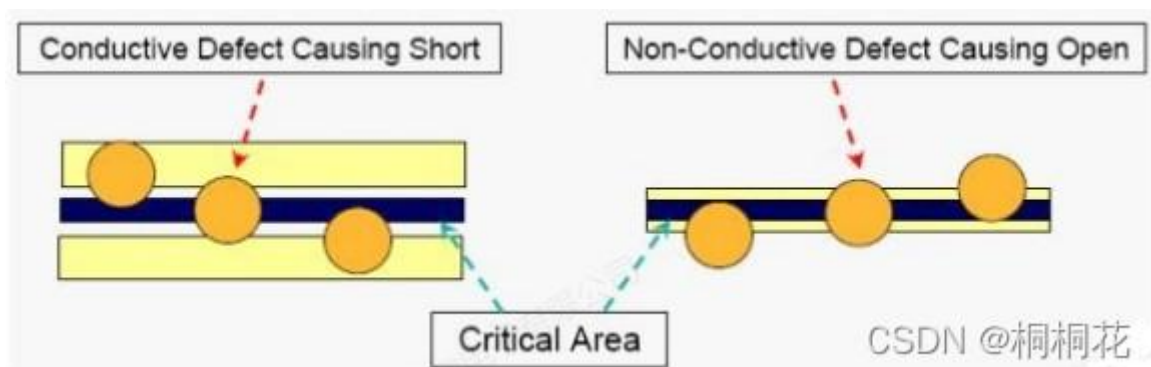
Metal antenna ratio = 500

Metal antenna ratio = 1100 **Why Wire**

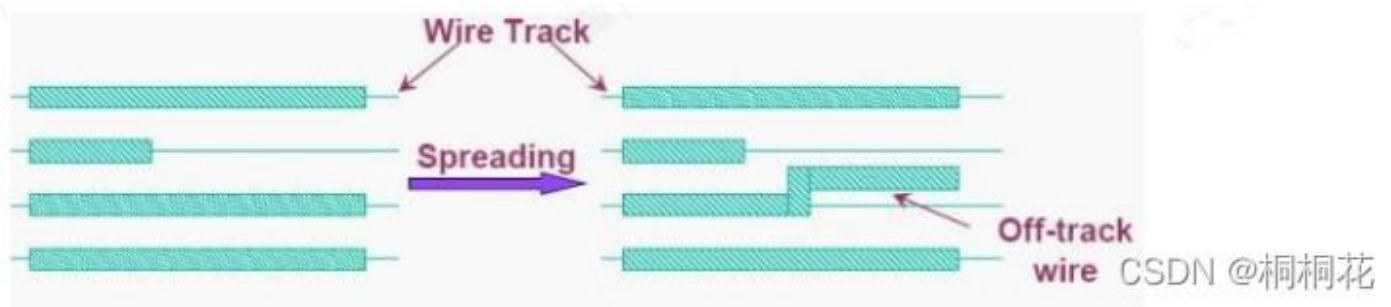
Spreading ?

Random particle defects in the manufacturing process can cause shorts/opens in lines, resulting in yield loss.

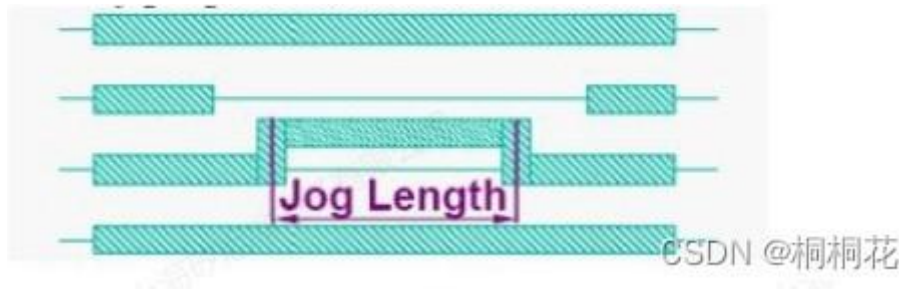
Such areas trimmed to shorts/opens are called critical areas.



- Improved yield for random particle defects
- Wire spreading leads to more evenly distributed wires
- Probability distribution of various defect sizes to calculate critical areas
- Distribution functions vary between different manufacturing processes
- Pushing routes 1/2 pitch from track
- Even reducing the critical area of "short" may cause the critical area of "open" to increase
- You can choose to widen the wires, so the critical area of "open" will not increase - Will not push frozen nets



Press pitch to push the route off the track, and the resulting jog metal length may exceed the minimum jog length.



Questions 111-120

111. Before repairing Antenna, why not run wire spreading?

Not recommended, antenna priority is ranked after DRC

Wire spreading [by pushing off-track] may not leave enough resources to repair the antenna

112. Will wire spreading switch layers?

If space allows, push the line away from the track, without cutting the jump layer to allow pushing the line apart (spreading)

However, after wire spreading, Search & Repair may result in minimal changes to resolve DRC

113. Will wire spreading cause Antenna violations?

The Antenna ratio may vary slightly with wire spreading depending on the antenna length.

New Antenna violations should not be introduced in most cases

114. Why should filler cells be inserted?

For better yield, the density of the chips needs to be uniform.

Some placement sites remain empty on some rows:

- ♦ Accepts two filler unit lists: with/without metal (with/without metal)
- ♦ DRC is not checked when inserting a metalless unit. You need to provide metal-free units (without metal)
- ♦ Cells with metal are inserted only if there is no DRC violation

Filler cell Insertion

It is recommended to insert units in the specified order (largest to smallest)

By default, respect hard/soft placement blockages

115. Why do we need to insert metal filling (Metal Fill Insertion)?

Uneven metal density can cause problems during the manufacturing process. Especially chemical mechanical polishing

Consider the extraction of a metal-filled environment:

- . Metal filling is not considered in extracting FILL view
- . Padding is not properly considered in extracting CELL view
- . Timing analysis does not consider padding

116. Do you know the method of reducing leakage controlled by input vector ?

The gate's leakage current also depends on its input. Therefore, find the **set of inputs** that leaks the least . By applying this minimum leakage vector to the circuit, you can reduce the leakage current when the circuit is in **standby mode** . This method is called the leakage reduction method of input vector control.

117. How to reduce dynamic power?

Reduce switching activity through well-designed RTL

Clock gating

Architectural improvements

Reduce supply voltage

Using multiple voltage domains - Multi vdd

118. What is the vector of dynamic power?

voltage and current

119. What is Partitioning?

Partitioning is the process of breaking a design into manageable parts. The purpose of partitioning is to divide complex design components into manageable parts for easier implementation. In this step, the model for timing and physical implementation is defined.

The **floorplan** defined during prototyping is pushed down to lower-level blocks, thereby preserving placement , power routing , and placement and routing - related obstacles. **Feed-through** can also be assigned to nets routed and buffered on the block by inserting **hole-punch** buffers or modifying the block **netlist** . Flat physical implementation does not require logical partitioning. Partition splitting is designed for both logical and physical implementation. For hierarchical physical implementation, logical partitioning directly affects the physical implementation stage.

Partitioning is a way to manage functional complexity from a logical design perspective.

Partitioning allows multiple design teams to work in parallel.

The bridge between flat and hierarchical physical implementations is:

Creation of timing budgets

Pin optimization

Feed-through or hole-punch buffer assignment

Floorplan push-down (Obstructions, Power routes) Floor plan push-down (Obstructions, Power routes)

Advanced netlist optimization: timing, clocking, power and signal integrity

120. Compare hierarchical and flattened design methods related to ASIC design ?

Flat Design Advantages

1. The flat design approach ensures that there are no issues with boundary constraints between different hierarchies.
2. Ability to analyze I/O and hard macro to block paths . You have more accurate timing analysis because no block modeling is required

Disadvantages:

1. Big amount of data
2. Long running time

Hierarchical Design Advantages :

1. Time can be saved by closing timing in parallel at the top and block level
2. Generate early top-level timing analysis
3. Smaller data sets and faster run times
4. Blocks can be reused after implementation.
5. If the design uses an IP block , it's easier to plug it into a layered modular design than trying to fit it into a flat design .

Disadvantages:

1. Preliminary block characterization is inaccurate and can produce false top-level and block-level timing violations, as well as masking timing violations that appear to make timing meet.
2. When the module changes, the module timing model needs to be updated frequently.
3. Due to boundary modeling, details are hidden or lost.

Questions 121-130

121. What parameters (or aspects) can distinguish chip design and block level design?

- ♦ The chip design has I/O pads; the block design has pins.

- Chip designs use all available metal layers; block designs may not use all metal layers.
- Chips are generally rectangular; blocks can be rectangular or rectilinear.
- Chip designs require multiple packages; block designs end with macros.

122. What inputs does StarRC require?

1. Milkyway or GDSII or LEF/DEF database
2. layer mapping file
3. nextgrd file (contains RC interconnect information)
4. StarRC command file
5. StarXtract

The included GDSII layer must be equivalent to the LEF database layer using the GDS_LAYER_MAP_FILE command.

If no GDSII layer is specified in the layer map file, it will not be translated for extraction and no parasitic parameters will be generated.

123. Among PMOS and NMOS used for power gating/power switches , which one do you prefer?

Header (PMOS):

- Higher resistance (due to lower mobility) so the slew rate/transition will be greater i.e. the switching activity is slower
- The short circuit power is greater due to the higher transition rate .
- Higher resistance, less leakage - Advantages
- Switch ON & Switch OFF takes longer because the conversion rate is higher

Footer (NMOS):

- . Due to higher mobility and drive strength, the resistance will be lower and the slew rate will be less
- . Short circuit power is smaller due to lower conversion rate
- . Since the resistance is lower, the leakage power will be greater
- . For the same amount of current, the footer gates are smaller (NMOS has twice the mobility of PMOS)
- . Switch ON & Switch OFF takes less time because the conversion rate is lower
- . We prefer PMOS header because it has less leakage (due to higher resistance) and slower switching rate. If the switching speed is fast, it will try to draw a huge inrush current to turn on the module at the same time, which will cause power integrity issues.
- . Therefore, power gating devices should be High VT cells to achieve slower switching.
- . NMOS is more leaky than PMOS, and the design is more sensitive to ground noise on the virtual ground (VIRTUAL_vss) coupled through the footer switch
- . The selection of footer & header depends on 3 parameters (switching efficiency, area efficiency & body bias)

- . Switch Efficiency: The ratio of drain current in ON and OFF states (I_{on}/I_{off}). The total leakage current of a power switch is primarily determined by the switching efficiency.
- . Area efficiency : Depends on product length*width ($L*W$). Switching efficiency in PMOS transistors decreases as W increases, so smaller W is preferred.
- . Body Bias : Applying reverse body bias on the sleep transistor can increase switching efficiency (body bias increases V_t , so the leakage current I_{off} will decrease) and significantly reduce leakage. The cost of reverse body biasing in header switches is significantly lower than in footer switches. This is because PMOS NWELLS are easily used for bias connections in standard CMOS processes. NMOS transistors do not have WELL in the standard CMOS process and require higher chip manufacturing costs and design complexity.
- . Conclusion: PMOS header is preferable in reverse body bias applications.

124. What is power gating , its integrity issues and the comparison between coarse grain power gating and fine grain power gating?

Power Gating

Effectively reduces leakage power in standby or sleep mode

Power Gating Overheads :

- The silicon area occupied by the sleep transistor.
- Cabling resources for permanent and virtual power networks .
- Complex power gating design and implementation process.

Power integrity issues

- IR drop on sleep transistor
- Ground bounce caused by inrush wake up current
- Wakeup latency

Compared with fine grain , coarse grain power gating has

- Not sensitive to changes in PVT
- Introduce fewer IR-drop changes.
- imposes smaller area overhead

125. After clock tree synthesis (CTS), there will be many timing paths ending with clock gate/ICG enable pins. Why are these paths not fixed in position and what should I do with them?

After clock tree synthesis, clock gates become critical because by default their clock pin arrival time has the same delay as the register clock pin. Once the clock tree is built, the clock gates will be in the middle part of the clock tree, not the leaf ends. Therefore, the clock arrives earlier than at the clock leaf pin, and timing is affected.

A simple example is shown below:

Pre-CTS, the register pin of the clock gates and the clock pin are 0 ns clock delay, which simulates the same arrival time for both.

Post-CTS, the clock gates are now in the middle of the tree, seeing 800ps latency. However, all registers see a 1.5ns arrival time for their clock pins because they are at the leaf level of the tree.

Any path from register to clock gate now sees a difference in clock arrival time, and pre-CTS slack is reduced by 700ps (1.5ns - 800ps). Since the clock gate should be at an intermediate point to allow for the closed portion of the clock tree, it is incorrect to assume that the clock gate's clock pins should be balanced with the registers.

These paths can be resolved by:

First, check the location of these ICGs in the post-CTS clock tree. Whether they are close to the root of the clock tree or the clock pin of the flip-flop affects how you handle them.

If the clock gate is approximately in the middle of the clock tree, you can benefit by splitting (duplicating) the clock gate. Splitting the clock gates creates parallel copies of the original drivers, resulting in more clock gate drivers with less load on each driver. If the split is done for pre-CTS, then we effectively push the clock gate further down the clock tree, increasing power but improving enable timing. See `split_clock_net` command.

If the clock gate is at the beginning of the tree or near the bottom of the tree, splitting the clock gate is unlikely to bring any improvement. In this case, you should add the pre-CTS clock delay value to the ICG clock pin in order to simulate the pre-CTS delay correctly. Using the example above, you would apply a -700ps delay to the clock gate's clock pin during `place_opt` but before clock tree synthesis. Applying a delay allows you to correctly model slack before you know the actual clock gate arrival time.

If the ICG was a single "top clock gate" fed by a relatively small logic cone, you could apply floating pin constraints to the flip-flops feeding the enable signal logic to get their clock earlier (useful skew). Sometimes this technique is the best solution for top-level clock gates because it does not impact power; splitting the top-level clock gates can have a very large power impact.

126. How should CRPR be handled in SI analysis? That is, during setup analysis using SI analysis or crosstalk analysis, how to remove the pessimism of cells affected by crosstalk? Why?

CRPR and Crosstalk Analysis (CRPR and crosstalk analysis)

(1) When you perform crosstalk analysis using Prime Time SI, the delay changes due to crosstalk on the common segment of the clock path may be pessimistic, but only for zero-cycle check. A zero-cycle check occurs when the same clock edge triggers both the launch and capture events of a path. For other types of paths, the latency changes caused by crosstalk are not pessimistic.

Because, for launch and capture clock edges, it cannot be assumed that the changes are the same

(2) Therefore, the CRPR algorithm eliminates the delay caused by crosstalk in the common part of the launch and capture clock paths only when the check is a zero-cycle check. In a zero-cycle check, the aggressor's switch direction affects both the launch and capture signals in the same way.

(3) The following are some situations where CRPR may apply to delays caused by crosstalk:

1. Standard hold check
2. Perform a hold check on a register that has a Q-bar output connected to a D input, such as in a divide-by-2 clock circuit.
3. Due to the parasitic capacitance between the Q-bar output and D input of the register, use crosstalk feedback for hold check
4. Do a hold check on a multi-cycle path set to zero, for example, a circuit that uses a single clock edge for startup and capture, design with a skew between startup and capture.
5. Some setup checks involve transparent latches

(4) There is an important difference between hold analysis and setup analysis related to crosstalk in the common part of the clock path .

For hold analysis ,

The launch & capture clock edges are usually the same edge. Clock edges passing through a common clock portion must not contribute differential crosstalk to the launch clock path and the capture clock path. Therefore, worst-case hold analysis eliminates the crosstalk contribution of the common clock path.

For setup analysis,

It will be done on a different clock edge, which will occur one clock cycle later. So on the common clock path, the crosstalk contributions from launch & capture paths are different. So we should not eliminate the crosstalk contribution from the common clock path.

127. What is uncertainty ? Why do we have different uncertainty for setup and hold before cts and after cts ?

Uncertainty:

Specifies a window within which clock edges can occur.

In physical design, uncertainty will be used to model several factors:

- ♦ Jitter (the deviation of a clock edge from its ideal position)
- ♦ Extra margins
- ♦ Skew (in pre-cts)

Specify different uncertainty for setup and hold.

Since the hold check is performed against the same clock edge, any deviation (jitter) in the clock edge will affect the launch flop and capture flop in the same way .

So **for hold uncertainty , there is no need to model jitter** , which is why we always see the value of hold uncertainty lower than the value of setup uncertainty .

Before CTS, uncertainty also models the expected skew after implementing the clock tree (post-CTS) . Therefore, **in the post-CTS stage, we will reduce the uncertainty value because there is an actual skew value.**

setup Uncertainty:

```
Pre-CtS = Jitter + Skew + Extra setup margin
```

```
Cts = Jitter + Extra setup margin
```

Hold Uncertainty:

```
Pre-CtS = Skew + Extra hold margin
```

```
cts = Extra hold margin
```

128. Why do we have different de-rating factors for clock cells and data cells ? What is the reason?

(1) The switching activity of clock cells is much greater than that of data cells , so it will cause more PVT changes. Therefore, clock cell latency changes due to OCV may cause more violations than the data path . This is why clock cells are derated more than data cells .

(2) The OCV impact is usually more obvious on the clock path, because in the chip, the clock path is longer,

(3) The clock unit has second-order effects, so it is derated more. Data units have first order effects and therefore less derating.

129. What are the advantages and disadvantages of buffers and inverters used in CTS ? Which one do you prefer to use when building a clock tree?

Inverter : Smaller area and can drive longer distances. But there are more switches. Suitable for pulse width and pulse cycle maintenance.

In other words: when you compare the same drive strength units, the current drive capacity of the Inverter is greater than the buffer , i.e. the Inverter is faster than the buffer .

Therefore, under the same net length, the number of Inverters it requires is less than the number of buffers .

Therefore, it would be better to use Inverter to insert delay (Insertion Delay). That is, it indirectly reduces the impact of OCV (OCV is proportional to the insertion delay) on timing.

Since there are more CTS switches based on inverter , OCV may be increased? (someone said). Maintains 50% duty cycle and inverter has regenerative property

i inverter has better noise reduction effect than buffer .

130. What are the uses of TIE cells ? What is the internal structure of a TIE?

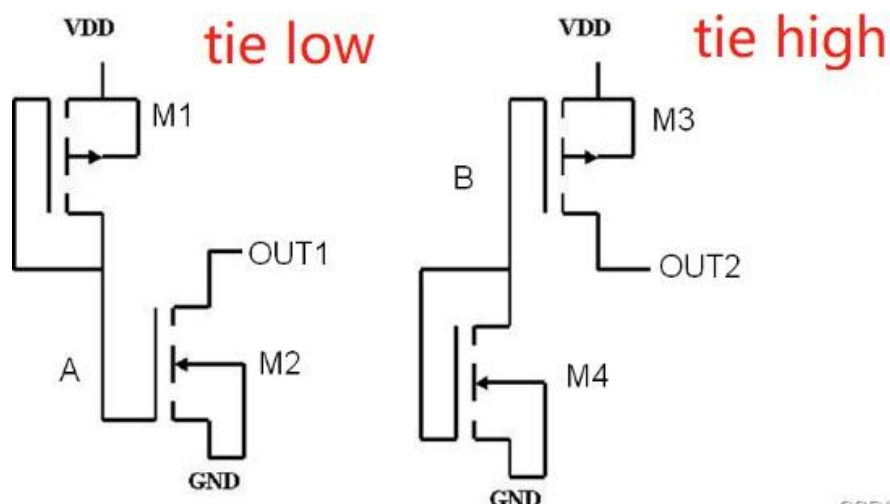
tie cell: voltage clamp unit. perform esd protection

(1) At lower technology nodes, the gate oxide layer of the transistor is very thin and is very sensitive to power supply voltage fluctuations. If the transistor gate is connected directly to the PG network, the transistor gate oxide may be damaged by supply voltage fluctuations. To overcome this problem, TIE cells are introduced b/w PG and transistor gates.

(2) Therefore, TIE cell is introduced to prevent ESD problems.

(3) These TIE cells can be easily converted from 0 to 1 and vice versa by simply replacing a metal layer .

(4) Suppose you only need to use a metal mask for ECO to change a 0 to a 1 on the input of one of the combinational logic gates, but you only have one tie down cell available . If this tie down cell was designed in such a way that you could easily change its functionality from 0 to 1 by using only one metal layer, it would be a cost effective change for local ECO .



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Questions 131-140

131. If we use ocv derating factors , why do we use clock uncertainties (setup uncertainty and hold uncertainty) after the post-cts stage ?

(1) **Jitter** is not part of OCV. This Jitter problem is caused by PLL noise. So we should keep uncertainties and ocv derating factors separately.

(2) **OCV derating** is based on the path margin (margin). It will only consider PVT changes OCV → process changes,

i.e. transistor channel length changes/gate oxide thickness changes due to mask changes, CMP changes and etching.

That is, if two instances of library cells with the same driving strength are located at different positions in the layout , the cell delay (cell delay) may be different due to these changes (cell delay changes due to process changes)

(3) **Temperature changes** : Junction temperature and clock cells switching activity and high - density areas may produce higher temperatures. So unit latency will vary

(4) **Voltage change** : For some cells, the voltage will decrease due to IR drop problem. This may be due to the higher density of these regions. IR drop margin depends on the IR drop you plan to implement in your design. If you meet the 3% IR drop, then you have the flexibility to reduce the flat margin of the OCV derating .

(5) **If you are not using ENDCAP cells** in your design , then you need to fill/add more margins in derating factors . Because the characteristics of each standard library cell assume that it is located in the middle of the chip (if a cell is located in the middle, the stress on the cell will be smaller, so it can run normally. If it is located at the end, then The stress will be greater, so the cell may not operate as expected). There are many factors like this. Foundries and companies decide to reduce or increase flat margins . **[I think the stress here is voltage stress]**

132. If base gets frozen , how do you solve the setup timing violation ?

1. See if there are any detours in the nets of that path . Then delete net and reroute.
2. Routing on higher metal layers , or layer promotion
3. Fix data path crosstalk (crosstalk) problem
4. Fix clock path crosstalk problem
5. Use buffer instead of fortune/spare cells
6. Logic restructuring . For example, rearrange the timing critical nets of the AND gate away from its ground and the timing critical nets of the OR gate , away from its power . Therefore, non-timing-critical nets come first and do not act as a load for timing-critical nets. Eventually the latency will be reduced.

133. What are the repair methods for antenna violations ?

1. Add antenna diodes near gate
2. Jump to a higher metal layer near the gate

3. If the path is not timing critical, insert a buffer near the input gate
4. Connect antenna violation net to buffer input pin, output pin float or connect dummy load

134. If the net is split by adding buffers , will the net delay be reduced?

Assume that net is L unit length or section, and use distributed RC model to represent each net section.

Assume that the resistance per unit length is R_p and the capacitance per unit length is C_p .

Net Total resistance $R_t = L * R_p$

Total Capacitance = $L * C_p$

Total net delay $D_t = R_t * C_t = (L^2) * R_p * C_p$

If buffer is inserted,

net length = $L/2$, net delay = $(L^2) * R_p * C_p / 4$

The derivation method of net delay is:

R 与 $(L/A = L/Wt)$ 成正比,

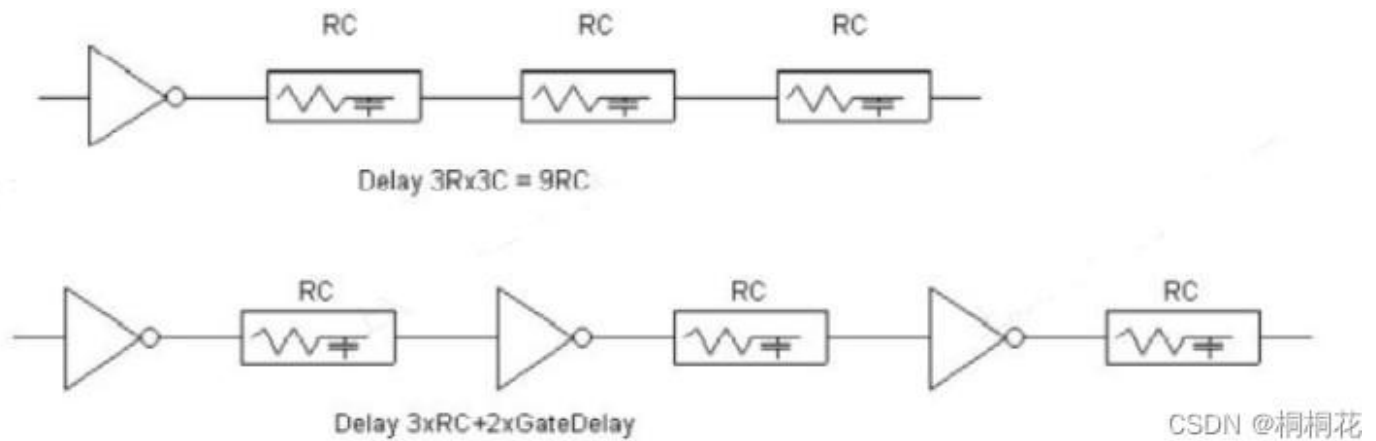
C 与 $(A/D = tL/S)$ 成正比;

RC 与 (L^2) 成正比

The concept of repeaters is the same as what I discussed in "Inserting the Buffer" (above). It's just that I'm trying to explain this in a different way, but the general concept is the same.

Long distance routing means a huge RC load due to a series of RC delays, as shown in the figure.

A good option is to use repeaters, which divide the line into segments. Why could this solution be better in terms of latency? Because gate delay is very small compared to RC delay.



In the case of a single inverter driven interconnect, the propagation delay becomes

$$T_{delay} = t_{gate} + nR \cdot nC = t_{gate} + (n^2)RC = t_{gate} + 9RC$$

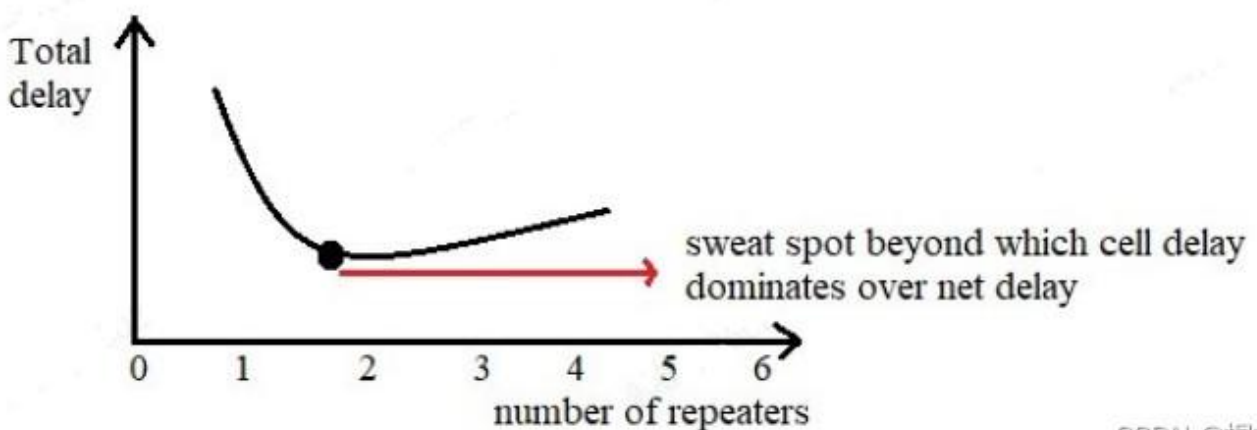
If two repeaters are inserted, the propagation delay becomes:

$$T_{delay} = t_{gate} \text{ (逆变器延迟)} + 2t_{gate} \text{ (中继器延迟)} + 3RC = 3t_{gate} + 3RC$$

This way you can see what the RC delay is like without the repeater in the circuit.

Therefore, if the gate delay is much smaller than the RC delay, the repeater can improve the switching speed performance, but at the cost of higher power consumption. As you keep adding repeaters to improve the transition on a fixed net of length L , then the total delay will decrease as you add repeaters to the network.

At a certain moment, gate delay is greater than RC delay, that is, gate delay is better than network delay. If you add repeaters beyond that point, the overall total delay starts to increase. So you shouldn't add buffers outside of that sweet spot. This is how we calculate how much net length specific buffers can drive.



135. Why can't we use PMOS as footer and NMOS as header ?

If we use NMOS as header (Drain D is connected to VDD, Source S is connected to load CL & SHUTDOWN block) , then NMOS will produce an output value of $V_{DD}-V_T$.

This means that we reduce the supply voltage of the "shut down" module connected to the NMOS source. This voltage reduction affects the performance of cells in the shutdown block .

If we use PMO as footer (Source S is connected to the SHUTDOWN block and Drain D is connected to ground) , then the PMOS will produce the output value of V_T at the source . This means that the shutdown block is not purely connected to the ground.

This arrangement attenuates the output voltage.

136. NLDM vs CCS?

CCS timing model:

(1) Solution to the problem described in the RC-009 warning message.

This warning occurs when the drive resistance of the driver model is much smaller than the network impedance to ground .

(2) Better processing of Miller Effect , dynamic IR drop, and multi-voltage analysis.

(3) With the emergence of smaller nanotechnology, CCS timing methods for modeling cell behavior have been developed to address the impact of deep submicron processes.

(4) The driver model uses a time-varying current source .

The advantage of this driving model is its ability to accurately handle high-impedance networks and other nonlinear behavior.

(5) The CCS timing receiver model uses two different capacitance values instead of a single lumped capacitance.

The first capacitor acts as a load before the input delay threshold. When the input waveform reaches this threshold, the load is dynamically adjusted to the second capacitance value.

This model provides a better approximation of loading effects in the presence of the Miller Effect

(6) The CCS timing model provides additional accuracy for modeling cell output drivers by using time-varying and voltage-dependent current sources .

Provides timing information by specifying detailed models of receiver pin capacitance and output charging currents under different scenarios

(7) The CCS model does not have long tail effect.

NLDM:

(1) The NLDM driver model uses a linear voltage ramp connected in series with the resistor (Thevenin model) .

The resistor helps smooth the voltage ramp so that the resulting driver waveform resembles the curvature of the actual driver driving the RC network.

(2) When the driving resistance is much smaller than the network impedance to ground, the smoothing effect will be reduced, which may reduce the accuracy of RC delay calculation.

When this occurs, Prime Time adjusts the driver resistance to improve accuracy and issues an RC-009 warning.

(3) The NLDM receiver model is a capacitor, representing the load capacitance of the receiver input.

Different capacitor values can be applied to different conditions, such as rising and falling transitions or minimum and maximum timing analysis.

However, a single capacitance value is suitable for a given timing check, which does not support accurate modeling of the Miller effect.

(4) NLDM timing models represent the delay through a timing arc based on output load capacitance and input transition time.

In fact, the load seen by the cell output consists of capacitance and interconnect resistance.

Interconnect resistance becomes an issue because the NLDM method assumes that the output load is purely capacitive

(5) NLDM has long tail effect

(6) Conventional STA with the NDLM library cannot consider the Miller effect and long tail effect.

(7) Timing analysis results can be more optimistic than Spice results.

137. How to repair the DRC of a specific area on the database after the route (which will be tape-out soon) ? Consider two situations, for example, the situation where the cell density of the area is high and the situation where the cell density of the area is low.

Cell Density is relatively high:

1. Collect all networks in the area and find non - critical timing nets with +ve slack margin exceeding 150ps .

Then, incrementally re-route those non-critical networks by turning off SI driven and timing driven options (delete these nets & delete global route & reroute these paths using route eco command route_zrt_eco.). (eco route)

Therefore, the tool directs those non-critical networks away from that area.

2. Collect all buffers/inverters from non-critical timing paths in this area and reduce their size. This way you can get some routing tracks or spaces in this area .

Collect all vias in the area and convert all multi-cut vias into single cut vias.

3. Based on DRC cleaning, area processing

4. Collect all nets from critical timing paths and incrementally re-route on the metal layer above the highest metal layer used in the block .

5. We can blindly apply cell padding or module padding. But it may affect timing because it disrupts all cells , including critical cells.

6. Finally, try to trim the PG straps by removing some vias without affecting the IR drop limit given by the foundry in this area. But this is less desirable since the cell density in this area is very high.

7. Add guide buffers to the nets that belong to non-critical timing paths and span the drc area , and place these guided buffers away from the drc area.

Cell Density is relatively low:

(Suppose DRC is higher because of some feed through or nets crossing from top to bottom . Vice versa)*

We cannot apply all the above techniques here because the cell density is very small.

The only thing we can do here is fix the PG. Even if PG is trimmed, the IR drop limit will not be affected because of the small number of cells.

138. In the pre-cts stage, how to solve the congestion problem in specific areas (core areas)?

1. Change the max density value & re-run the placement step to see if congestion is under control
2. If density (cell density and pin density) is greater, apply cell padding or module padding or partial density to those cells .
3. Check whether there is a floor plan problem, causing the module to separate
4. Due to some floor plan problems, check whether there is a buffer/inverter chain entering this area

139. There are 10 macros , they should be placed in a 5x2 (10 macros should be placed in 2 columns) array.

How many vertical channels will you leave for routing all the macro pins ?

Assume that each macro 's 10 metal layers are designed with 200 pins , and the macro is blocked to metal 4 (the maximum layer used by the block level is M8)

The total number of macro pins is

$$10 \times 200 = 2000$$

The available vertical metal layers are

M3、M5 和 M7。

At the 28nm technology node,

M1-M6 的track间距为 $0.05\mu\text{m}$,
M7-M8 的track间距为 $0.1\mu\text{m}$ (2x),
M9-M10的track间距为 $0.8\mu\text{m}$ (8x)

Assuming that the minimum space required is H, then the total space required is

$$H/0.05 + H/0.05 + H/0.1 = 2000 \text{ (for M3, M5, M7)}$$

$$H = 2000/50 = 40\mu\text{m}$$

This means we have to leave $40\mu\text{m}$ of space from bottom to top for these macros .

If you like, the bottom is a lot of wasted routing tracks because the 2 macros at the bottom only have 400 IO pins .

So if you want to use that space efficiently, don't leave $40\mu\text{m}$ on the bottom side, just keep it equal to the VDD-VSS spacing.

IO pins will increase as you go to the top side, and you need $40\mu\text{m}$ space on the top side (routing track requires 2000 IO pins at the top).

This means that macros should be placed in a V-SHAPE manner to effectively utilize regions and routing tracks .

140. In the presence of crosstalk, will you remove the CRPR on the setup half-cycle timing path & hold half-cycle timing path? (i.e. when you plug in the inverter on the capture clock

pin)?

no.

Crosstalk contributions from the common clock path are different during capture clock path & launch clock path calculations .

Because the clock edges of launch flops & capture flops are different. That is, the launch edge & capture edges are separated by half a period and are used for setup & hold calculations. So we should not remove these crosstalk values from the timing analysis of setup & hold

Questions 141-150

141. How to improve insertion delay?

(1) By using appropriate clock drive strength cells .

That is, it is not a low drive strength cell and prefers to use clock inverter instead of clock buffers.

(2) Use double width for clock nets (double width)

Because it reduces the resistance by half ($R'=R/2$) and increases the capacitance to ground slightly.

So overall the insertion delay will be improved due to the main effect of the resistor

(3) Place the clock port on any core edge.

This way, this should be more or less equidistant from all corners .

(4) Place the first level clock gating element in the center of the design and build the clock tree from there.

(5) Slightly relax the max transition limit and skew limit to obtain insertion delay.

(6) multi-point CTS.

That is, divide the entire design area into 4 equal parts and then build.

From the main clock port to the Hclock tree at these 4 points, then add 1 large clock buffer in each area.

1. Disconnect all CP pins from the master clock port
2. Collect all registered CP pins in each region ,
3. Connect the CP pins back to the output of the large clock buffer located in this area
4. Then build a regular clock tree from the output pins of that big clock buffer .

(7) clock mesh

But the wiring resources and power consumption will be more.

(8) Before entering the CTS step, congestion should be minimal.

Otherwise congestion may detour clock nets , so a large number of cells may be needed to repair DRV.

So it will reduce insertion latency.

(9) Floor plan problems, such as lack of density in some macro channels .

Because some registers may be placed in these areas.

Therefore, the CTS engine will try to balance this register with all other leaf pins by adding a large number of clock cells

(10) fish-bone CTS

(11) Only use a single buffer/inverter with appropriate drive strength for CTS .

This would be good in MCM designs since the OCV effects of different corners would be minimized.

This technique does not improve insertion latency. However, since the impact of OCV is small, it indirectly helps reduce the number of violations .

(Using only one buffer/inverter is overly optimistic.

The CTS must drive varying amounts of load at the spine root. A large number of cells must be used to process. Otherwise, if you use low drive , it will add high drive cells in some places even if it is not needed . Too many cells will be added .)

142. How to solve the problem of antenna violations ? What are the possible solutions to these problems?

Antenna Ratio = Metal area connected to gate/gate area.

Antenna violation occurs if the antenna ratio exceeds the value specified on each metal layer .

Solution:

(1) The layer jumps to a higher metal layer (the metal area will be reduced)

(2) Add an antenna diode near the gate (the gate area will increase)

Antenna effect

When a metal line in contact with a transistor's gate is plasma etched, it can charge to a voltage high enough to eliminate the thin gate oxide. This is called plasma-induced gate oxidation damage, or simply the antenna effect.

It increases gate leakage, changes threshold voltage, and reduces transistor life expectancy. Longer wires will accumulate more charge and are more likely to damage the gate.

During high-temperature plasma etching, diodes formed by source-drain diffusion can conduct large amounts of current. These diodes release charge from the wire before the gate oxide is damaged.

143. Can you use buffer to repair antenna violation ?

Yes, it can be fixed with buffer ,

i.e. replace antenna diode (output float and input pin connected to gate) with buffer .

This will increase the gate area. In this way antenna violation will decrease

144. There are three modules with different voltage domains (assuming V1, v2, V3).

V1 is "always on", V2 is "ON/OFF (ONO Block)", V3 is "shut down"

V1 is placed at the top, V2 is placed in the middle, and V3 is placed at the bottom.

If the signal passes from V3 through V2 to V1, how many isolation units are needed? vice versa.

Signal from V1-V3:

When the signal is transmitted from V1 to v2, no isolation unit is required,

Because V1 is always on.

When V2 is off and V3 is on: 1 isolation unit is required between V2 and V3

When V2 is on and v3 is on: there is no isolation unit between V2 and V3

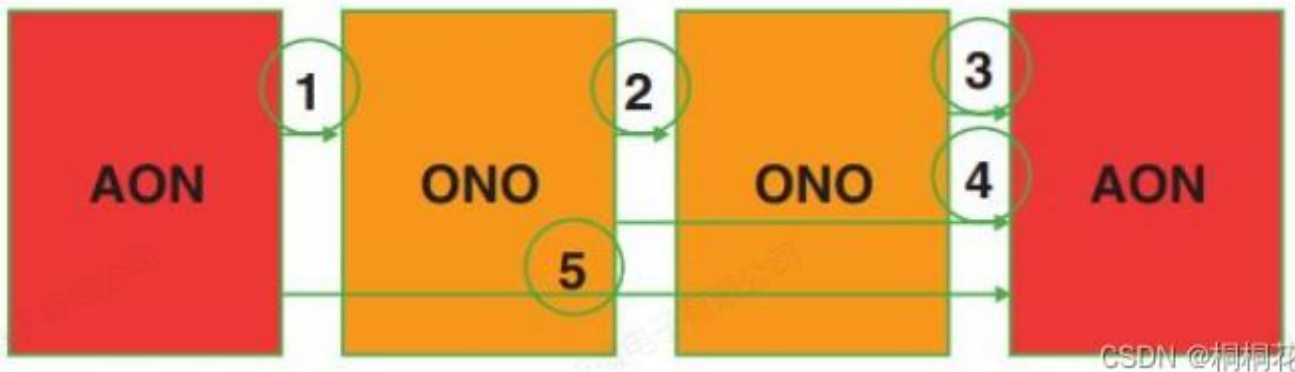
Signal from V3-V1:

When the signal is passed from V3 to V1, 2 isolation units are required.

between a V1 and V2

One is between V2 and V3.

For better understanding: The following figure describes the power domain crossing scenario of AON and ON0



Power domain crossover scenario.

The example picture is:

1. AON drives ONO (no isolation required)
2. ONO drives ONO (no isolation required)
3. ONO drives AON (requires isolation)
4. ONO block ONO feedthrough
5. AON feedthrough in ONO block

145. How to use conformal LEC to generate functional ECO ?

By comparing the synthesized netlist implemented by functional eco with the routed netlist with conformal , a functional ECO patch is generated .

146. What will happen if I add clock slew/transition in the pre-cts stage?

- (1) Clock slew/transition is used to model the ck to q delay of the flip-flop and the library setup check in advance in the pre-cts stage, instead of waiting for the CTS step.
- (2) In the pre-cts stage, the clock transition constraint on clock pins is nothing more than modeling the library setup margin, which can be seen after cts.
- (3) The Library setup check on the flip-flop will increase (Library setup check will change according to the slew on the clock pin and the slew on the data pin). Therefore, the data path has less time period and the tool will work harder to fix the timing. violations.

147. What happens if you have multiple clocks passing through the MUX? How do you build a clock tree?

1. To establish a "clock tree" through the MUX's D0 pin from its functional clock port to the functional clock of all register pins .
2. **Set set_dont_touch_network** on MUX/Z pin , then build CTS for test_clock.
3. Then, fix the "DRC only" clock tree on the Test clock . The Test clock is connected to the D1 pin of the MUX, that is, only connected to the D1 pin.

148. Why focus on clock skew rather than timing closure in the clock tree stage ?

That is, if the timing requirements are met during CTS , why do we need to pay attention to skew ?

Why can't we focus on timing instead of satisfying skew in the CTS stage?

Reducing clock skew is not only a performance issue, but also a manufacturing issue.

Scan-based testing requires minimal skew to allow error-free movement of scan vectors to detect stuck-at faults and delay faults in the circuit . (Scan-based testing, which is currently the most popular method of structurally testing chips for manufacturing defects,)

In these circuits, Hold failures at the best-case PVT Corner are common. Because there are usually no logic gates between the output of one flip-flop and the scan input of the next flip-flop in the scan chain .

In this case, dealing with and reducing clock skew can often resolve these hold failures

149. There are 3 flops .

In the pre-cts stage, the setup time from A to B is +200ps, and from B to C is -50ps.

In the CTS stage, the skew constraints given are 50ps.

(That is, A to B is 50ps skew , B to C is +50ps skew)

How will you solve it?

Will borrow timing on either side , that is, advance the clock pin of B flop by 50ps

150. In the post route stage, if I have to slightly over-constrain the design, would you prefer to adjust clock frequency or clock uncertainty ?

It's always better to adjust the clock frequency. Because it changes/affects the calculation of crosstalk arrival windows.

This way the EDA tool can see these crosstalks and fix it appropriately without over-fixing the design and the success rate on the silicon is always higher.

And if you change uncertainty, it will not affect the calculation of crosstalk arrival windows.

That is, you don't see an increase in timing path violations due to crosstalk windows . But you do see a lot of timing path violations due to increased uncertainty , you need to blindly fix them, and it's like over-fixing the design. Designs on silicon may fail and target performance may not be achieved.

Because when we try to run the design at targeted/changed frequency , we may see cell delay changes/noise bumps due to crosstalk

Questions 151-160

151. If clock uncertainty is adjusted , will it affect SI?

No. The clock uncertainty setting does not affect the calculation of crosstalk arrival windows .

152. If the clock frequency is adjusted (the clock period is smaller), will it affect the SI?

Yes.

Changes in clock frequency will affect the calculation of the crosstalk arrival window. If a stable signal passes near a clock edge, crosstalk noise increases.

As a result, the latency of cross-influenced cells will change and more setup/hold violations will occur.

If there is an opportunity for optimization during the route phase, the tool will work to fix these timing violations.

153. What are the most commonly used top-level commands in EDA?

```
placeOpt
clockOpt
routeOpt
ecoRoute
ecoPlace
. . .
```

154. How to obtain the options/default settings in EDA?

```
get*Mode,
//其中*表示 ECO, trailRoute, detailRoute
```

155. How to get macro's llx & ury

```
set llx [lindex [lindex [dbGet [dbGet -p top.instance.name $macro].box] 0] 0]
set lly [lindex [lindex [dbGet [dbGet -p top.instance.name $macro].box] 0] 1]
```

156. b/w What is the difference between regular OCV and AOCV? Do you think regular OCV derating factors are more pessimistic than AOCV?

Yes, Regular OCV is more pessimistic when we have deep logic level depth .

In Regular OCV , flat derating is applied to all cells regardless of logic level, so we will see a lot of timing violations.

AOCV (advanced OCV): As the logic depth increases, the derating coefficient will decrease. For example, the farther the cell is from the bifurcation point, the derating coefficient will increase.

Longer paths with more gates tend to have less total changes. Because random variations from door to door tend to cancel each other out. Therefore, AOCV applies higher derating values to shorter clock paths and lower derating values to longer clock paths.

AOCV determines the derating factor based on **a measure of the path's logical depth and the physical distance traversed by a specific path** . Longer paths with more gates tend to have less total variation because random changes from gate to gate tend to cancel each other out. Paths that span larger physical distances across the chip tend to have larger system variations. AOCV is less pessimistic than traditional OCV analysis, which relies on constant derating factors that do not account for path-specific metrics.

157. What does " location based Derating " in AOCV mean? What is your reference for deciding the number of cells to derate?

OCV derating will increase as the unit's position from the clock bifurcation point increases

158. Why can't we route the design first and then do clock tree synthesis? Is there any reason?

Generally routing is driven based on timing, which is only possible after the clock is established.

If the design is routed first, you won't get proper routing resources for the clock tree, so clock routing will take a detour and will affect insertion delay& skew.

159. What will happen if PMOS and NMOS in a CMOS inverter are interchanged?

PMOS: ON ($v_{gs} < -V_{tp}$) OFF ($V_{gs} > -V_{tp}$)

NMOS: ON ($v_{gs} > v_{tn}$) OFF ($v_{gs} < V_{tp}$)

Assume $V_{tn}=V_{tp}=V(t)$

After PMOS & NMOS exchange positions, the connection is as follows:

NMOS: Drain D is connected to V_{dd} , source S is connected to load CL

PMOS: Drain D is connected to ground, source S is connected to load CL

The output voltage will pass through the load CL (assuming the initial voltage on CL is 0v)

Vin = Vdd:

For NMOS:

$V_{gs} = V_{dd} - 0 = V_{dd}$, greater than $V(t)$. Therefore NMOS is on. [i.e. $v_{gs} > v(t)$]

It starts charging the load capacitor CL to Vdd.

When the output voltage VO across CL reaches $V_{dd} - V(t)$ [i.e. $VO = V_{dd} - V(t)$],

Then the gate and source voltage difference of the NMOS will drop to $V(t)$ [i.e. $V_{gs} = V_g - V_s = V_{dd} - (V_{dd} - v(t)) = V(t)$].

Then NMOS will close.

So when $V_{in} = V_{dd}$, the output $VO = V_{dd} - V(t)$. The output decays with $V(t)$.

For PMOS:

$V_{gs} = V_g - V_s = V_{dd} - 0 = V_{dd}$, which is greater than $-V(t)$, so PMOS is turned off.

Vin = 0:

For NMOS:

$v_{gs} = V_g - V_s = 0 - (V_{dd} - V(t)) = -(V_{dd} - v(t))$, so NMOS is turned off

For PMOS:

$v_{gs} = v_g - V_s = 0 - (V_{dd} - v(t)) = -(V_{dd} - V(t))$, less than $-V(t)$, that is, $v_{gs} < -V(t)$, the condition is met. So PMO is on.

Therefore the voltage across the load capacitor CL will start to discharge towards 0v through the PMOS and it will stop discharging when the voltage across the load CL reaches $V(t)$.

So at this time $V_{gs} = V_g - V_s = 0 - v(t) = -V(t)$. So PMOS will be off at this time.

Therefore, when V_{in} is 0v, the output voltage on load CL will be $v(t)$.

****Summarize:****

When $V_{in} = v_{dd}$ $VO = V_{dd} - V(t)$

When $V_{in} = 0v$ $VO = V(t)$

So this circuit does not act as a pure buffer, but as a partial buffer.

160. If the multi cycle value of hold is 2, at which edge do you verify/check the hold violation ? Does this hold check depend on frequency?

By default, setup MCP (MCP multi-cycle) will be checked against the capture edge

By default, the hold MCP will be checked against the launch edge .

Timing checks for edges will change based on the -start or -end option specified in the MCP definition.

```
create_clock -name CLKM -period 10 [get_ports CLKM]
set_multicycle_path 3 -setup -from [get_pins UFF0/Q]-to [get_pins UFF1/D]
//setup多周期约束指定 从 UFF0/CK 到 UFF1/D 的路径, 最多可能需要三个时钟周期才能完成setup检
```

```
| set_multicycle_path 2 -hold -from lget_pins UFF0/Q]-to lget_pins UFF1/D]
```

Specify hold multi cycle as 2 to obtain the same hold check behavior as the single cycle setup case .

This is because without this hold multi cycle specification, the default hold check is done on the active edge before the setup capture edge , which is not what we want.

We need to move the hold check two periods before the default hold check edge , so two holds are specified for multiple periods.

The number of cycles represented on multicycle hold specifies the number of clock cycles to move backward from its default hold check edge, which is the one active edge before the capture edge is set.

Since this path has a multicycle setting of 3, its default hold check is on the active edge before the capture edge.

In most designs, if the maximum path (or setup) requires N clock cycles, it is not feasible to implement a minimum path constraint greater than (N-1) clock cycles.

By specifying a multicycle hold of two cycles, the hold check edge is moved back to the start edge (at 0ns). Therefore, in most designs, a multicycle setting specified as N (cycles) should be accompanied by a multicycle setup specified as N-1 (cycles). Multicycle retention constraints.

What happens when a multicycle setting of N is specified but the corresponding N-1 multicycle hold is missing?

In this case, the hold check is performed one cycle before the capture edge is set. For the hold check, the capture edge is moved back to 0ns and the transmit edge is also at 0ns, then the hold is not frequency dependent at all.

Questions 171-180

171. What is timing window and explain it?

STA obtains this information from the timing windows of aggressor nets .

During the timing analysis process, the earliest and latest switching times of nets were obtained. These times represent the timing windows within which the network can switch within one clock cycle . Switching windows (rising and falling)

provide necessary information about whether the aggressor nets can be switched together.

Timing window : The difference between the latest and earliest arrival times for a particular network is the time window for that network.

The timing window is a window within which the signal can change at any time within the clock cycle.

172. Why not fix the dynamic peak power? Why only fix the RMS power?

RMS (Root Mean Square) power rating: The meaning of RMS power is defined as root mean square, which is a statistical way of expressing DC voltage or AC voltage. It doesn't use peaks, but averages, so you get a better idea of its true performance and power handling capabilities.

in dc circuit

Power is always calculated as RMS power that produces the same heating effect as DC power

The current drawn is always rms current..... $I = \sqrt{\text{Power}/\text{Resistance}}$

173. How to fix text short in LVS ? Can we use text short for tapeout? What is text short ?

text short: The same net shape or pin shape or substrate layer with two different labels (label).

(Ideally no risk since it's only technically label short).

But I wouldn't tapeout, it could hide two different nets with the same label , or indeed it's an open case.

174. What is +ve unateness, -ve unateness & non-unate? Do you see Unateness in the library? Do we see unateness in DFF? What kind of Unateness will you see in DFF?

+ve Unateness:

If the output signal direction is the same as the input signal direction or the output signal does not change, +ve unate is used to represent a timing arc.

Example : AND, OR

-ve Unateness:

If the output signal direction is opposite to the input signal direction or the output signal does not change, a timing arc is said to be -ve unate.

Example : NOR, NAND, inverter

Non-Unate:

In a non-unate timing arc, the output transition cannot be determined based solely on the direction of change of the input, but also depends on the state of other inputs.

Example: XOR

Because DFF has non-unateness for the timing arc **CP**→**Q** , since it not only depends on the CP transition, but also on the transition on D Pin. See example below:

```
pin(Q){
  direction :      output ;
  max_capacitance : 0.404;
  function :       "IQ";
  timing(){
    related_pin : "CP";
    timing_sense: non_unate;
    timing_type : rising_edge;
  }
}
```

175. What is the value of -ve lib in library ?

library hold margin value can be negative.

Therefore, a negative hold check, means that **the data pin of the flip-flop** can change **before the clock pin** and still satisfy the hold time check .

The trigger's library setup margin value can also be negative.

This means that on the flip-flop's pin, **the data** can change **after the clock pin** and still satisfy the setup time check .

Can both setup and hold be negative numbers?

No; in order for the **setup check** and **hold check** to be consistent, the sum of the **setup** and **hold** values should be positive.

So if **the setup (or hold) check contains a negative value** - the **corresponding hold (or setup)** should be positive enough so that the

setup plus hold value is positive

For flip-flops, it is helpful to have a negative hold time on the scan data input pins.

This provides flexibility in terms of clock skew and can eliminate the need for almost all buffer interpolation to fix hold violations in scan mode .

```
pin (D){
    direction : input; timing ) {
        related_pin : "CK";
        timing_type : "hold_rising";
        rise_constraint ("setuphold_template_3x3") {
            index_1("0.4,0.57,0.84"); /* Data transition*/
            index_2("0.4,0.57,0.84"); /* Clock transition*/
            values(/* 0.40.570.84*/\
                /*0.4*/"-0.220,-0.339, -0.584",\
                /*0.57*/"-0.247,-0.381,-0.729",\
                /*0.84*/"-0.398,-0.516,-0.864");
        }
    }
}
```

176. What is antenna violation and how to solve it? What kind of antenna violation are you seeing at the 28nm technology node ? Why does the accumulation area/gate area need to be repaired when the wafer foundry releases all the charges after each mask is manufactured?

There are two antenna violations at the 28nm technology node:

(1) Metal area/gate area (metal area/gate area)

(2) Cumulative metal area/gate area (cumulative metal area/gate area)

Because etching occurs layer by layer, even if you remove the charge after each lower layer, there is still a chance that a small portion of the charge will remain or accumulate again, destroying the gate when the charges accumulate together. Because at lower technology nodes, the gate length is very minimal and is sensitive to slight charge accumulation.

In cumulative area mode, the tool considers metal segments on the current layer and all metal segments on lower layers . In this mode, the antenna ratio is calculated as

Antenna ratio = all connected metal areas / total gate area (antenna_ratio = all connected metal areas / total gate area)

177. What is the difference between nxtgrd file and ICC TLUPlus file? Why can't we use nxtgrd in ICC to match RC delay?

Both the **nxtgrd file** and the **TLUPlus file** were generated from the same **ITF file (with the same type of version)** using the **grdgenxo utility** in STARRC .

Both files contain similar types of **RC interconnect related information** and **cap tables** . But the file format is different.

The ICC extract engine (rc_extract) may not adopt the format in the nxtgrd file

178. If vdd/freq is not involved, how to reduce the short-circuit current of the standalone inverter ?

If the output load capacitance is low and the input rise/fall times are long, the short circuit current will be larger.

To reduce short circuit power dissipation, the input/output rise and fall times should be of the same order of magnitude t

$$PAvg(\text{short-circuit}) = \frac{1}{12} [k_t f (VDD - V_{thn} - |V_{thp}|)^3]$$

Generally speaking, the short-circuit current is proportional to frequency f and voltage.

(Short-circuit current is proportional to frequency:

That is, short-circuit current usually occurs when the clock changes from 0 to 1 or 1 to 0.

So if the clock switches more due to clock frequency, the short circuit current will be more and vice versa)

179. What is die/scribe/sealer line/mask/die/corner cell?

Special corner cells are used to turn on the power signal around the corners of the shutdown block.

180. Can we increase the size of GRC cell?

There is no user control over the size of the GRC in ICC, it is dynamically calculated by the tool and is not constant.

By default, the width of the GRC is equal to the height of the standard cell row.

Questions 181-190

181. What measures have you taken to prevent SI problems in your design?

Placement

Reduce congestion in the design or do SI aware placement

Avoid areas with high cell density

Use **the place_opt -congestion -area_recovery** command

Do not use cells with lower drive strength . It acts as a victim net. (influenced by others)

Keep high drive strength cells in the unused list. It will act as aggressor nets (influence others)

Help prevent crosstalk by controlling the maximum transition constraints defined in the design . The maximum transition constraint depends on the technology and library. You need to find the best compromise between lower maximum transition constraints and congestion. During post route optimization, the maximum transition constraint can be relaxed.

Use the maximum net length constraint in the IC Compiler tool to minimize crosstalk effects by preventing very long wires .

CTS

- Apply NDR rules to the clock network . Therefore, the clock network is less sensitive to crosstalk effects.
- Apply spacing b/w clock network from signal nets
- Because clock networks are typically high-frequency networks, they are often strong attack networks. You can prevent crosstalk by shielding the clock network with a ground wire.
- Try to avoid placing clock gators very close together by adding some padding to these gators . Because, they act as attackers for adjacent signal nets

Route

- Do SI aware routing or crosstalk aware detail route
- Route can perform the following signal integrity tasks

(1) Prevent crosstalk (during global routing and track assignmen)

(2) Fix crosstalk violations during post route optimization

- Crosstalk prevention during track allocation:

Enable crosstalk prevention by running **set_si_options -route_xtalk_prevention true** and using the **-xtalk_reduction** option when running **route_opt** .

182. How will you determine the sign off requirements for static IR drop analysis and dynamic IR drop analysis ?

It comes from top level , usually based on 10-20% of SOC.

- Static IR Drop : 2.5 to 3% of (VDD + VSS)
- Dynamic IR Drop : 3 times that of static IR drop

183. Do we have to consider timing margins in IR drop target ?

Yes. It takes into account timing information.

Redhawk takes a **timing window file** as input for better results. You can find more information in the Redhawk manual.

[**timing window file** contains conversion and load information on each pin]

184. What is a zero -bit retention flop ?

All retention flops (**retention flops**) need to be isolated on their clock pin and reset pin .

These isolations can be implemented as part of the **retention flop** , or by connecting a separate isolation cell to the CK/RST pin .

The advantage of the first implementation is that it reduces complexity.

The second implementation takes up less area because we can **use a common isolation cell for multiple retention flops** . But it increases implementation complexity.

The second method is called **zero -bit retention flops** .

Retention cell: Retention cell, a special cell that can maintain its internal state when the power is turned off.

Retention cell is sequential logic and has two types:

1. retention flip-flop;
2. retention latch.

A Retention cell is composed of an ordinary flip-flop (or latch) plus an additional save-latch. save-latch can save the state when the power is turned off, and restore the normal flip-flop state when the power is turned on again.

Retention flip-flop

The difference from ordinary flip-flop is that there is an extra save-latch.

1. Save-latch is usually an HVt cell to save static power consumption;
2. Save-latch is powered by backup power supply;

Under normal circumstances,

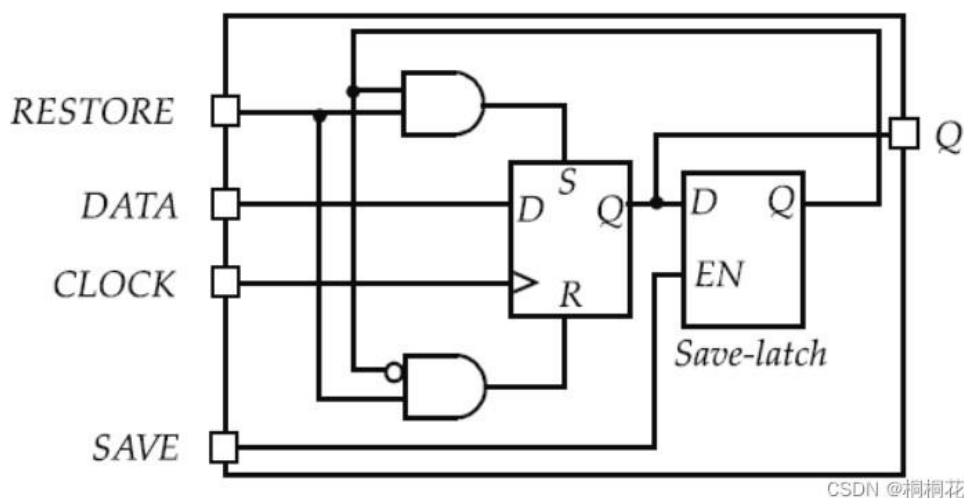
Retention flip-flop has the same function as ordinary flip-flop, but the output will be latched in Save-latch

When the power is turned off,

Since Save-latch is powered by backup power, Save-latch still maintains its original state;

When the RESTORE signal is pulled to 1,

Save-latch will send the output to the previous flip-flop, and it can immediately restore the state when it was powered off.



Retention flip-flop

184. Generally speaking, explain the implementation method of zero-bit retention flop (zero-bit retention flop)?

First, if there are any isolation cells in the incoming Netlist on the retention flop 's CK/RST pin , we remove them.

After HENS, we connect the last buffer to the reset pin of retention flop and convert it to iso high cell . This will take care of resetting pin isolation .

Before the CTS phase, we get all fan -in of CK pins for all retention flops . These will be ICG outputs . On these outputs we will insert an isolation (iso low) cell. We add a don't touch on the output of these isolation cells . Now we let the tool do the CTS. During CTS, the tool will clone isolation cells as needed . This will take care of clock isolation .

185. What are the precautions for wiring the retention flop secondary pin ?

Throughout the design, there is a secondary power stripe between the primary power stripes .

During placement , we must ensure that all RFFs are aligned with secondary power stripes . This is to reduce the resistance connection of RFF secondary power .

Apply the route-as-signal attribute on the RFF's secondary power pin .

Then, apply a 3x width NDR rule to the signals on these routes on the power nets . These signals will be routed before clock nets and signal nets.

186. What is the difference between destination isolation cell and source isolation cell ?

Isolation is required when going from switchable domain to AON domain .

Isolation cells can be placed in switchable domains (source isolated) or AON domains (destination isolated) .

If it is in a switchable domain (source isolated) , it requires auxiliary power from the AON domain.

187. When do we need level shifters ?

Level shifters are required when there is a significant (if above noise margin) voltage difference between two domains .

188. If we reduce the frequency (increase the clock cycle), what impact will it have on setup and hold ?

1. It will improve setup timing for full-cycle and half-cycle timing paths .
2. If it is a full clock cycle path (because the start and capture edges arrive at the same time), it will not affect the hold; but it improves the hold timing of the half-cycle path because the capture is half a cycle earlier than the start clock.

189. Does hold depend on frequency?

1. For full-cycle timing paths, hold does not depend on the frequency of
2. But the hold depends on the frequency of the half-cycle timing path because the startup and capture edges occur at different times.

190. What types of EM violations have you solved during your career ?

1. First I would try to increase the width of the metal. If it's crowded then I would go to a higher level to get more room to increase the width of the metal.
2. If violation appears on violation, add more vias .

Questions 191-200

191. If I randomly select a unit in my design, what is the power of that unit in static and vectorless IR drop analysis ?

Static power applies to the average power calculation algorithm and assumes everything is switching since the power is evenly distributed.

Vector IR analysis is applicable assuming that the switching rate is 20%, then the probability that a specific unit will switch is also 20%.

192. Why does antenna violation occur on signal net and not on power net?

Power nets are not connected to the gate.

193. How would you repair a semi-circular path?

Hold repair of half cycle path is easy, Setup is the key to half cycle path.

194. What are the techniques to solve cross talk?

Improve Layer , reduce net length, shield, enhance driver, and increase buffer .

195. How does body biasing affect timing ?

The threshold voltage V_t will decrease as the body/substrate bias voltage increases.

As a result, the device runs faster, timing is improved, and setup timing convergence is easier. More power consumption.

196. On post signoff DB (database), what will happen if we increase the frequency?

If we increase the frequency, the timing window for each timing arc in the design changes.

As a result, the overlap of timing windows may change and therefore may increase/decrease crosstalk effects in the design.

197. Does noise glitch always affect device functionality?

No, **noise glitch** does not always affect functionality unless it is caught by flop .

If there is a noise bump or glitch on the set/reset pins of a clock or flip-flop , it will affect the functionality of the design.

If the noise bump height is greater than the noise threshold & the noise bump width is greater than the delay of the fanout cells , the noise bump on the victim network will propagate to the output of the fanout cells

As long as this noise bump doesn't propagate through the combined units, there's no problem and no functionality changes.

If this noise bump propagates, and eventually reaches the D pin of the flip-flop, and is captured by a register, it will change the functionality.

198. What is the relationship between timing window and design frequency?

The timing window is nothing but the difference between the maximum and minimum arrival time on any timing arc .

If we change the frequency, the arrival time of the Timing window will change.

199. If you wanted to improve performance, which item would you change in design uncertainty or frequency ?

Frequency .

If you increase uncertainty , uncertainty 's close timing does not guarantee the required performance because it does not solve noise- related problems.

But changing the frequency will change the arrival windows of crosstalk and affect the timing. So if we can use it to close timing , we can guarantee the desired frequency.

200. How do you improve dynamic power consumption in your design without considering architectural changes?

Multibit

clock gating clock gating

xor self-gating on ungated registers xor self-gating on ungated registers

power aware placement using SAIF

reduce insertion delayreduce insertion delay

don't use huge uncertainty values unnecessarily

etc.

Questions 201-210

201. What is threshold voltage? How does it affect cell propagation delay?

Threshold voltage is the minimum voltage required to establish a channel between source and drain in CMOS.

The delay of the cell is inversely proportional to the threshold voltage.

202. What is Power aware placement?

Low-power placement attempts to shorten the length of high-activity nets based on available switching activity .

Low-power placement does not perform any optimizations, including resizing drivers . However, it goes hand-in-hand with timing, power, DRC, and congestion optimizations of sizing the cells .

203. HVT and ULVT are scaling under corners . If the path setup is also critical, which one would you rather choose to fix the hold ?

Due to the large variation in **driving voltage ($V_{gs}-V_t$)** , the cell delay variation of the HVT unit is higher at different PVT corners compared to the ULVT unit . So I use ULVT for this purpose.

PVT corners (for different operating voltages and different temperatures due to temperature inversion)

204. How to repair dynamic voltage drop (Dynamic voltage drop)?

(1) Add additional power/ground straps

Make the power grid denser by adding additional power/ground straps to increase current conductivity .

(2) Cell padding:

Add cell padding to cells that switch simultaneously to reduce the peak current demand of the power grid .

(3) Reduce cell:

Reduce the drive strength of cells in non-critical timing paths to reduce instantaneous current demands at local hot spots or as a preventive measure.

You can use the **set_clock_cell_spacing** command to expand clock cells (clock cells have more switching activity compared to data path cells).

This simply changes the timing window for non-critical timing path cells

(4) Insert decap cells:

The Decap acts as a charge reserve, providing current to the standard unit when batteries are switched simultaneously in hot spots.

However, decaps are leaky, which increases the leakage power in the design.

(5) Splitting output capacitance:

The amount of current drawn from the power grid is proportional to the output capacitance being driven.

Load shunting can reduce the peak current demand of the power grid . Therefore the dynamic IR drop problem will be solved.

(6) Use MIM (Metal-Insulator-Metal) to stabilize the power supply

205. What are the factors that affect V_t ?

(1)VDD(VDS)

As VDS increases, the depletion region around the drain will increase. Therefore the channel length is reduced. So the threshold voltage will change or drop.

(2)Substrate body voltage:

As the substrate body voltage increases from 0V, the threshold decreases.

$$V_t = V_t(s_b=0) - K [\text{Sqrt} (P_{hy} + V_t(s_b) - \text{sqrt}(p_{hy}))].$$

Channel length: Threshold voltage changes proportional to channel length

(3) Gate oxide layer thickness:

As the gate oxide thickness decreases, the threshold voltage decreases. For smaller V_{gs} voltage, if the thickness is smaller, a channel will be formed.

(4) Temperature:

As temperature increases, the threshold voltage decreases.

$V_t(T) = V_t(T_r) - K(T - T_r)$; $T_r \Rightarrow$ room temperature, where K is a factor

(5) Channel doping concentration :

Threshold voltage decreases with increasing channel doping

(6) Substrate doping will increase V_T

(Assuming that the P-type substrate doping of NMOS will increase V_T)

206. What happens to V_t when the temperature increases? Why?

V_T decreases with increasing temperature.

$V_t(T) = V_t(T_r) - K(T - T_r)$; ($T_r \Rightarrow$ room temperature, where K is a factor)

207. What happens to mobility when the temperature increases? Why?

As temperature increases, mobility decreases because increased temperature induces more charge carriers that collide with other charge carriers. This will reduce carrier mobility.

208. Does the mobility continue to decrease as the temperature increases?

Yes

209. PMOS (holes) vs NMOs (electron) mobility?

Electron mobility is always higher than hole mobility. The velocity of electron mobility is 2 to 2.5 times higher than that of holes.

210. What does cell delay depend on?

Input slew ,

output load,

input signal vector sequence, input signal vector sequence

Multiple input switching (MIS), multiple input switches

V_T ,

Mobility,

temperature ,

channel-length ,
VDD gate oxide thickness,

Questions 211-220

211. What happens to power and timing when clk transition is not good ?

1. Dynamic power consumption (short circuit power consumption will increase),
2. Increased clock unit delay. Increased clock unit delay. Will affect timing (timing depends on the location of the unit in the clock network)
3. Affects design performance (operating frequency will decrease)
4. If the clock transition on the CK pin of the capture register is poor, the library setup & hold time will increase. This will complicate setup & hold timing repairs.

212. What happens to setup and hold when clk transition is not good ?

We need to consider different points here.

Case 1

If the clock transition on the launch clock pin is bad and the clock transition on the capture register clk is good, the clk to q delay of the launch register will increase. So, it will reduce the setup window .

So, in this case, setup will be worse and hold will be better.

Case 2

If the clock transition on the capture clk is bad and the clock transition on the launch clock pin is good, then this will increase the capture path latency and worsen the setup library margin .

You then have to judge a setup based on the combined effect of library setup margin plus increased capture path delay due to bad clock transitions .

Case 3

A different situation arises here if there are clock transitions on the clk inverter (other than the ck pin of the start and capture registers).

A

If a bad clock transition occurs on the common clock path, then it affects both launch & capture paths in the same way . Therefore, it does not affect setup or hold violation .

B

If the launch clock path has a bad clock transition , the launch path will be delayed. setup gets worse, better for hold violation .

If a bad transition occurs on the capture clock path , the capture clock will be delayed. This is good for setup , but bad for hold timing .

213. Why should meshing be performed on conventional CTS ?

Better skew , lower latency, helps achieve higher performance or frequency, better timing closure

214. Will you place your clock gates near the sink or root ?

In terms of timing,

Placing the ICG close to sinks can better solve the ICG enable timing problem, but this will lead to poor power consumption.

For power efficiency,

Placing the ICG near the root reduces power consumption better but the timing on the Enable pin will be poor.

215. What is power gating ?

Power gating is a technique that turns off a module when no operation is being performed. This can save a lot of electricity.

There are two types of power gating technology available;

(1) header, implemented using PMOS, will be used to disconnect VDD from the block.

(2) footer, implemented with NMOS, will be used to disconnect VSS from block.

216. What is isolation cell ? How do you decide to use an AND gate or an OR gate to implement an isolation cell ?

(1) The isolation cell is placed at the interface where the signal goes from the switchable power domain to the AON power domain , and both work at the same voltage.

(2) The main purpose of placing the isolation cell is to prevent unknown logic signals from propagating from the switchable power domain to the AON domain when the switchable power domain is turned off.

(3) Another reason is that if the isolation cell is not inserted , unknown logic will reach the AON domain, causing metastable problems (logic level between 0 and 1), thus consuming short-circuit power.

(4) The AND gate with an isolated control signal (which is 0) comes from the power

management module to prevent unknown signals from entering the AON domain.

(5) **The OR gate with an isolated control signal (is 1)** comes from the power management module to prevent unknown signals from entering the AON domain. .

217. pre-CTS, the source of Clock Skew ?

In-chip process, voltage, temperature (PVT) changes

- Different clock buffers with different channel lengths
- Local voltage drop causes buffer delay to increase
- Hot spots cause increased gate and line delays
- Device mismatch for cross-chip clock jitter

218. What is Jitter and its sources?

Clock jitter is the clock edge inaccuracy of a clock signal generation circuit relative to an ideal clock.

Clock jitter can be viewed as the statistical variation in the clock period or duty cycle.

Sources of clock jitter:

Temporary power changes

1. Varying activity can change the supply voltage in different cycles affecting global or local clock buffers. ,

Phase locked loop jitter

1. Power supply changes to the PLL affect the oscillator frequency
2. PLL elements have no zero response time
3. PLL multiplied reference clock jitter
4. Global clock distribution may increase the jitter of the PLL due to jitter in the feedback clock signal caused by power supply noise.

line coupling

1. Changing the data can change the coupling of different cycles

dynamic offset circuit

219. What is the difference between clock buffer and regular buffer ?

Clock buffers

advantage:

- Since the on-resistances of PMOS and NMOS transistors are equal, the rise and fall transition times are equal. (This can be achieved by increasing the width of PMOS by 2.5 times the width of NMOS.)
- Clock buffer maintains pulse width**
- The clock buffer delay will be smaller than the regular buffer because the resistance of the regular buffer is 2.5 times the NMOS resistance. Therefore, the rise time of a regular buffer will be longer - this is not true (see spreadsheet below) The clock buffer actually has more latency than a regular buffer .

Cell	Process	Temperature	Voltage	Delay (s)	Leakage (W)	Internal Energy (J)	Avg. Pin Cap (F)
a16lvt16_inv_a8	SSG	0	0.72	1.5269E-11	3.1805E-10	2.9314E-15	7.31284E-15
a16lvt16_ckinv_a8	SSG	0	0.72	1.8572E-11	3.2959E-10	3.5544E-15	8.61849E-15
Ratio				1.216	1.036	1.213	1.179

CSDN @桐桐花

shortcoming:

- Due to the increased PMOS width, the area of the clock buffer is higher than that of the regular buffer , that is, the area loss is greater
- Since the on-resistance of PMOS is lower, the leakage current is more. Therefore, the leakage power of the clock buffer will be greater.

220. What is the Miller Effect ?

Miller effect

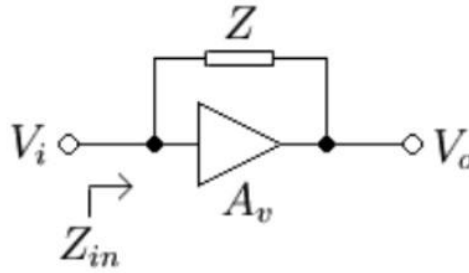
In microelectronics, in the inverting amplifier circuit,

Due to the amplification effect of the amplifier, the distributed capacitance or parasitic capacitance between the input and the output will increase its equivalent capacitance value to the input end by $1+K$ times, where K is the voltage amplification factor of the amplifier circuit.

Although generally the Miller effect refers to capacitive amplification, the impedance between any input and other high-amplification sections can also change the input impedance of the amplifier through the Miller effect.

[A brief discussion on the Miller effect in MOSFET - Zhihu \(zhihu.com\)](#)

假设一个增益为 $-A_v$ 的理想反向电压放大器



在放大器的输出和输入端之间连接一个阻值为 Z 的阻抗。容易得到,

$$Z_{in} = \frac{1}{j\omega C(1 + A_v)} = \frac{1}{j\omega C_M} \quad \text{where} \quad C_M = C(1 + A_v).$$

把阻抗 Z 替换为容值为 C 的电容,

$$Z = \frac{1}{sC}$$

$$Z_{in} = \frac{1}{sC_M} \quad \text{where} \quad C_M = C(1 + A_v)$$

由此可见, 反向电压放大器增加了电路的输入电容, 并且放大系数为 $(1+A_v)$ CSDN @桐桐花

The internal physical structure of a CMOS inverter has an internal feedback capacitor between the output drain and input gate, called C_{gd} .

According to Miller's theorem, C_{gd} appears at the input, multiplied by the amplifier gain $A+1$ (i.e., $C_{gd}(A+1)$).

This reduces the maximum operating frequency of the amplifier compared to without C_{gd} .

Then as mentioned before, C_{gd} can greatly limit the bandwidth of the amplifier.

If a CMOS inverter is used as a logic gate, the transistor acts as a switch.

In the on and off states, they are in a quasi-static state and the effect of C_{gd} is negligible. capacitance. The result is that the inverter switching speed is slowed down and the propagation delay time is increased.

In short, the MILLER effect of C_{gd} reduces the maximum operating frequency of the CMOS inverter.

We can use cascade connections (common sources in series with common gates) to reduce the Miller effect. Cascaded common gate improves input-output isolation (or reverse transmission) because there is no direct coupling between the output and the input. This eliminates the Miller effect and therefore helps improve bandwidth.

Device engineers can reduce this capacitance by reducing the overlap area between the gate

and drain. That is to technically minimize Cgd.

The load cell/receiver is a high drive strength cell (single-stage cell , such as an inverter) with a light output load (short net), so fast switching on the output, strongly coupled back to the input interconnect via Miller capacitance (similar to crosstalk) and cause substantial distortion at the input signal, such as delayed signal transitions.

Therefore, even if there is no external crosstalk, the Receiver here also acts as an aggressor driver . Therefore, it will affect the working frequency of cells .

Questions 221-230

221. Pre & post-route correlation

(1) In the pre-route stage (pre-route stage), the elmore delay engine is used by default (in the ICC compiler) to calculate the interconnect RC delay, and in the post-route stage (post-route stage), the Arnoldi delay engine is used Calculate interconnect RC delays.

So we should check the type of delay engines we use in the pre-route stage .

For better correlation with post-route , we must use AWE (Asymptotic Waveform Evaluation) delay engine in the pre-route stage.

(2) In the pre - route stage , the coupling capacitance is not considered, so there is no crosstalk effect in the pre-route stage. In the post-wiring stage, the impact of crosstalk appears. So there will be timing correlation problems.

From the route stage & post-cts optimization stage, the timing of the same path was reported and it was seen that crosstalk was the main cause of bad correlation .

If so, then try to reduce congestion in your design and you will see improved timing correlation.

(3) Increasing the uncertainty value in the post-cts stage and performing post-cts optimization may improve the correlation.

But this approach can also over-optimize non-critical timing paths. This is like unnecessarily increasing area overhead by optimizing non-critical timing paths

Before that, if you see a large number of timing violations against the post cts DB, find the timing violation path in the route phase and increase the uncertainty value.

If the number of violating paths is small, check the reason for the violation. Routing all those timing-critical nets using NDR, or routing using higher metal layers, may help you resolve these violations.

222. Post-route & Signoff timing correlation

(1) . Prime-time always uses the Arnoldi delay engine to calculate the interconnect RC delay. So for post-route designs, always turn on the Arnoldi delay engine to get better timing correlation (ICC)

(2) Sometimes the derate used in the implementation tool is inconsistent with the signoff PT . Use derating to report timings for the same path from the PT and implementation tools, and then check the derating consistency between the two tools. If they do not match, use appropriate derating in the implementation tool for timing correlation .

(3) If you find no correlation by directly comparing the timing path of ICC Compiler with that of PrimeTime, try to narrow down the problem by reading the same parasitics file in ICC Compiler , and then find out the RC scaling factors . Timing correlation may be improved using these new RC scaling factors .

ICC Compiler execution command:

```
extract_rc  
report_timing
```

PrimeTime execution command:

```
read_parasitics  
report_timing
```

(4) Report all time series variables in Signoff PT and compare them with variables in the implementation tool. Try modifying these variables in the implementation tool to get better timing correlation.

You will see the same variables in ICC and PT-SI, but we may not see similar variables in other implementation tools, but we may find variables that provide similar functionality as in PT.

(5) The implementation tool uses GBA by default. This is a more pessimistic approach. And PT runs on GBA and PBA. So if the PT uses PBA, change it to GBA. This way you'll see better timing correlation.

GBA (Graph Based Analysis):

A unit's latency will be calculated based on the worst-case slew propagation through one of its input pins .

PBA (Path Base Analysis):

Cell delay will be calculated based on the actual propagation through the input pins

(6)

✘ Check the correlation between IC Compiler and PrimeTime tools in ICC, and the correlation between ICC Compiler and StarRC tools, including timing and signal integrity related variables, commands and SDC settings in IC Compiler and PrimeTime tools.

Run command:

```
check_signoff_correlation
```

✘ To check only the correlation between IC Compiler and Prime Time settings, use the command

```
check_primetime_icc_consistency_settings
```

rather than a command

```
check_signoff_correlation
```

(7) Use the same input data b/w ICC & PT, such as netlist, SDC, reference library, operating conditions

223. ICC Vs Extraction Correlation

(1) Make sure to use the same **ITF file** to generate **TLUPlus** and **nxtgrd** files

(TLUPlus files will be used for nxtgrd files in ICC and StarRC)

(2) Ensure that both **nxtgrd** and **TLUPlus** files are generated using the same **grdgenxo** version

(3) Use **OPERATING_TEMPERATURE** to pass the operating temperature information to StarRC.

ICC Compiler automatically obtains this information from operating conditions.

(4) Make sure to provide pin capacitance information to StarRC.

IC Compiler obtains pin capacitance information from .db file

StarRC obtains pin capacitance information from the db file existing in the LM view.

If LM views do not exist, specify the pin capacitance information through **SKIP_CELL_PORT_PROP_FILE** in StarRC .

(5) Turn off vIRtual shield extraction

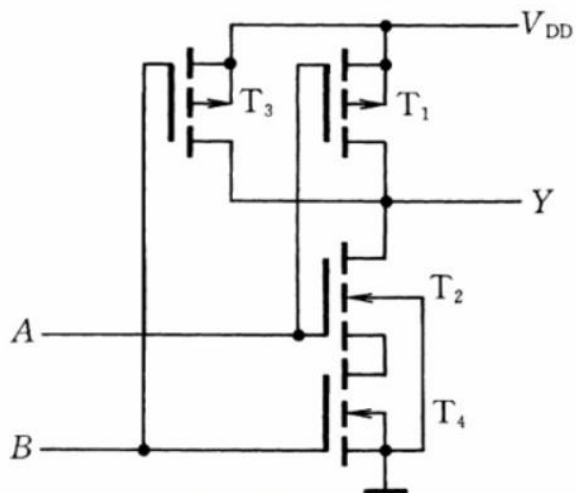
set_extraction_options -vIRtual_shield_extraction false

(6) Ensure that the metal filling settings between IC Compiler and StarRC are consistent.

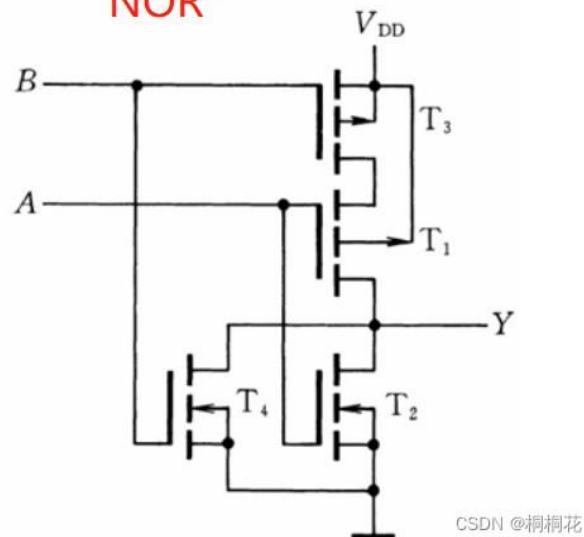
It is recommended to perform correlation without metal filling in ICC and StarRC.

224. Use CMOS to draw NAND/NOR gate

NAND



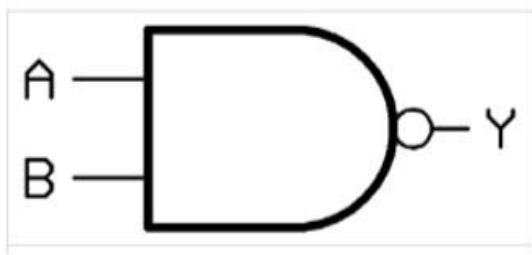
NOR



CSDN @桐桐花

NAND gate NAND:

逻辑符号



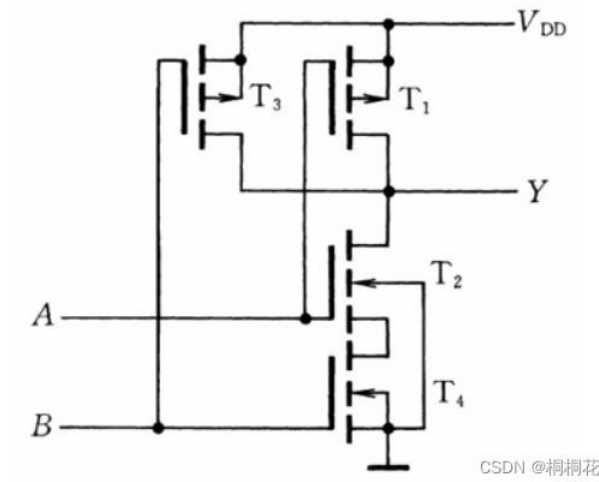
真值表

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

逻辑表达式

$$Y=(A \cdot B)'=(A')+(B')$$

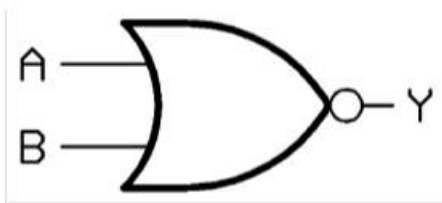
CSDN @桐桐花



CSDN @桐桐花

NOR gate NOR

逻辑符号



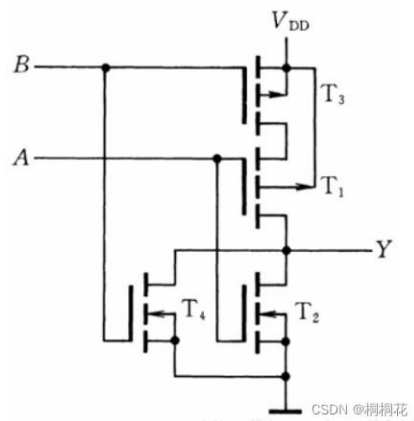
真值表

输入A	输入B	输出Y
0	0	1
0	1	0
1	0	0
1	1	0

逻辑表达式

$$Y = \overline{A + B}$$

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CSDN @桐桐花

225. Why do the PMOS and NMOS in the transmission gate always have the same area?

CMOS transmission gate:

(1) A CMOS transmission gate can be formed by a parallel combination of NMO and PMOS transistors with complementary gate signals.

(2) Compared with NMOS transmission gates, the main advantage of CMOS transmission gates is that they allow the input signal to be transmitted to the output without threshold voltage attenuation.

The advantage of using a complementary pair, rather than a single NMOS or PMOS device, to implement a transmission gate is that the gate delay time is almost independent of the voltage level

of the input variable of the CMOS transmission gate.

(3) In some cases, you will find that the reason why the beta ratio is equal to 1 is that when the transmission gate is open, both PMO and NMOS are open and in parallel.

Although only PMO is good at passing '1' and NMOS is good at passing '0', they are indeed weak at passing opposite logic levels.

Therefore, the total average resistance ratio is only one Moson. So you don't need $\beta=2$ like a CMO inverter, but a value like 1.5 or 1 is enough.

Using a transistor as a switch between a driver circuit and a load circuit is called a transmission gate because the switch can transmit information from one circuit to another.

The bias applied to the transistor determines which terminal acts as the drain and which acts as the source.

****NMOS transmission gate:****

- NMOS gate connected to variable V_g
- One terminal is connected to the load CL
- One terminal will be connected to the input voltage V_{in} .
- The output will be obtained through load CL .

When $V_{in} = V_{dd}$

(1) $V_g = 0$

The terminal of $V_{in}=V_{dd}$ will be the drain and the other terminal of the load CL will be the source.

So $V_{gs} = 0 - 0 = 0$. NMOS is off and $V_o = 0$

(2) $V_s = V_{dd} - 0 = V_{dd}$,

So the NMOS turns on (i.e. $V_{gs} > V_t$) and starts charging the load CL to V_{dd} .

Once the output reaches $V_{dd} - V_t$, charging will stop because at this point the voltage V_{gs} will become V_t .

So for $V_{in} = V_{dd}$, $V_o = V_{dd} - V_t$, it is attenuated by V_t .

This is one of the shortcomings of the NMOS transmission gate when $V_{in}=V_{dd}$.

When $V_{in} = 0$:

(1) One end of $V_{in}=0$ serves as the source, and one end of the other load CL serves as the drain D .

(2) If $V_g = V_{dd}$,

Then $V_{gs} = V_{dd} - 0 = V_{dd}$, so NMOS is on (because $V_{gs} > V_t$),

So the voltage across load CL will start discharging through NMOS & source

So when $V_{in} = 0$ the output $V_o = 0$.

This means that when $V_{in} = 0$ the NMOS transistor provides "good" logic 0

That's why we are not using NMOS as transmission gate because it is producing output value $v_{dd} - V_t$ when $V_{in} = V_{dd}$

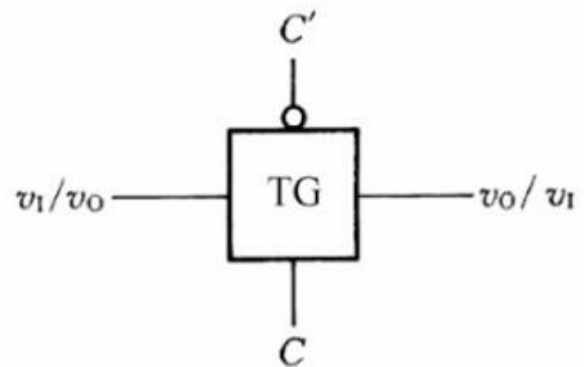
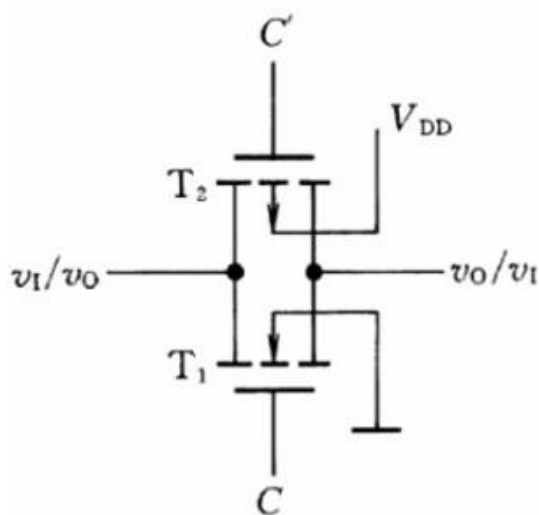
CMOS transmission gate :

The CMOS transmission gate is composed of a PMOS and an NMOS tube connected in parallel, which has a very low on-resistance (a few hundred ohms) and a high off-resistance (more than 10^9 ohms).

T1 is an N-channel enhancement type MOS tube, and T2 is a P-channel enhancement type MOS tube.

The sources and drains of T1 and T2 are connected as the input and output terminals of the transmission gate respectively. **(TP and TN are structurally symmetric devices, and their drains and sources are interchangeable.)**

C and C' are complementary control signals.



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Since the structure of the CMOS transmission gate is symmetrical, the output and input terminals can be interchanged, making it a bidirectional device.

Basic knowledge of digital circuits - CMOS gate circuit (NAND gate, NOR gate, NOT gate, OD gate, transmission gate, tri-state gate)_The blog of ferrying vicissitudes-CSDN blog_cmos gate circuit

Set the turn-on voltage of TP and TN $|V_T| = 2V$, the variation range of the input analog signal is -5V to +5V.

Prevent current from flowing directly into the substrate from the drain, causing a PN junction to be reversely biased between the substrate and the drain-source:

- The substrate of TP is connected to +5V voltage,
- The substrate of TN is connected to -5V voltage.

The gates of the two tubes are controlled by complementary signal voltages (+5V and -5V), respectively using C and ! C means.

The transmission gate works as follows:

When terminal C is connected to low voltage, the switch is open.

When the C terminal is connected to a low voltage of -5V, the gate voltage of TN is -5V.

When V_i takes any value within the range of -5V to +5V, TN does not conduct. At the same time, the gate voltage of TP is +5V, and TP is not conducting.

In order to turn on the switch, the C terminal can be connected to the high voltage +5V.

At this time, the gate voltage of TN is +5V, V_i is in the range of -5V to +3V, and TN is turned on.

At the same time, the gate voltage of TP is -5V, and TP will be turned on when V_i is in the range of -3V to +5V.

It can be seen from the above analysis that

- When $V_i \leq -3V$, only TN is turned on,
- When $V_i \geq +3V$, only TP is turned on
- When V_i is in the range of -3V to +3V, both TN and TP tubes are turned on.

226. Explain CEL and FRAM view?

CEL is a complete view of the design including all layers (such as GDS), and FRAM is only a schematic view of the design (such as lef)

CEL view

A complete layout view of the physical structure, such as through-hole, standard cell, macro, or entire chip; includes cell placement, routing, pinout, and netlist information.

1. All cell information required for placement, routing, and mask generation.
2. Placement information, such as tracks, site rows, and placement blocks;
3. routing information such as netlist, pinout, route guide, and interconnect modeling information, as well as all mask layer geometries used for final mask generation.

FRAM view:

The FRAM view is an abstraction of the cell and contains only the information required for placement and routing: areas of metal blockages that are not allowed to be routed, areas of vias that are allowed, and pin locations.

The process of creating a FRAM view from a CEL view is commonly called blockage, pin, and via (BPV) extraction.

The FRAM view is used for placement and routing, while the CEL view is used only to generate the final mask data flow for chip manufacturing.

227. What is the reason for connecting n-well to VDD and p-substrate to VSS?

Prevent forward biasing drain to n-well junction and source to p-substrate junction.

228. Well Edge Proximity (WEP) effect

Because ions scatter from the photoresist mask to the edge of the well, transistors close to the retrograde well (eg, within about 1 nm) may have a different threshold voltage than transistors farther from the edge. This is called the edge proximity effect.

229. Why is Metal Density Rule needed?

We must maintain the minimum and maximum density of a specific layer within a designated area. The etch rate has some sensitivity to the amount of material that must be removed.

For example, if the polysilicon density is too high or too low, the transistor gate may end up being over-etched or under-etched, causing channel length changes. Similarly, CMP processes can cause pitting (over-removal) of copper when the density is uneven.

To prevent these problems, the metal layer may need to have a minimum density of 30% and a maximum density of 70% in an area of 100m X100m. Diffused polysilicon and metal layers may have to be added manually or through a fill procedure after the design is complete.

Fills can be grounded or left floating:

- Floating fill: Helps reduce total capacitance, but has greater coupling capacitance to nearby wires.
- Grounded fill: A ground mesh needs to be routed to the fill structure.

Metal fillings can come in a variety of shapes and sizes.

Typically, there are two types of metal-filled structures in designs:

- Grounded metal fill: connected to power or ground through vias
- Floating metal fill: No connection to signal, power or ground networks

Both types may exist in the same layout.

When running a StarRc extraction, you can specify whether you want simulated or real metal filling.

These two methods produce different results based on accuracy requirements.

Simulation fill is used only in the early stages of the place and route process and should not

be used in conjunction with the place and route process.

LEF/DEF There are two different forms of syntax for specifying metal fill.

Floating metal filled polygons are specified in the "FILLS" section of the DEF file.

If filled polygons are connected to power and ground nets, they are specified in the SPECIALNETS section of the power and ground nets (**part of the special WIRing, shape defined as FILLWIRE**) .

Metal fillings can be processed in two ways:

- **as ground metal filler**
- **floating metal filling**

Ground metal filling

During signal network extraction, filled polygons are treated as polygons belonging to power and ground networks.

No special treatment was performed on these polygons during ion extraction.

floating metal filling

In this mode, the capacitance between the signal and filled polygons and between different filled polygons is calculated.

Filled nodes are reduced on the fly and the equivalent capacitance between the signal nets and the ground capacitance of the signal nets is calculated.

A metal fill is said to be floating if it is not connected to any circuit element in the netlist.

The potential on the filled mesh is effectively determined by setting the charge on the filled mesh to zero.

Even if filler networks are not electrically connected, they can introduce capacitive coupling effects between other networks

****Gate and Diffusion Capacitance: ****

Diffusion capacitance depends on the size of the source/drain regions. Wider transistors have proportionally greater diffusion capacitance. Increasing the channel length increases the gate capacitance proportionally but does not affect the diffusion capacitance.

230. Explain ECO Extraction

ECO extraction is a technology that extracts only the parts of the design that are different from the reference design.

The StarRC ECO extraction process only performs extractions on networks that have been changed by ECO fixes, significantly reducing the overall run time. During this flow, the StarRC tool maintains two parasitic netlists:

- One for whole chip extraction
- One for ECO-affected networks

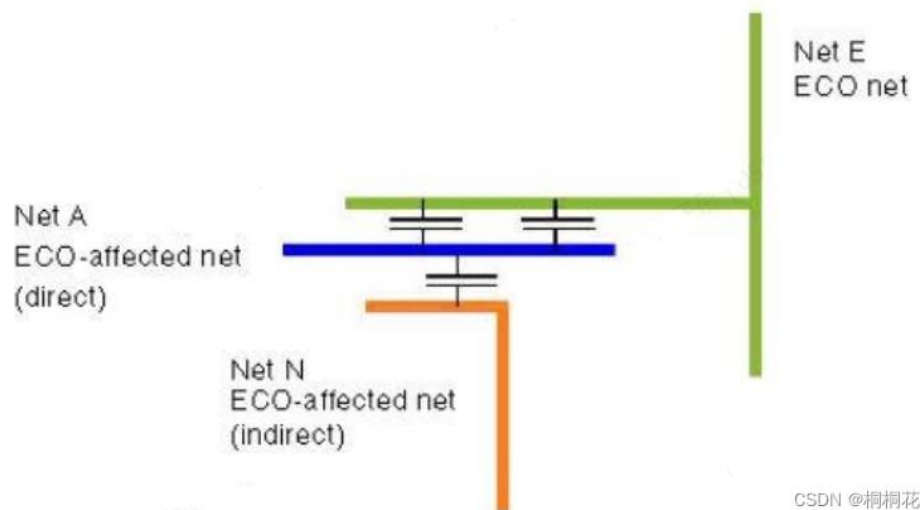
ECO extraction performs re-extraction through an intelligent selection network, achieving the same extraction accuracy as full-chip extraction. In addition to the nets directly modified by ECO, the tool also selects neighboring nets based on their coupling capacitance to the ECO net.

The COUPLE_TO_GROUND command should be set to NO in the ECO process.

Net E, ECO net, is a net that was modified as part of a timing violation repair.

Net A is coupled to Net E; this type of net is an ECO-affected net.

Net N is not directly coupled to Net E, but to Net A. This type of net is a net indirectly affected by ECO.



The StarRC tool includes net E and net A, but not net N, in the ECO netlist (the file specified by the NETLIST_ECO_FILE command). The coupling capacitance between net A and net N is displayed as a floating capacitor under net A in the ECO netlist.

The PrimeTime tool can read the ECO netlist and adjust the coupling and total capacitance of network N accordingly.

If you do an LVT swap on a unit, the StarRC tool will not re-extract the connected wires or replace them in the netlist.

The netlist from the full-chip extraction is written to the file specified by the NETLIST_FILE command. The netlist from ECO extraction is written to the file specified by the NETLIST_ECO_FILE command.

The ECO_MODE command controls ECO extraction. Options are YES, RESET and NO (default).

The first StarRC run is always a full-chip extraction, since a reference run must exist for ECO extraction. Subsequent StarRC runs can be either full chip or ECO extraction, depending on the number of ECO-affected nets compared to the design size.

The StarRC tool maintains the complete netlist and the ECO netlist; the PrimeTime tool can read both netlists and use them appropriately. .The

netlist from the full-chip extraction is written to the file specified by the `NETLIST_FILE` command. The netlist from ECO extraction is written to the file specified by the `NETLIST_ECO_FILE` command.

An ECO extraction will be performed unless one of the following conditions applies: The StarRC tool does not perform any extraction, nor does it update the netlist, if the design database has not had any logical or physical changes since the last extraction.

If the StarRC directory is missing, the tool will perform a full chip extraction since there is no reference run.

Questions 231-240

231. STAR-RC& PT ECO Flow.

STAR-RC Flow.

StarRC command file for ECO Extraction of MW DB.

```
MILKYWAY_DATABASE: CPU.mw
BLOCK: top_block_rev0
ECO_MODE: YES
STAR_DIRECTORY: star
NETLIST_FILE: pre_eco_full_chip.spef
NETLIST_ECO_FILE: post_eco_incr. spef
SUMMARY_FILE: pre_eco.star_sum
```

Execute StarRC run

The first run is a full chip extraction, and the generated netlist is saved under the name specified by the **NETLIST_FILE** command.

An empty netlist is saved to the file specified by the **NETLIST_ECO_FILE** command.

PT ECO Flow

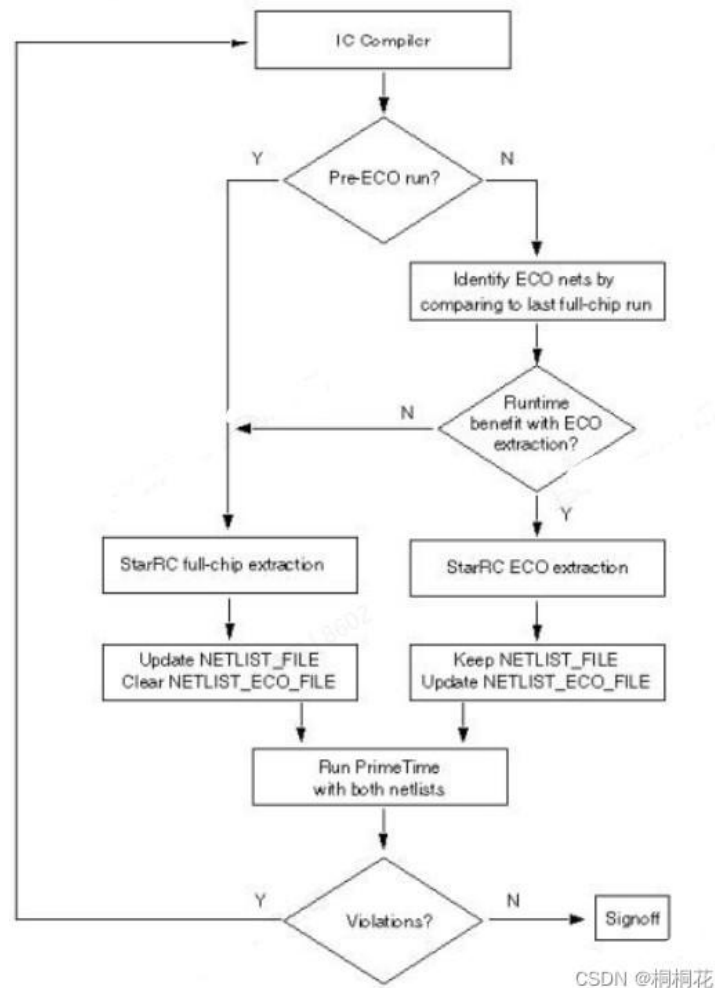
Modify the read_parasitics command in the PrimeTime script to include the full and ECO netlist names as follows:

```
read_parasitics -format spef -keep_capacitive_coupling \
```

```
./pre_eco_full_chip.spef \  
-eco ./post_eco_incr.spef
```

Execute PrimeTime run

Fix timing violations and save changes in the design database.



232. Do we perform dynamic/static IR drop analysis for all modes (function, scan capture, scan shift)?

We take the use case scenario of a chip that consumes major power.

Therefore, we adopt this VCD and will close it. There can also be multiple VCDs running in parallel.

233. At how many corners will we conduct IR drop analysis?

Dynamic power depends on "load capacitance"

Therefore, we choose the corner with more impedance. (rcworst corner or cworst corner)

234. What is the static IR drop analysis/dynamic data switching rate?

It just depends on the design.

235. Is there any relationship between dynamic IR drop targets and static IR drop targets?

(Dynamic IR drop target $\leq 3 \times$ static IR drop target).

Static is based on average power, dynamic is based on peak/average/rms current waveform analysis.

236. What is ramp up voltage? How will it change? How to calculate? How does it affect IR drop analysis?

Ramp/wake up time: The time required to bring the voltage from 0 to peak during system startup time.

In the ramp up analysis process we will get the wake-up time and voltage, and how many cells are switching and not switching.

If all designs switch at the same time, then demand will exceed supply and the IR drop will be larger, so we prefer to connect all Power Switch units in a chain and start in series.

237. off_state_leakage current? How will the IR drop be affected?

By using header power switches we can get relief when they are off.

If we are doing power gating, then the leakage current will be controlled, so the leakage power will be automatically controlled in the design.

238. What are the advantages and disadvantages of multi bit flip-flop (multi-bit flip-flop) design?

Replacing single-bit cells with multibit cells has the following benefits:

1. Area reduction due to shared transistors and optimized transistor level layout.

Due to transistor-level optimization of cell layout (which may include shared logic, shared power connections, and shared substrate wells), the area of a 2-bit cell is smaller than that of two 1-bit cells. The register bits assigned to the bank must use the same clock signal. and the same control signals, such as preset and clear signals.

2. Reduce the total length of the clock tree network.

Due to the lower net length, the combined flip-flop reduces the dynamic power consumption by approximately 23.68% and the total power consumption by approximately 8.55%. It was also found that the global clock buffer dropped to 37.84%.

3. Reduce clock tree buffers and clock tree power consumption

4. This should also reduce the clock skew of sequential logic gates because the clock paths are balanced within the entire multi-bit cell.

5. SoC implementation using multi-bit flip-flops should result in smaller SoC area because the number of total clock buffers is reduced, thereby reducing congestion.

6. Due to shared logic (clock gating or set-reset logic) and optimized multi-bit circuits and layout from library team, the use of multi-bit should improve timing numbers.

shortcoming:

1. IR Drop problem

2. EM Violations

3. The LEC difficulty is the same as the LEC difficulty when comparing names. So you need to have svf files to map multibit fops to their single bit flops

239. Will jitter affect hold violations?

Jitter is the time variation of a periodic signal.

Since we are checking hold violations on the same clock edge (start and capture edges of the same flip-flop), jitter will not affect hold violations if there are no unusual clock paths.

If there is an unusual path between the transmit and capture clocks (due to distributed jitter), jitter may cause hold violations.

240. Will jitter affect setup violations?

Jitter affects setup violations because we do check for setup violations on the clock edges of launch flops and capture flops.

Questions 241-250

241. For complex or scattered floorplans, what are the recommended settings?

For designs with complex or scattered floor plans with narrow passages, use the following settings:

(1) The place_opt command enables the high fan-out synthesis function based on global-route, which can improve routing and reduce congestion.

```
set_app_options -list {place_opt.initial_drc.global_route_based 1}
```

(2) The place_opt command enables two-pass flow, which will generate better initial placement.

```
set_app_options -list {place_opt.initial_place.two_pass true}
```

(3) Enable global routing during initial clock tree synthesis, which can improve congestion estimation and perform the construction of congestion aware clock tree.

```
set_app_options -list {cts.compile.enable_global_route true}
```

Notice:

Steps (1) and (2) only apply if the Synopsys physical guidance (SPG) process is not used

242. What kind of netlist processing is done when carrying out floor plan? (What will you look for netlist while floor planning?)

1. looking for macros
2. Group macros acc into hierarchy
3. Separate macros based on number of pins
4. Make sure the macros in the substrate and well have VDD/VSS set.
5. Place soft blockages in channels to avoid placing flops.
6. If necessary, add density screen.

243. TCL palindrome processing (TCL proc for palindrome)

```
proc check_palindrome {in_str} {  
    set str_len [string length $in_str]  
    set str_mid [expr $str_len/2]  
    for {set i 0} {$i < $str_mid} {incr i} {  
        if {[string index $in_str $i] != [string index $in_str end-$i]} {  
            puts "$in_str is not a palindrome"  
            return  
        }  
    }  
    puts "$in_str is a palindrome"  
}
```

244. How to open a file and print the lines starting with an error?

```
set f [open x.txt r]  
while { [gets $f line] > 0 }  
{  
    if { [regexp {^Error} $line]}
```

```

|         { |           puts $line
|     else {
|         continue
|     }
| }

```

245. Why only cell delays have aocv but wire delays do not? (wire delays are flat derates, no aocv).

What about wire delays? They depend on factors like OPC, which is also random? (Line width changes based on OPC technology, such as etching, etc.). This etching process is not uniform across the entire chip and varies from location to location depending on the metal density in that area. This means that the metal width is due to less etching and if it is over etched its width will be reduced. Wire delays change from one place to another, right? Is this also random? Is this the reason why we use flat derates?

(1) Because cells are doped, this is affected by the process. AOCV is basically a function of (V_t , doping) or process.

(2) The etching effect is handled by the nxgrd characterization, and the etching values are listed in the form of tluplus/nxtgrd (delays of wire changes relative to thickness (delays of wire changes wrt thickness, net length and temperature, etc.)).

(3) Why do we apply flat derates or ocv to wires, why do we not apply aocv or variable derates?

Wire delays vary only due to local process and temperature (not voltage). If the timing path spans a huge distance, the local temperature of one net n1 may be different than that of net n2 at a different location (assuming the thickness of nets n1 & n2 and the length of the nets at both locations are the same). To solve this problem, we have to use flat derates or OCV for wires.

246. If you have an always on domain and a switching domain, where would you place the isolation cell?

In normally open domain (AON domain)

247. What is the composition of power consumption?

Dynamic Power:

- Short circuit power
- Switching power due to external capacitance
- Switching power due to internal capacitors within the boundary of a cell

Static or Leakage Power:

- When the NMOS tube is connected to VSS, and the PMOS tube is connected to VDD, the sub-threshold current flows from drain to source when NMOS is at VSS, PMOS is at VDD.
- Reverse saturation current through PN junction diodes formed b/w N-well & p substrate
- Gate tunneling leakage current

248. How to reduce dynamic power consumption (Dynamic power)?

technology:

(1) Multiple power supply voltages or reduced power supply voltages

- Multi-frequency or DVFS or reduced frequency
- Reduce switching activity on nets

(2) Reduce load or wire capacitance by minimizing net length

(3) Use the **place.coarse.icg_auto_bound** command to place the register near its driver ICG to improve transitions on nets.

(4) Use decoupling capacitors: It helps reduce power transients on the die and reduces dynamic or active power in the design, but it increases leakage power.

(5) Use clock gating, xor self-gating, power gating and multi-bit flops to handle dynamic power.

(6) Area Recovery optimization: Reduce the driving intensity of non-critical timing paths. Make their input pin capacitance decrease (CL is the sum of wire cap, driver inherent capacitance and fan-out load input pin capacitance). So dynamic power consumption will be reduced.

(7) If we provide a SAIF file based on gate-level or RTL simulation by enabling **power_low_power_placement** and **set_dynamic_optimization**, the tool optimizes the dynamic power by minimizing the length of the net of high-intensity switches to improve power QoR. (Or annotate **set_switching_activity** on the design without SAIF file)

The SAIF file has the static probability and toggle rates for each signal network in the design, and the sdc will have the toggle rates and static probabilities for the clock network.

(8) Reduce unnecessary pessimism in setup/hold uncertainty (few people run less timing scenarios by maintaining huge uncertainty) & use POCV (this will reduce the number of instances (instance count) and its internal short circuits power).

(9) Use logical reconstruction to decompose high-activity networks: Boolean factoring decomposition can be applied to high-activity networks to decompose high-activity networks. This minimizes the amount of logical fan-out for high-activity networks.

(10) Pin exchange: In some cells, pins with equivalent functions can have different input capacitances. In this case, it is beneficial to move the high-activity net to a pin with lower capacitance.

249. Where to get activity factor?

VCD or SAIF file

250. How to determine activity factor f for input ports?

This information is from vcd or based on IO constraints we follow 30% of the clock cycle

Questions 251-260

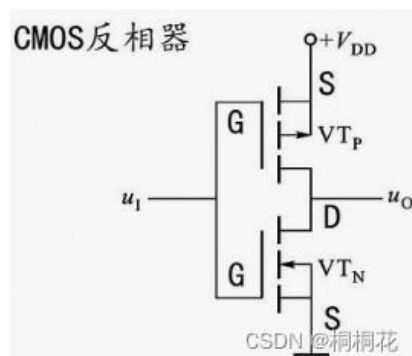
251. If the transition of a net you take on the macro fails. How do you fix a transition with minimal rewiring?

- ImproveLayer
- increase driver strength
- Reduce sinks size
- split load
- Remove the nets/nets around the violating net that have positive setup/hold margins and slew margins, reroute it globally, see if there are any alternatives to routing the macros around the net, and then buffer it.

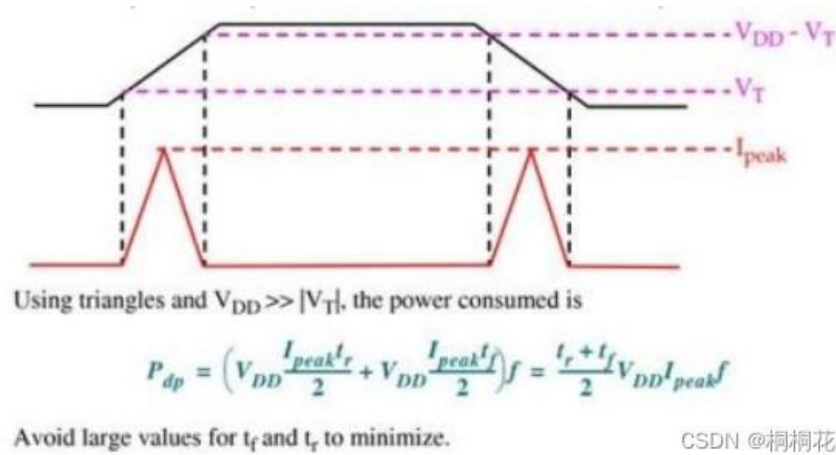
252. What is short-circuit current? How does transition affect short circuit current?

Short circuit current:

Add a rising signal to the input of the CMOS inverter. As the signal transitions from low to high, when $V_{gs} > V_{tn}$, the P-type transistor turns off and the N-type transistor turns on.



During the signal conversion process, when $V_{tn} < V_{gs} < V_{DD} - V_{tp}$, the P-type and N-type transistors can be turned on at the same time in a short time. During this period, current I_{sc} flows from V_{dd} to GND. This current is called a short-circuit current and causes the dissipation of short-circuit power. Or when the input voltage is between V_{tn} and $V_{DD} - |V_{tp}|$, both PMOS and NMOS are open circuit for a short time, causing current to flow from V_{DD} to V_{SS} . This is called short-circuit current, and the power consumed is called short-circuit power.



For circuits with fast transition times, the short-circuit power can be very small. However, for circuits with slow transition times, the short-circuit power can account for up to 30% of the total power dissipated by the gate. Short circuit power is affected by transistor size and gate output load capacitance.

253. What is AWP (advance wave propagation)

Due to the Miller effect and the long tail effect, the signal waveform will be distorted when the receiver inputs it. If we do not consider these effects, Cell delays are optimistic. We need to enable AWP in time.

delay_calc_waveform_analysis_mode for better and accurate unit delay values. For this, we need to provide ccs timing models and ccs noise models.

254. In some cases, you only have the timing report after the route. What factors would you consider to improve the setup timing?

First, check whether the timing path is a true path or a false path by checking the clock.

If both emission and capture belong to the same clock group, then see if their clock pins are well balanced and whether the skew is smaller. If there are fewer common clock paths, the skew will be larger and the ocv derates will make timing closure difficult.

Likewise, if launch and capture belong to different clock domains, check with the designer whether this path is valid. If it is valid, then these clocks are unbalanced and not in the same skew group, then try to use better skew improvements to build CTS. After this, the data path is checked. Check the data path for any low drive strength cells with bad transitions, meaning that the tool is not doing a better job or the tool cannot increase the size of these cells due to high cell density. If the data path has a large buffer/inverter chain, detours occur due to congestion in the design and DRV is fixed by adding buffer chains. Use this method to resolve congestion and improve timing.

(1) Check the delta derate delay. If the delay value is relatively large, reduce the net length during optimization. If you see a lot of clocks, use better NDR rules.

(2) By avoiding the use of cells with low driving intensity, more cell delays can be improved.

(3) Check the skew. If the skew is relatively large, it can be improved during CTS.

(4) You need to check the setup time and clk->q access time of flop or memory.

255. If there is a larger Input transition, what are the design problems?

(1) Larger output transition

(2) Larger cell delay

256. What is the difference between drive strength and fanout?

a. Driving capability:

i. What is the maximum capacitance a device can drive?

ii. max_capacitance in .lib file

b. Fanout:

i. These are the gate numbers of the driver

ii. Device models will determine fan-out

iii. Technological changes also determine fan-out

257. What are the requirements for models in CMOS design?

a. Faster calculation speed

b. Accurate calculation

258. What is the input of .lib?

a. Input transition

b. Output load

259. What is the output of .lib?

a. Output Transition

b. Cell delay

260. Why are the input capacitances of larger drivers similar?

a. All STD cells are designed as two stages [1stage => 2stage]

b. The first stage is mainly for the function of the cell

c. 2 The second stage is for the driving capability of the cell.

- d. For larger drive cells, only the drive capability is enhanced in stage 2
- e. Since there is not much change in the first stage, the input capacitance is similar

Questions 261-270

261. Why does .lib use LOOKUP table?

- a. easy to obtain
- b. Memory memory consumption is less
- c. Lower precision for out-of-range inputs

262. What information can be obtained from the lib file?

- a. Input capacitance
- b. Leakage power consumption under all conditions
- c. Cell function
- d. Output transition time displayed in the form of query table (Output transition)
- e. Cell delay table
- f. Insertion delay table for macros [Always verify this during the CTS phase]

263. What information does the lef file contain?

- a. SIZE
- b. TILE
- c. Symmetry
- d. OBS
- e. PIN BOX with layer
- f. Origin

264. Why does a thicker metal layer require more space?

Thick metal layers have more potential for changes during manufacturing.

265. What is the function of TLU-PLUS file?

The TLU-PLUS file contains RC models, and ICC uses the TLU-PLUS file to calculate RC delay.

266. What is a mapping file and where is this file needed?

- a. The mapping file is an aliasing file that helps other tools understand the metal layers used in the PD tool.
- b. STAR-RC uses this file.
- c. Dump the GDS file from the tool. (To dump GDS file from tool.)

267. How to create a basic SDC for any new design?

- a. Import Verilog files in ICC
- b. Get all clock ports in the design
- c. Define clock in these clock ports
- d. Run the check_timing command and clearly report the error
- e. Define IO conditions
 - i. set_input_deriving_cell
 - ii. set_input_transition
 - iii. set_load
 - iv. set_input_delay
 - v. set_output_delay

268. How to verify SDC?

a. Multi-cycle paths

- i. Setup 3 cycles
- ii. Hold 2 cycles

b. 2 synchronizers (2 flop Synchronizer)

- i. Always have 2 cycles of Multi-cycle to capture data
- ii. Hold check will have multiple 1 periods
- c. Check all false paths in SDC

269. What will happen if an AND gate does not have a timing arc in the library and is used in the design?

- a. For these paths through AND cells, the design will fail because the path will not be seen and optimized.
- b. There is a warning in the log file about this problem of missing timing arc.

270. Create_generated_clock -invert/-combination

- a. By default **create_generated_clock** goes through sequential cells
- b. By specifying combinational options, the generated clock is tracked through the combinational logic unit.

Questions 271-280

271. Things that need to be reset in the design

a. POWER related errors

- i. `reset_upf`
- ii. `load_upf ORCA_TOP.upf`

b. operating condition

- i. `get_app_var continue`
- ii. `set_app_var continue_on_operating_mismatch true`

c. After placemet, cells overlap

- i. `remove_sdc`
- ii. `remove_scenario -all`
- iii. `source mmmc.scenario.tcl` (found in prj1.FP.manual.tcl)
- iv. `remove_pnet_options`

272. Limitations of power via

- a. According to metal width, select vvia array
- b. Depending on the technology, via staking will be selected.

273. Why is routing blockage defined at the chip edge (DIE edge)?

- a. This is mainly for tools to avoid any routes routing out of the chip area (DIE are).

b. This also helps to keep routes within the scope of the design to avoid drc at chip (DIE) boundaries.

274. Why are TAP cells needed?

- a. LATCH problem.
- b. 65nm process, TAP cells started to be used.
- c. For 65nm and above processes, each STD cell has its own body connection.
- d. It is difficult to shrink STD cells with body bias contact.
- e. Body bias connection needs to be at um interval (interval of um)
- f. Shrinking CMOS is easier by maintaining body bias contact.
- g. TAP cell provides this body bias connection.
- h. By placing them in checker board fashion, the total cells requirement is reduced to about half.

275. What are the types of physical only cells in design?

- a. ENDCAP: end nwell ending
- b. TAP cells: body bias contact
- c. DECAP: local power source
- d. SPARE cells / programmable cells
- e. FILLERs: nwell continuity
- f. ESD cells: used as MACRO in flipchip design
- g. Navigation marker cells
- h. Foundry cells used to test metal layer and CMOS layers

276. Why do all edges need to leave space between core and DIE?

- a. PORTs
 - b. Avoid shorts b/w blocks between substrate and well (Avoid shorts b/w blocks)
 - c. Avoid noise interference between the substrate and the well (Noise b/w blocks)
- b/w: base/well

277. What are the factors that limit Marco's direction in design?

POLY manufacturing accuracy

278. Does Macro need to be aligned with STD rows?

unnecessary

STD ROW is a reference grid provided for the placement engine to place STD cells, not for MACROS.

279. Can macros be placed at DIE boundaries?

Yes, if the macro has a space rule from core to DIE boundary (CORE -DIE boundary spacing rule), it can be placed like this.

280. In FULL chip, can there be macro in the IO placement area?

It is possible for IO pads to have proper power planning

Questions 281-290

281. How to determine the metal layer for power planning?

- a. Frequency
- b. Architecture (CPU, DSP): switching
- c. IR drop

282. Top down vs bottom -up

- a. Priority is design closure
- b. Data comes from top to bottom
- c. Feedback goes bottom to top
- d. FROM TOP
 - i. DIE initial shape
 - ii. All initial ports placement
- e. To TOP;
 - i. Final design shape
 - ii. Final ports placement